

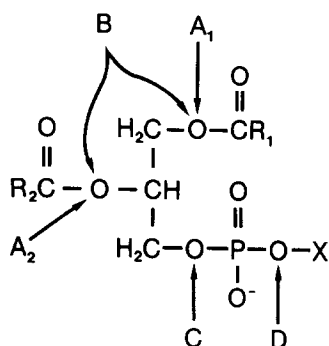


Fig. 2. Sites of hydrolysis by phospholipases.

cellular membranes requires the action of one or more phospholipase. Their functions go beyond their role in membrane homeostasis; they also function in such diverse roles from the digestion of nutrients to the formation of bioactive molecules involved in cell regulation. There are indications that a few phospholipases may carry out a biological function independent of their catalytic activity by binding to a regulatory membrane receptor. Phospholipase-like proteins with toxic properties, yet which lack a functional catalytic site, are found in venoms. It is of interest that most, but not all, phospholipases studied in detail thus far are soluble proteins. The soluble nature of many phospholipases suggests that their interaction with cellular membranes is one of the regulatory mechanisms that exist to prevent membrane degradation or to precisely control the formation of phospholipid-derived signalling molecules.

The classification of the phospholipases, based on their site of attack, is given in Fig. 2. The phospholipases A (PLAs) are acyl hydrolases classified according to their hydrolysis of the 1-acyl ester (PLA₁) or the 2-acyl ester (PLA₂). Some phospholipases will hydrolyze both acyl groups and are called phospholipase B. In addition, lysophospholipases remove the remaining acyl groups from monoacyl(lyso)phospholipids. Cleavage of the glycerophosphate bond is catalyzed by phospholipase C (PLC) while the removal of the base group is catalyzed by phospholipase D (PLD). The phospholipases C and D are therefore phosphodiesterases.

1.2. Assay of phospholipases

The subject of phospholipase assays has been concisely reviewed [1–4]. The simplest procedure is to measure protons released during hydrolysis and is usually made by continuous titration with the aid of a pH Stat. However, this technique lacks the sensitivity and specificity of other types of assays. A second commonly used approach employs synthetic substrates for spectrophotometric or fluorometric assays [3,4]. These substrates permit a continual assay well suited for kinetic studies and are of reasonable sensitivity. The major drawback is that the substrates are not natural substrates and as such, their use should be considered as a model that may or may not reflect the enzyme's kinetic properties in biological systems. With fluorescent substrates the product is

physically removed from the substrate aggregate, resulting in a spectral change in order to achieve kinetic analysis. This separation can be accomplished by the addition of albumin to absorb the product or by partitioning the product of a short-chain substrate into the aqueous phase. The thioacyl ester analogs of phospholipids provide a sensitive spectrophotometric assay for some PLA₁ or PLA₂ assays based on the reaction of released thiol with Ellmann's reagent.

An alternative fluorescence approach that does not involve synthetic substrates is to use a fluorescent displacement assay where the normal product of hydrolysis such as a long-chain fatty acid displaces a fluorescent probe from a protein that binds long-chain fatty acids. The change in fluorescence is monitored and the assay has the advantage that any source of natural phospholipid substrate can be used including cell membranes and lipoproteins (D.C. Wilton 1990, 1991; G.V. Richieri, 1995).

A third approach employs phospholipids with radioisotopes incorporated into specific positions in the molecule. The products are separated from the substrate by partitioning or chromatographic procedures. By the appropriate choice of labeling, the specificity of the enzymes can readily be established and as little as a few picomoles of product can be detected. The use of isotopes has been helpful in the measurement of phospholipase activity using the membranes of whole cells or isolated subcellular fractions previously labeled with radioactive phospholipid precursors.

A fourth and elegant method of phospholipase assay employs a monomolecular film of phospholipid [5]. With this technique the interfacial properties of the lipid substrate are carefully controlled and zero-order kinetics are obtained. This technique provides important data on the effect of lipid surface pressure and enzyme penetration on catalysis.

1.3. Interaction of phospholipases with interfaces

The catalytic turnover of phospholipases at the interface distinguishes them from the general class of esterases (Fig. 1). Therefore, the study of phospholipases must include an understanding of their interaction with the lipid interface. The increased enzyme activity seen when phospholipids are present above their cmc (Fig. 1) implies that phospholipases have an interfacial binding surface and that interfacial binding will precede normal catalysis. Interfacial binding is crucial because the cmc of normal cellular phospholipids is very low ($\ll 10^{-9}$ M) and hence the concentration of monomeric substrate in the aqueous phase is sub-nanomolar. Moreover, the half-time for desorption of a long-chain phospholipid from the bilayer interface is of the order of hours. What this means in practice is that the phospholipase must first bind to the interface and that the interfacial binding step has a profound effect on overall catalysis. This concept is illustrated in Fig. 3 and interfacial binding ($E \rightarrow E^*$) often provides the basis for the physiological regulation of these types of enzyme. Unless the phospholipase can bind productively to the phospholipid surface, it cannot access substrate and express activity.

The nature of interfacial binding and the rate enhancements that are achieved are controversial areas. However, it is clear that both polar and non-polar interactions are involved and the precise contribution of each must depend on the nature of the

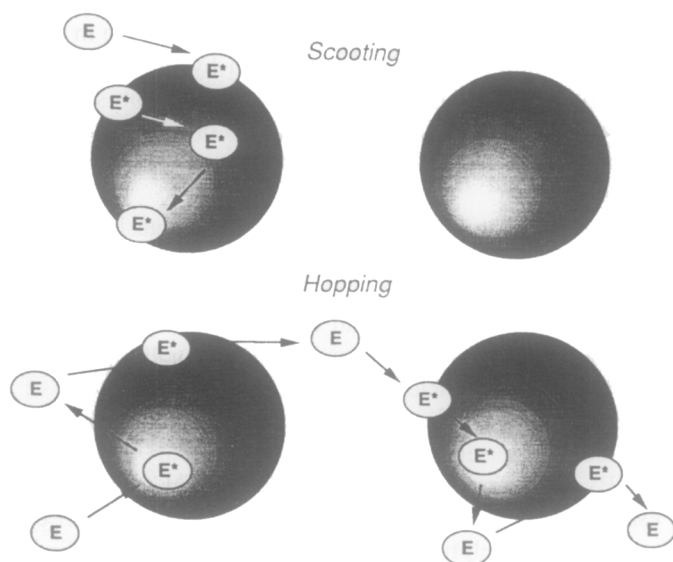


Fig. 3. Schematic illustration of the two modes of interfacial catalysis with vesicles. In the scooting mode (top) the enzyme bound to the interface does not dissociate and exhibits first-order-type kinetics on that vesicle. The excess vesicles which do not contain enzyme are not hydrolyzed. In the hopping mode (bottom) the enzyme desorbs from the interface after each or a few turnover cycles. All vesicles, therefore, are accessible for hydrolysis. E, enzyme in solution; E* enzyme bound to the vesicle. (Adapted from Berg et al. [7].)

phospholipid interface and the interfacial binding surface of the phospholipase. A number of factors can be considered that could make a major contribution to the enhanced hydrolysis at interfaces and these relate primarily either to the *substrate* or to the *enzyme*.

1.3.1. Substrate effects

Phospholipids in aqueous solution form aggregates the nature of which is determined by the structure of the phospholipid. Long-chain phospholipids normally form structures containing bilayers whereas shorter-chain phospholipids form micelles. Once the phospholipid is present as an aggregate it is the concentration of substrate in the surface of the aggregate that is critical and is defined as mole fraction, being unity with a pure phospholipid. Therefore, provided the enzyme can bind to the interface, the effective concentration of substrate will rise dramatically producing a corresponding rate enhancement.

Several assay systems have been developed to describe kinetic events at the interface. Each system has certain advantages yet have limitations that need recognition. An elegant kinetic analysis of secreted PLA₂s has been developed by Jain and coworkers (reviewed in Refs. [6,7]). This group of PLA₂s in general have a high affinity for anionic phospholipids such as the model system, phosphatidylmethanol. The high dissociation constant for the phosphatidylmethanol interface, $\ll 10^{-10}$ M, allowed the identification of the scooting mode of action of the porcine pancreatic phospholipase (Fig. 3). Under

those conditions the enzyme remains attached to the vesicle interface thus promoting hydrolysis of the outer phospholipid monolayer without loss of vesicle integrity or release of enzyme from the vesicle. Care must be taken, however, to demonstrate that the integrity of the vesicle is maintained (low Ca^{2+} concentrations) and that the number of enzyme molecules is substantially less than the number of vesicles present. Under these 'scooting' conditions, hydrolysis can be measured in the absence of interfacial effects and has provided direct evidence that interfacial binding is separate from catalysis.

Scooting conditions are the preferred assay method the assessment of potential enzyme inhibitors [6,7]. This is because the inhibitor or other factors will not affect interfacial binding of the enzyme and only the effect of the inhibitor on classical catalysis ($\text{E}^* + \text{S} \rightarrow \text{E}^*\text{S} \rightarrow \text{E}^* + \text{P}$) will be measured. By contrast, in the 'hopping' model (Fig. 3) the enzyme is able to leave the vesicle during successive rounds of catalysis and will rebind to different vesicles thus producing a continuum of substrate composition and structure as hydrolysis proceeds, making kinetic analysis more difficult [6,7].

Another useful assay system is the Triton mixed-micelle substrate. This model developed by Reynolds et al. [3] can employ a wide range of substrates and is an effective way of measuring enzyme activity. Since in this system there is an exchange of substrate molecules between the micelles that is more rapid than the catalytic rate (M.J. Thomas, 1999) a large excess of substrate is desirable. As a result, linear kinetics can be obtained until 40–50% of total substrate is degraded. One limitation that needs consideration is the affinity of the enzyme for Triton. This, however, can be determined (R.A. Burns, 1982).

Important factors when considering the enhanced hydrolysis at interfaces are the substrate environment in the monolayer and the need to transfer a substrate molecule from this monolayer to the active site. Interfacial disorder may provide an important parameter that facilitates such transfer of substrate to the active site. Phospholipase activity is enhanced under conditions that affect phospholipid fluidity, packing density of the phospholipids and polymorphism of the aggregate. A highly ordered structure seen with phosphatidylcholine phospholipids either above or below the transition temperature tends to give low rates of hydrolysis. Discontinuities in such ordered structures such as assays at temperatures close to the transition temperatures and the presence of other lipids such as anionic lipids or non-bilayer forming phospholipids, promote catalysis by perturbing the interface. In the case of anionic lipids it may be difficult to distinguish the contribution to catalysis due to enhanced interfacial binding from effects on the structure of the interface that promote phospholipid transfer to the active site.

1.3.2. Enzyme effects

An important question is if binding of the phospholipase to the interface promotes a conformational change in the structure of the enzyme that facilitates catalysis compared with the enzyme structure in free solution. This is clearly the case with many lipases where lid opening is seen on binding to the aggregate. In the case of the porcine pancreatic secreted PLA_2 , NMR analysis of the enzyme when bound to anionic micelles

compared to the enzyme in solution has revealed significant differences compared with the X-ray structure. In particular, the N-terminal region of the enzyme is disordered in free solution but in crystal structures is condensed as an α -helix when bound to an anionic micelle (B. van den Berg, 1995).

The effect of specific phospholipid molecules on overall protein conformation and hence activity may be considered as an example of allostericity. Such allosteric behavior has been discussed in terms of a second phospholipid-binding site or as discrete interactions between the head groups of phospholipids and specific residues on the interfacial binding surface of the enzyme. The emerging evidence for secreted PLA₂s would support a model involving multiple interactions between enzyme and interfacial lipid [7].

2. The phospholipases

Many types of phospholipases have now been purified and characterized including full crystal structures. Further, the cellular function of many phospholipases has been examined using various techniques of molecular biology including gene transfection, gene knockouts and antisense strategies. Since it is impossible to cover all phospholipases that have been characterized, only examples will be discussed in detail where significant information relating structure to function is available. The roles of phospholipases in signal transduction are dealt with in Chapter 12.

2.1. Phospholipase A₁

The phospholipases A₁ comprise a large group of 1-acyl hydrolases, some of which also degrade neutral lipids (lipases) or remove the acyl group at position 2 in addition to that at position 1 (phospholipase B) and thus must have lysophospholipase activity. Where the enzyme appears to show low selectivity for the *sn*-1 or *sn*-2 positions, the term phospholipase A is used. The term phospholipase B should be restricted to those enzymes where the mechanism involves minimal accumulation of lysophospholipid product. In this section we also consider various enzymes of the PLA type that do not fit a more precise definition in terms of acyl chain selectivity.

2.1.1. *Escherichia coli* phospholipases A

Two phospholipases A have been purified from *Escherichia coli* based on their differential sensitivity to treatment with detergents [2]. A detergent-insensitive enzyme is localized in the outer membrane, whereas a detergent-sensitive enzyme is found on the cytoplasmic membrane and in soluble fractions. The outer membrane enzyme, known as outer membrane phospholipase A has broad substrate specificity, and demonstrates PLA₁, PLA₂, lysophospholipase A₁ and lysophospholipase A₂ activity as well as hydrolyzing mono- and diacylglycerols. The recent crystal structure allows a more detailed discussion of what is an integral membrane phospholipase [8].

The protein is a 12-stranded anti-parallel β -barrel with amphipathic β -strands travers-

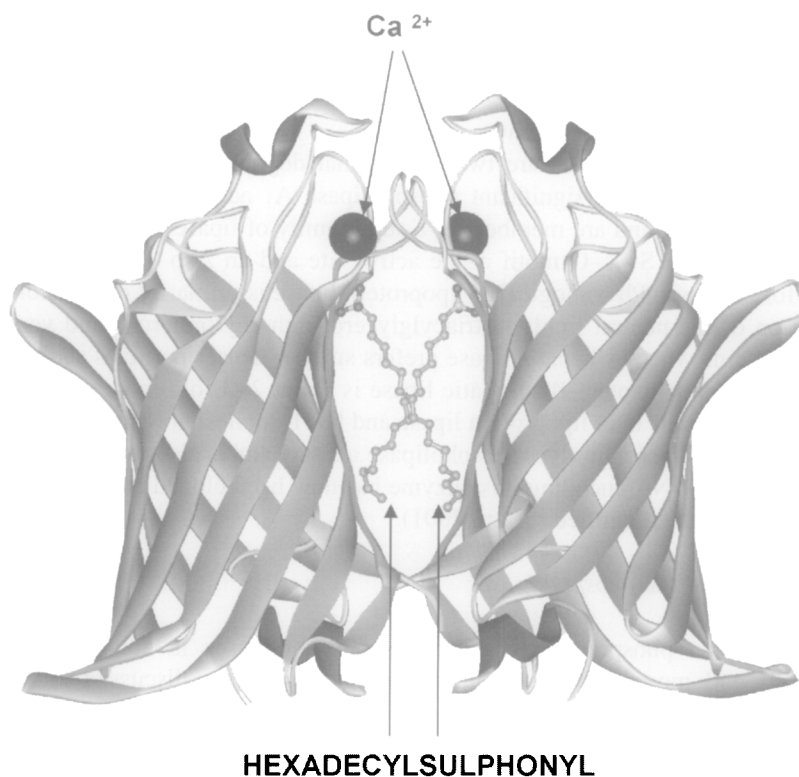


Fig. 4. Crystal structure of the outer membrane phospholipase A dimer from *E. coli* shown in the plane of the membrane. The top half of the molecule is located in the lipopolysaccharide monolayer facing the exterior of the cell. The phospholipid monolayer of the outer cell membrane would be located around the bottom half of the protein. Two calcium ions are shown at the active sites while Ser-144 of each active site is covalently modified with a hexadecylsulphonyl moiety represented in a ball and stick format. Structure is adapted from Snijder and Dijkstra [8] using MSI Weblab Viewer.

ing the membrane (Fig. 4). The active site catalytic residues are similar to a classical serine hydrolase triad except that in addition to the serine (Ser-144) and histidine (His-142) there is an asparagine (Asn-156) in place of the expected aspartic acid. Calcium at the active site is predicted to be involved in polarization of the substrate ester carbonyl group and stabilization of the negatively charged reaction intermediate.

The active site is located at the exterior of the β -barrel at the outer leaflet side of the membrane (Fig. 4). The lipopolysaccharide outer leaflet does not normally contain phospholipid and this structural isolation of active site from substrate present in the inner leaflet is believed to be important in preventing uncontrolled phospholipase activity. It is proposed that appropriate bacterial stimulation results in phospholipid moving to the outer monolayer and enzyme activation, including dimerization (Fig. 4), allows phospholipid hydrolysis. This hydrolysis enhances the permeability of the outer cell membrane producing the appropriate adaptations such as the release of bacteriocins. The fact that *E. coli* mutants deficient in this phospholipase A have normal growth

characteristics and phospholipid turnover is consistent with the specialized role of this enzyme in bacterial adaptation.

2.1.2. Lipases with phospholipase A₁ activity

Lipoprotein lipase and hepatic lipase are two lipases that degrade triacylglycerols in lipoproteins but also demonstrate significant phospholipase A₁ activity. The enzymes have 50% sequence identity and are members of a superfamily of lipases and phospholipases that share the G-X-S-X-G motif at the active site and an Asp-His-Ser triad that is required for catalysis. Physiologically, lipoprotein lipase is primarily responsible for the degradation of the neutral lipids in triacylglycerol-rich chylomicrons and very low density lipoproteins whereas hepatic lipase prefers smaller denser particles such as high density lipoproteins (Chapter 20). Hepatic lipase is about 2–3 fold more efficient at hydrolyzing phospholipids than lipoprotein lipase and has lysophospholipase activity. Another lipase, intestinal lipase, has lysophospholipase activity and is also referred to as a phospholipase B. The cDNA encoding this enzyme has now been cloned and essential catalytic residues have been identified (T. Lu, 2001).

2.2. Phospholipase B and lysophospholipases

The distinction between phospholipase B and lysophospholipases is not clear [2] since both diacyl- and monoacyl-phospholipids are substrates. As discussed above (Section 2.1) a working definition of these enzymes can be based on the extent to which the lysophospholipid accumulates upon hydrolysis of the diacyl substrate. Those phospholipases B for which there is significant structural and functional information are discussed. In addition mammalian lysophospholipases are reviewed.

2.2.1. Phospholipase B from microorganisms

The amino acid sequence of the phospholipase B from *Penicillium notatum* was deduced from its cDNA (N. Masuda, 1991). The details of the overall catalytic mechanism are unknown, but intriguing, as substrate specificity is affected by the presence or absence of detergent as well as the glycosylation state of the enzyme. Phospholipase B activity has subsequently been identified in other fungi including *Cryptococcus neoformans*, *Saccharomyces cerevisiae* and *Candida albicans*. In the case of *C. albicans*, gene disruption has identified this phospholipase B as a major factor in host cell penetration and hence pathogenicity (S.D. Leidich, 1998). It is likely that this secreted enzyme has a similar role in other fungi. Phospholipase B activity has also been isolated from *Mycobacterium phlei* and *Mycobacterium lepraemurium* where product analysis indicates that the initial activity is that of a PLA₁ (S. Maeda, 1996).

2.2.2. Mammalian lysophospholipases

Lysophospholipids [9] are generally found in very low concentrations (0.5–6% of total lipid membrane weight) in biological membranes. High concentrations of lysophospholipids affect membrane properties and membrane enzymes even leading to cell lysis. Increased lysophospholipid levels are associated with atherosclerosis while lysophosphatidylcholine is highly abundant in atherogenic lipoproteins such as oxidatively

modified low density lipoprotein where it constitutes up to 40% of the total lipid. Other diseases where there are links with lysophospholipid levels include inflammation, hyperlipidemia and lethal dysrhythmias in myocardial ischemia. At low concentrations lysophospholipids such as lysophosphatidylcholine and lysophosphatidic acid have the properties of second messengers. Against this background the role of lysophospholipases in mammalian tissues assumes a considerable significance.

There are two small mammalian lysophospholipases (I and II) which despite their similar size appear to be the products of separate genes rather than splice variants or post-translational modifications [9]. The enzymes, which are calcium-independent, have been purified from a variety of tissues and lack PLA₁, PLA₂ or acyltransferase activity. Their mechanism of action appears to involve the catalytic triad characteristic of serine proteases and lipases. Using an NMR based assay, lysophospholipase I hydrolyzed 1-palmitoyl-lysophosphatidylcholine and 2-palmitoyl-lysophosphatidylcholine at similar rates [9].

In addition to the small lysophospholipases, there are a few high molecular weight enzymes that display lysophospholipase activity. Two such enzymes, the group IV cytosolic PLA₂ and the 85 kDa calcium-independent PLA₂ are discussed in Sections 2.3.2.1 and 2.3.2.2. Similarly, a number of PLA₁s also display significant lysophospholipase activity including enzymes that are specific for phosphatidylserine (Y. Nagai, 1999) and phosphatidic acid (M.H. Han, 2001).

2.3. *Phospholipase A₂*

The phospholipases A₂ were the first of the phospholipases to be recognized. Over a century ago, Bokay (1877–1878) observed that phosphatidylcholine was degraded by some component in pancreatic fluid that is now known to be the pancreatic phospholipase A₂. At the turn of the century, cobra venom was shown to have hemolytic activity directed toward the membranes of erythrocytes (P. Keyes, 1902). The lytic compound produced by the venom phospholipase was identified a decade later and termed lysocithin (later, lysolecithin). These studies spurred further investigation of this intriguing class of enzymes and their mechanism of attack on water-insoluble substrates.

An increasing number of PLA₂s have now been identified within this expanding super-family and this has required a re-evaluation of classification criteria. Recently, Six and Dennis have categorized these enzymes into two types based on catalytic mechanism [10]. In one group (Table 1) are those enzymes that utilize a catalytic histidine as the primary catalytic residue whereas the other group involves a catalytic serine (Table 2) and normally an acyl-serine intermediate. Within these two categories individual enzymes retain their historic grouping.

2.3.1. *The 14 kDa secreted phospholipases A₂*

These enzymes now include secreted PLA₂s from such diverse sources as venoms, mammalian and plant tissues (Table 1). They are characterized by a requiring mM concentration of calcium for activity and involving an active site histidine and aspartate pair. They are typically extra-cellular enzymes with a large number of disulfide bonds. The enzymes in groups I–III have provided the foundation for our understanding of

Table 1
Characteristics of secreted PLA₂s using a histidine residue for catalysis^a

Group	Sources	Size (kDa)	Di-S (no.)	Unique Di-S	C-term extension (no. of residues)	Molecular characteristics
I	A Cobra/krait venom	13–15	7	11-77	None	5 residue pancreatic/elapid loop, propeptide
	B Mammalian pancreas	13–15	7	11-77	None	
II	A Human synovial fluid/platelets, rattlesnake venom	13–15	7	50-137	7	Lacks disulfide at C61–C94
	B Gaboon viper venom	13–15	6	50-137	6	
	C Rat/mouse testes	15	8	50-137, 86-92	7	
	D Human/mouse spleen/pancreas	14–15	7	50-137	7	Extra cysteine in C-terminal extension
	E Human/mouse brain/heart/uterus	14–15	7	50-137	7	
	F Mouse testes/embryo	16–17	7	50-137	30	
III	Bee, lizard, scorpion, human	15–18	5	N/A	N/A	Human form (55 kDa) has novel C- and N-terminal domains
V	Mammalian heart/lung/macrophage	14	6	None	None	
IX	Marine snail venom	14	6	N/A	N/A	
X	Human spleen/thymus/leukocyte	14	8	11-77, 50-137	8	
XI	A Green rice shoots	12.4	6	N/A	N/A	
	B Green rice shoots	12.9	6	N/A	N/A	

^a Adapted from Six and Dennis [10]. N/A, not available.

these secreted enzymes and sufficient quantities of natural and mutant enzymes from these groups have been obtained for X-ray crystallographic analysis. Also, the sequences of a very large number of these phospholipases are now known and have been used to demonstrate their structural, functional, and evolutionary relatedness [11].

The conserved active site residues, His-48 and Asp-99 (pancreatic enzyme numbering), provide the catalytic dyad and, with the availability of the crystal structure of the pancreatic enzyme, produced the proton-relay mechanism (Fig. 5) in 1980 [12]. In this mechanism a water molecule directly replaces the serine found in the classical protease/lipase catalytic triad mechanisms. More recently, an alternative mechanism has been proposed (J. Rogers, 1996) that involves two water molecules (W5 and W6) seen at the active site of the crystal structure (Fig. 6). In this mechanism, proposed by Jain and referred to as a 'calcium coordinated oxyanion' mechanism, the attacking nucleophile (W5) is coordinated to the calcium thus enhancing its nucleophilicity. This water is connected to His-48 by a second water molecule (W6). Thus the major formal difference between the two mechanisms is that the latter mechanism involves a second water molecule while the first water molecule is activated by coordination to

Table 2
Characteristics of PLA₂s using a serine residue for catalysis^a

Group	Alternative name	Sources	Size (kDa)	Ca ²⁺ requirement/role	Molecular characteristics
IV	A cPLA ₂ α	Human U937 cells/ platelets, rat kidney, RAW 264,7	85	< μM Membrane translocation	C2 domain, α/β-hydrolase, phosphorylation
	B cPLA ₂ β	Human, liver/pancreas/ brain/heart	114	< μM Membrane translocation	C2 domain, α/β-hydrolase
	C cPLA ₂ γ	Human heart/skeletal muscle	64	None	Prenylated, α/β-hydrolase
VI	A-1 iPLA ₂ -A	P388D ₁ macrophages, CHO cells	84–85	N/A	8 ankyrin repeats
	A-2 iPLA ₂ -B	Human B-lymphocytes	88–90	N/A	7 ankyrin repeats
	B iPLA ₂ -γ/2	Human heart/skeletal muscle	88	N/A	Membrane-bound
VII	A PAF-AH	Mammalian plasma	45	N/A	Secreted, α/β-hydrolase, Ser/His/Asp triad
	B PAF-AH (II)	Human/bovine liver/kidney	40	N/A	Myristoylated, Ser/His/ Asp triad
VIII	A PAF-AH Ib α ₁ (subunit of trimer)	Human brain	26	N/A	G-protein fold, Ser/His/ Asp triad, dimeric
	B PAF-AH Ib α ₂ (subunit of trimer)	Human brain	26	N/A	G-protein fold, Ser/His/ Asp triad, dimeric as hetero- or homodimer

^a Adapted from Six and Dennis [10]. N/A, Not available; c, cytosolic; i, (calcium) independent; PAF-AH, platelet-activating factor acetylhydrolase.

the calcium. It is argued that the rate limiting step lies during the decomposition of the tetrahedral intermediate whereas in the originally proposed mechanism the formation of this tetrahedral intermediate is rate limiting [7].

2.3.1.1. Group I secreted PLA₂s. This group historically contains group IA enzymes such as those from cobra and krait venom while the IB enzymes are the mammalian pancreatic enzymes. Both the cobra and pancreatic enzymes were early models for structure–function analysis [14] including crystal structures of both apo-enzyme and enzyme with bound phospholipid inhibitors. However, at this time there are no crystal structures of enzymes bound to a lipid interface or with a phospholipid substrate at the active site.

A space-fitting model of substrate bound to the active site of the cobra venom enzyme (Fig. 7) gives insight into how the enzyme functions even though bulk interactions with the lipid interface are missing. The most obvious feature is existence of the active-site tunnel into which the substrate enters. However, the enzyme interacts loosely with the first 9–10 carbons of the acyl group at position 2 of the glycerol that may account for the enzyme's lack of acyl specificity. This model also suggests that the substrate molecule is not completely withdrawn from the bilayer and significant

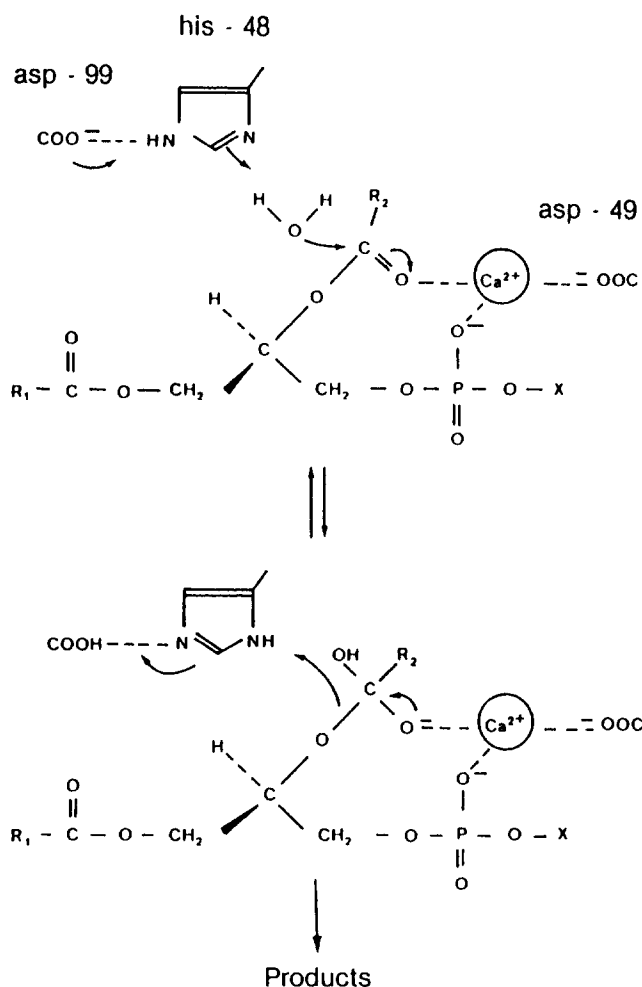


Fig. 5. Proton-relay mechanism of hydrolysis proposed by Verheij et al. [12] for secreted phospholipases A₂.

hydrophobic interactions of the molecule undergoing hydrolysis and the interface are maintained. A particular feature is the ability of the cobra venom enzyme to hydrolyze phosphatidylcholine in vesicles and cell membranes, a feature that may in part reflect the presence of tryptophans and other aromatic residues on the interfacial surface. Such residues, particularly tryptophan, are able to partition into the interfacial region of a phosphatidylcholine interface (W.M. Yau, 1998) promoting interfacial binding and catalysis [15]. The presence of such residues in the presumptive interface region can be clearly seen in Fig. 7.

The mammalian pancreatic enzymes (group IB) have primarily a digestive role and are secreted as the pro-enzyme (zymogen) that requires subsequent proteolytic cleavage to remove a hexapeptide at the N-terminus. The pro-enzyme is unable to bind to the phospholipid interface. Unlike the cobra venom enzyme, the pancreatic

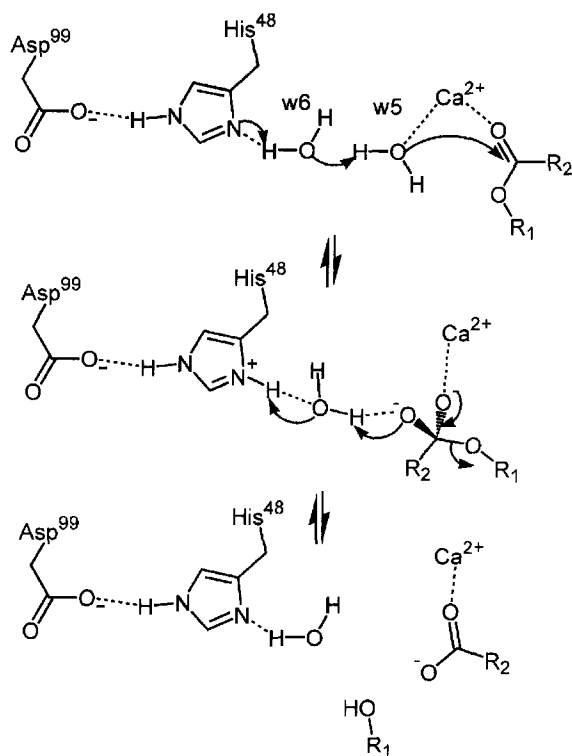


Fig. 6. Calcium coordinated oxyanion mechanism of hydrolysis proposed by Jain for secreted phospholipases A₂ (J. Rogers, 1996). The two water molecules at the active site that are implicated in catalysis are shown as W5 and W6.

enzyme expresses low activity with a zwitterionic interface such as that provided by phosphatidylcholine. The enzyme has a considerable preference for anionic interface that can be provided by anionic phospholipid per se or by the inclusion of other anionic lipids in the phosphatidylcholine interface. Presumably the anionic bile salts provide such negative charge in the mixed micelles produced in the intestine to allow lipid digestion. The expression of the pancreatic enzyme in other tissues and the presence of cell surface receptors for this enzyme (as well as other secreted phospholipases A₂) suggest additional physiological roles [11,16].

2.3.1.2. Group IIA–F secreted PLA₂s. Historically, the group IIA secreted PLA₂s included venom enzymes from rattlesnakes and vipers. They are characterized by a C-terminal extension while they lack a surface loop region (elapid loop) present in group I enzymes. The most interesting member of this group IIA is the mammalian enzyme that was isolated from the synovial fluid of patients with rheumatoid arthritis and from platelets. This was the first mammalian non-pancreatic enzyme to be identified and was implicated as a key enzyme in arachidonic acid release from phospholipids, the first step in the production of the inflammatory eicosanoids. However, it has subsequently been



Fig. 7. X-ray crystal structure of cobra venom (*Naja naja naja*) phospholipase A₂ with bound Ca²⁺ showing a space-filling model of dimyristoyl phosphatidylethanolamine bound in the catalytic site. The ends of the fatty acid chains stick out of the enzyme and are presumably associated with the micelle or membrane [13].

established that it is the intracellular group IV enzyme that plays the dominant role in eicosanoid production (see Section 2.3.2.1) and hence the precise physiological role of this IIA enzyme in the inflammatory response remains to be clarified.

It is clear that the human IIA enzyme behaves as an acute phase protein (R.M. Crowl, 1991) and extra-cellular levels increase dramatically in acute inflammatory conditions such as septicemia where blood levels can rise over 100-fold. The detailed structural properties of the enzyme are unusual and appear to reflect at least some of its physiological roles. The enzyme is highly cationic ($pI > 10$) and has a net positive charge of +19 due mostly to cationic residues distributed across the surface of the protein, a feature that is linked to the high affinity of the protein for heparin. An unusual characteristic of the enzyme is its marked preference for anionic phospholipid interfaces and very low affinity for a zwitterionic interface as provided by phosphatidylcholine.

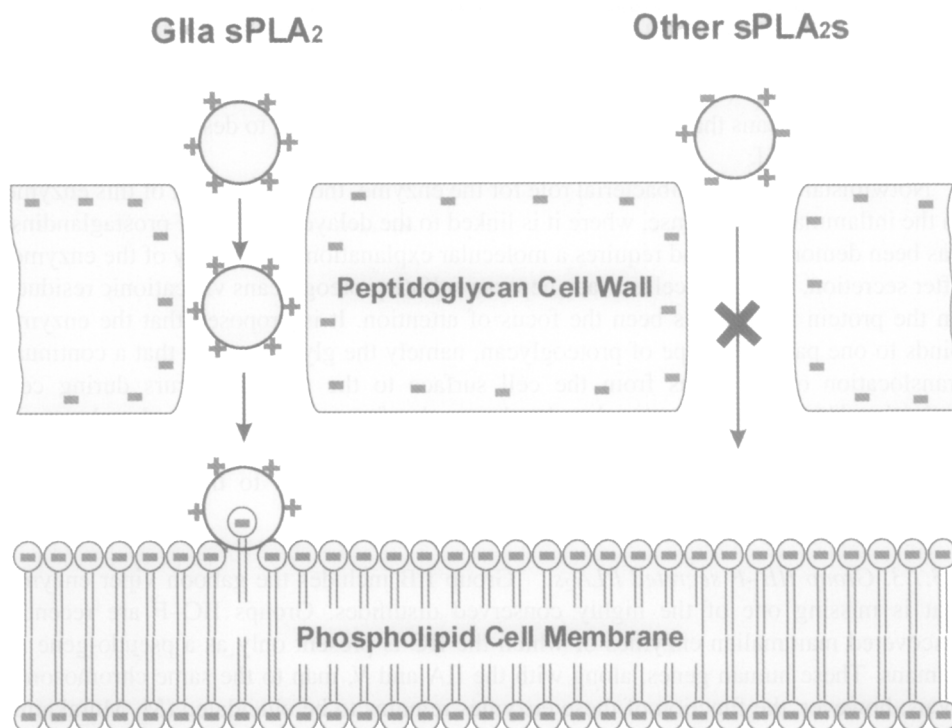


Fig. 8. A model of bacterial cell wall penetration and phospholipid hydrolysis by human group IIA secreted PLA₂ (gIIA sPLA₂). The extreme and global nature of the positive charge of this secreted PLA₂ has been proposed to allow the enzyme to move through the negatively charged cell wall mediated by the continual making and breaking of electrostatic bonds. For simplicity bacterial phospholipids are shown as being negatively charged, although the bacterial membrane will contain some zwitterionic phospholipids, normally phosphatidylethanolamine.

This preference is expressed at the stage of interfacial binding while active site substrate preference is modest and favors phospholipids such as phosphatidylglycerol over phosphatidylcholine. These surface properties are consistent with one particular physiological role for the enzyme, namely as an antibacterial protein that shows specificity towards gram-positive bacteria [17]. The highly cationic nature of the protein allows the enzyme (but not other more neutral secreted PLA₂s) to penetrate the highly anionic bacterial cell wall. Moreover, the anionic bacterial cell membrane that is rich in phosphatidylglycerol provides an optimum substrate for this enzyme (Fig. 8). In contrast, this enzyme is essentially inactive against the host cell membranes that are normally zwitterionic being rich in phosphatidylcholine and sphingomyelin. This lack of activity against host membranes (which is partly due to lack of an interfacial tryptophan residue (S.F. Baker, 1998)) is important as the serum levels of the enzyme can rise to above 1 $\mu\text{g/ml}$.

The enzyme is released from cells with known antibacterial activity (macrophages and Paneth cells) and is present at very high concentrations ($\sim 10 \mu\text{g/ml}$) in human tears

(X.D. Qu, 1998) along with lysozyme. The increased sensitivity of mice lacking the group IIA enzyme to gram-positive bacteria provides further support for the antibacterial role of the enzyme (V.J.O. Laine, 1999). The preference of this enzyme for anionic interfaces also means that this enzyme may be active in helping to destroy apoptotic and damaged cells [16].

Notwithstanding an antibacterial role for the enzyme, the involvement of this enzyme in the inflammatory response, where it is linked to the delayed release of prostaglandins, has been demonstrated and requires a molecular explanation. The ability of the enzyme, after secretion, to bind to cell surface heparan sulfate proteoglycans via cationic residues on the protein surface has been the focus of attention. It is proposed that the enzyme binds to one particular type of proteoglycan, namely the glypicans and that a continual translocation of glypicans from the cell surface to the nucleus occurs during cell activation [16]. This is proposed to be the result of a caveolae-mediated endocytotic event called potocytosis which transfers the group IIA secreted PLA_2 within calcium-rich vesicles to the nuclear region of the cell in proximity to the cyclooxygenase (Chapter 13) [16].

2.3.1.3. Group IIB–F secreted PLA_2 s. Group IIB includes the gaboon viper enzyme that is missing one of the highly conserved disulfides. Groups IIC–F are recently discovered mammalian enzymes of which the IIC is present only as a pseudo-gene in humans. These human genes, along with the IIA and V, map to the same chromosome locus; however, the function of these enzymes remains to be elucidated (M. Murakami, 2001).

2.3.1.4. Group III secreted PLA_2 . The bee venom enzyme was historically placed in group III and is the primary allergen in bee venom. The enzyme is significantly larger than other secreted PLA_2 s with less structural homology. The availability of a crystal structure has allowed extensive mutagenesis. In particular, the charge reversal mutagenesis of 5 of the 6 cationic residues on the interfacial binding surface produced a mutant with minimal effects on binding to anionic interfaces. This result highlighted the importance of non-electrostatic interaction in interfacial binding (F. Ghomashchi, 1998). The incorporation of spin labels on the protein surface following cysteine mutagenesis provided a method for defining the topological relationship between the protein and the membrane interface (Y. Lin, 1998). The results demonstrated the interaction of hydrophobic residues but not cationic residues with the interfacial region of the membrane.

A group III human enzyme has recently been identified that possesses long N- and C-terminal extensions but as yet no function has been ascribed to this enzyme.

2.3.1.5. Group V secreted PLA_2 . This enzyme is the most studied human secreted PLA_2 after the group IIA enzyme. The enzyme has a tryptophan on the interfacial binding surface, Trp-31, which is partly responsible for its ability to hydrolyze cell membranes. Exogenously added group V enzyme can catalyze phosphatidylcholine hydrolysis in the cell surface with subsequent activation of group IV cytosolic PLA_2 and enhanced leukotriene biosynthesis. Removal of Trp-31 by mutagenesis greatly reduces

the effectiveness of added enzyme [18]. Like the IIA enzyme, the group V enzyme binds to heparan sulfate proteoglycans on the cell surface and is internalized. However, the dynamics of this process and also mutagenic studies (K.P. Kim, 2001) indicate that internalization is linked to degradation of the enzyme and not to eicosanoid production in human neutrophils and mast cells. In contrast, internalization results in eicosanoid production in human embryonic kidney 293 cells [16,21] (Y.J. Kim, 2002). Thus, the apparent differences between the cellular fate of the group IIA and V enzymes may reflect the types of cells being used for these studies.

2.3.1.6. Group X secreted PLA₂. This enzyme is effective in the hydrolysis of phosphatidylcholine vesicles and the plasma membrane when it is added exogenously to adherent mammalian cells. Hydrolysis is accompanied by enhanced prostaglandin E₂ production. The enzyme does not bind to heparin so heparan sulfate proteoglycan-associated internalization and degradation seen with the group IIA and V enzymes, respectively, cannot occur with this enzyme (S. Bezzine, 2000). The crystal structure of the group X enzyme is now available (Y.H. Pan, 2002).

2.3.1.7. Group IX, group XI and other secreted PLA₂s. At this time these groups are represented by proteins from a snail venom (group IX) (conodipine-M) and green rice shoots (group XI) respectively (Table 1) while an increasing number of secreted PLA₂s are being identified from analysis of genomic data bases in both the plant and animal kingdoms [11]. The cloning and expression of these enzymes and related proteins will be the first step in understanding their physiological functions.

2.3.2. Phospholipases A₂ that involve a catalytic serine residue

2.3.2.1. Group IV cytosolic PLA₂. Over the past decade or more the group IV cytosolic PLA₂ has taken center stage as the phospholipase that is primarily involved in the regulation of prostaglandin and leukotriene biosynthesis (Chapter 13) as part of the inflammatory response [19]. Even though this enzyme translocates to a membranous fraction, it is recovered from the cytosolic fraction of the cell and hence it is termed cytosolic. This enzyme is distinct from the mammalian secreted phospholipases A₂ in its size (85 versus 14 kDa), stimulation by Ca²⁺ (at micromolar versus millimolar concentration), specificity at the *sn*-2 position of the substrate (arachidonate versus no specificity) and catalytic mechanism. This enzyme shows minimal head-group specificity and phosphatidylcholine is the normal substrate. The discovery of the cytosolic phospholipase A₂ has provided new insights into the cell signalling events that initiate the 'arachidonate cascade' described in Chapter 13.

The sequence of the cytosolic PLA₂ was deduced from the cDNA isolated from a number of species and is now referred to as cytosolic PLA₂ α as the result of the more recent discovery of further isozymes (β and γ). The human enzyme, mapped to chromosome 1, is highly conserved amongst mammalian species but the sequence can differ up to 20–30% between mammals and non-mammalian vertebrates. Distinct regions of the enzyme have been identified including two domains, an N-terminal C2 domain and a larger C-terminal catalytic domain (Fig. 9). The γ -isozyme lacks the C2 domain and is prenylated at the C-terminus. The complete crystal structure of human

cytosolic PLA₂ α (A. Dessen, 1999) has been reviewed [20] and provides a logical basis for discussion of the function and regulation of this important enzyme.

The catalytic domain. The enzyme shows lysophospholipase and transacylase activities that are consistent with the formation of an acyl-serine intermediate characteristic of lipases, and Ser-228 has been identified as the involved residue. In addition Asp-549 has been demonstrated to be the second member of the predicted catalytic triad. However, histidine has not been identified as the third member and none of the 19 histidine residues in the protein has been shown to play any catalytic role. At present a dyad mechanism must be invoked while Arg-200 is in a position to stabilize the oxyanion intermediate. Interestingly, 1-palmitoyl lysophosphatidylcholine is degraded at a rate comparable to 2-arachidonoyl phosphatidylcholine which raises the possibility that the enzyme serves multiple functions in the cell.

The C2 domain. The enzyme has a Ca²⁺-dependent phospholipid binding domain (CaLB or C2 domain) common to many proteins that translocate to membranes from the cytosol in the presence of Ca²⁺. The activity of the cytosolic PLA₂ increases several fold as the Ca²⁺ concentration is increased to concentrations found in activated cells (300 nM). Since Ca²⁺ is not involved in the catalytic event, Ca²⁺ promotion of enzyme-membrane interaction probably accounts for Ca²⁺'s stimulatory effect. The molecular mechanism by which Ca²⁺ binding promotes membrane interactions has been the subject of intensive investigations. The C2 domain consists of an anti-parallel β -sandwich composed of two four-stranded sheets (Fig. 9) while the structure is capped

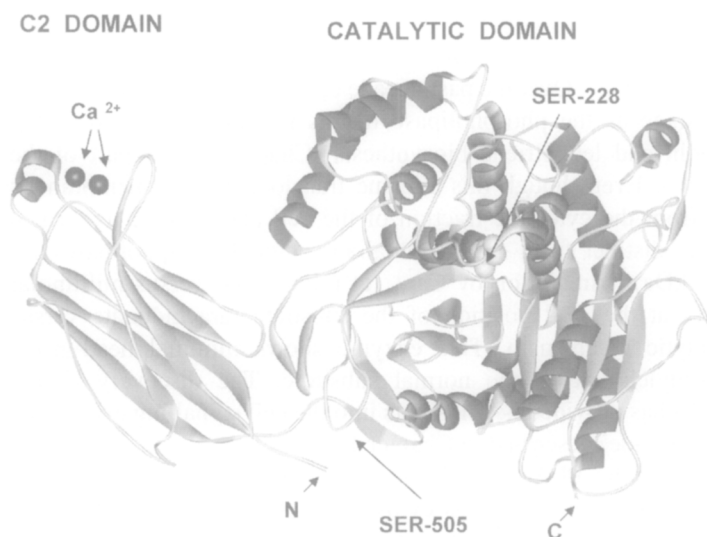


Fig. 9. Crystal structure of group IV cytosolic PLA₂ showing the C2 and catalytic domain. The two calcium ions bond to the C2 domain and the active site Ser-228 are highlighted. The phospholipid interface would be parallel to the top surface of the molecule as shown. A flexible region between residues 499 and 538 is not seen in the crystal structure but contains Ser-505, the approximate position of which is indicated by the arrow. The N- and C-terminals of the protein are indicated. Adapted from A. Dessen (1999) and Dessen [20] using MSI Weblab Viewer.

by three Ca^{2+} binding loops known as the calcium-binding regions. It would appear that the binding of Ca^{2+} , allows penetration of two of the loops into the membrane providing a stable membrane–protein interaction (O. Perisic, 1999; L. Bittova, 1999). The net result is that the enzyme translocates to the membrane surface to allow catalysis. Immunofluorescence studies have revealed that the enzyme preferentially translocates to the nuclear envelope region, a location that also contains the cyclooxygenase and 5-lipoxygenase, presumably allowing facile transfer of the released arachidonic acid to these first enzymes in the prostaglandin and leukotriene pathways. The mechanism by which cytosolic PLA_2 selectively translocates to this specific nuclear region is unclear in view of the lack of obvious phospholipid head-group selectivity of the enzyme. It has been proposed that the 60 kDa protein vimentin acts as an adaptor for membrane targeting and interacts directly with the C2 domain in a Ca^{2+} -dependent manner [21].

Phosphorylation of cytosolic $\text{PLA}_2 \alpha$. There are multiple sites for phosphorylation on the cPLA_2 and a number of protein kinases appear to use the enzyme as substrate. A critical site is Ser-505, the site phosphorylated by mitogen-activated protein kinase. Phosphorylation at this site increases the activity of the enzyme both in vitro and in vivo. However, other phosphorylation sites may be involved; in particular, Ser-727 may also have an important role in some cell systems (M.A. Gijon, 2000). The Ser-505 is located in a flexible loop that connects the C2 and catalytic domains (Fig. 9). It is possible that phosphorylation of this residue produces the optimum orientation of the two domains with respect to the membrane interface or affects the interaction of these two domains with another membrane protein such as vimentin [20].

Gene knockouts. The very large number of enzymes that have now been discovered with PLA_2 activity increases the difficulty of being able to unambiguously define the role of a regulator of the inflammatory response involving arachidonic acid to one particular enzyme. Gene knockout studies in mice involving cytosolic $\text{PLA}_2 \alpha$ have been particularly successful in both defining the primary importance of cytosolic PLA_2 in inflammation and identifying specific physiological roles [22]. Thus cytosolic PLA_2 knockout mice have revealed important roles in normal fertility, generation of eicosanoids from inflammatory cells, brain injuries and allergenic responses. Other forms of PLA_2 cannot replace these functions and hence the enzyme becomes a prime pharmacological target.

2.3.2.2. Group VI phospholipase A_2 . This group consists of the intracellular calcium-independent PLA_2 s and at this time it is the group VIA enzyme for which most information is available [23]. An additional calcium-independent PLA_2 has recently been described (D.J. Mancuso, 2000) and has been categorized as group VIB.

The group VIA enzyme is 85–88 kDa and consists of multiple splice variants that contain 7 or 8 ankyrin repeats. The enzymes are widely distributed and have both a cytosolic and membrane location within the cell. They exhibit lysophospholipase and transacylase activity while their ability to be inactivated by hydrophobic serine-reactive inhibitors is indicative of a catalytic serine and acyl-serine intermediates. A variety of evidence indicates that these enzymes are involved in basal phospholipid fatty acid remodeling as a general housekeeping function within most cell types. It is possible that these enzymes also have roles in signal transduction and other physiological functions [23].

2.3.2.3. Group VII and group VIII PLA₂s. The group VII and VIII enzymes (Table 2) are better known as platelet-activating factor acetylhydrolases [24]. These enzymes catalyze hydrolysis of the *sn*-2 ester bond of platelet-activating factor (see Chapter 9) and related pro-inflammatory phospholipids, and thus attenuate their bioactivity. They exist as both secreted (plasma) and intracellular forms and the plasma form has attracted most attention. The plasma form not only hydrolyzes platelet-activating factor but also a range of oxidatively damaged phosphatidylcholines from cell membranes and lipoproteins that originally contained arachidonic acid at the *sn*-2 position. Such oxidative damage can be significant in reperfusion injury, cigarette smoking and inflammation and as such their formation is essentially uncontrolled. Therefore, the sole mechanism for regulating the biological impact of these compounds lies in their degradation by the acetylhydrolases.

The clinical importance of this enzyme is highlighted by the lack of platelet-activating factor acetylhydrolase activity in a significant proportion of the Japanese population and in the majority of cases this is due to a mutation, V279F. This mutation is associated with asthma, stroke, myocardial infarction, brain hemorrhage and non-familial cardiomyopathy. It is possible that treatment with recombinant platelet-activating factor acetylhydrolase may be beneficial [24].

Intracellular forms of the enzyme all hydrolyze platelet-activating factor acetylhydrolase but some are more selective and do not hydrolyze oxidized phospholipids [24].

2.4. Phospholipase C

2.4.1. Bacterial phospholipases C

Phospholipases C have been known to be associated with bacteria since the classic demonstration by Macfarlane and Knight (1941) that α -toxin in *Clostridium perfringens* was a phospholipase C (reviewed in Waite [2]). The most extensively studied phospholipases C are those from *Bacillus cereus* and provided the first crystal structure for a PLC (E. Hough, 1989). Another PLC that is specific for phosphatidylinositol is also secreted in large amounts by *B. cereus* and has provided a crystal structure (D.W. Heinz, 1995).

2.4.2. Mammalian phospholipases C

These phospholipases are primarily involved in signal transduction and are reviewed in detail in Chapter 12. The structure and mechanism of action of phosphatidylinositol-specific PLCs from both mammalian and bacterial sources has been presented [25,26]. The structure of mammalian phosphatidylinositol-PLC δ is shown in Fig. 10 and highlights the domain structure. This protein was crystallized in the absence of its PH domain that is at the N-terminal. A 'tether and fix' model of membrane association is suggested (L.-O. Essen, 1996) whereby initial specific interaction is via the PH domain that will target membrane phosphoinositides followed by the binding of the C2 and catalytic domains. A two-step catalytic mechanism is proposed for both mammalian and bacterial enzymes [26] involving a cyclic phosphodiester intermediate which in the case of the bacterial enzyme can be the major product of the reaction.

The major pathway of sphingomyelin degradation involves a special phospholipase

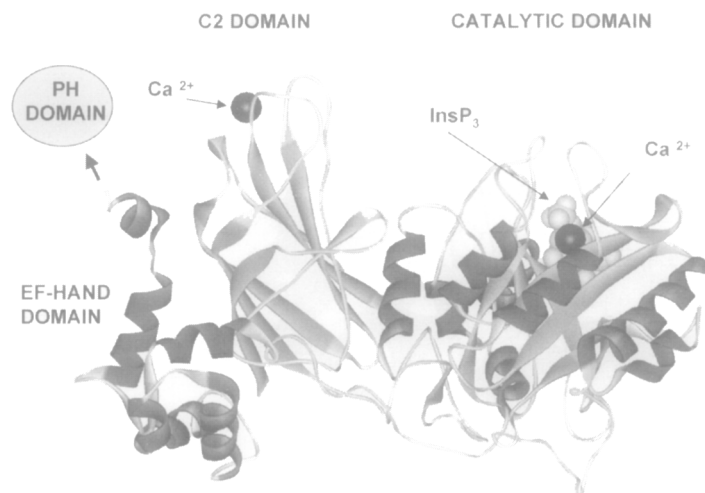


Fig. 10. Crystal structure of phosphatidylinositol-PLC δ showing the C2 and catalytic domains. The position of the PH domain that would be attached to the EF-hand domain is indicated. The membrane surface would be parallel to the top surface of the molecule as shown. Calcium ions are shown bound to the active site and to the C2 domain. The position of inositol trisphosphate in space-filling format at the active site is indicated. Adapted from L.-O. Essen (1996) and Heinz et al. [26] using MSI Weblab Viewer.

C, a sphingomyelinase. The enzyme is secreted by *B. cereus* while several mammalian sphingomyelinases are achieving a prominent role in signal transduction in mammalian systems. The description of sphingolipids and their metabolism is covered in Chapter 14.

2.5. Phospholipase D

Classically, plants and bacteria have been the major sources for the purification of phospholipases D. The function of phospholipase D in plants is not known although it may be involved in cell turnover and energy utilization during different cycles in plant life. Bacterial phospholipases D in some cases are toxins and can lead to severe cellular damage either alone or in combination with other proteins secreted from bacteria. These bacterial enzymes may also serve to help provide nutrients for the cell such as inorganic phosphate, as do the bacterial phospholipases C [2].

All phospholipases D characterized thus far act by a phosphatidate exchange reaction that has a covalent phosphatidyl-enzyme as an intermediate [27]. For this reason, the enzyme can catalyze a 'base-exchange' reaction in which alcohols can substitute for water as the phosphatidate acceptor. In fact, alcohols are better than water as phosphatidate acceptors; about 1% of an alcohol (e.g., ethanol) in water yields phosphatidylethanol almost exclusively and this principle is the basis of some PLD assays.

In the case of the member of the PLD family, Nuc, when the enzyme was incubated with ^{32}P -labeled inorganic phosphate, a phospho-enzyme intermediate was produced and identified as phospho-histidine (E.B. Gottlin, 1998). Recently, the first crystal structure of a PLD from *Streptomyces* sp. has been published (I. Leiros, 2000) and

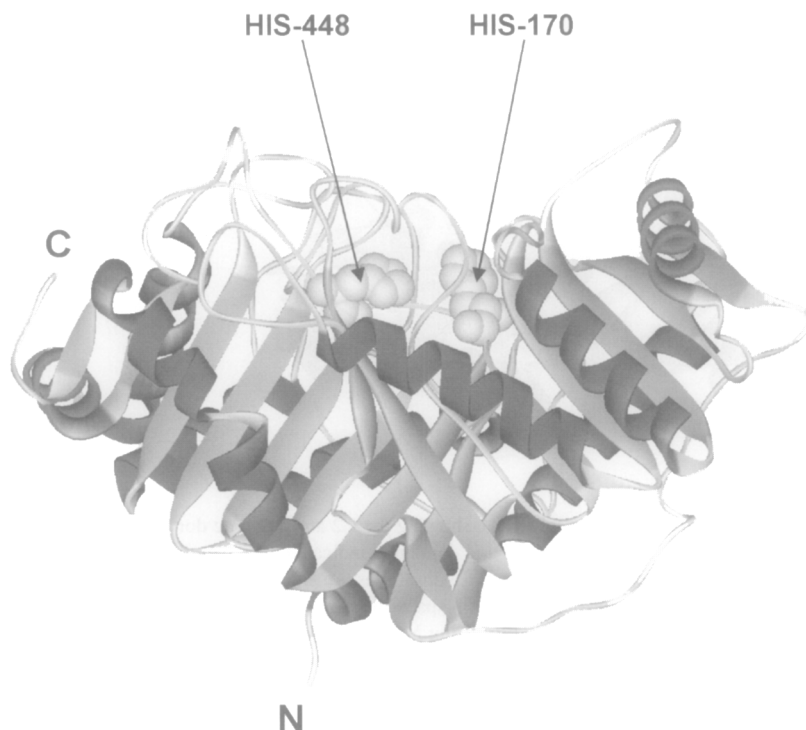


Fig. 11. Crystal structure of the PLD from *Streptomyces* sp. showing two active site histidines in space-filling format together with the N- and C-termini of the protein. Adapted from I. Leiros (2000) using MSA Weblab Viewer.

is shown in Fig. 11. The positions of two active site histidines, His-170 and His-448, are indicated and it is proposed that His-170 is the more likely residue to act as the nucleophile and thus be covalently modified during catalysis.

Mammalian PLDs are now the subject of intense interest as they appear to be intimately involved in signal transduction. Two mammalian PLDs have been identified and these contain several domains often associated with signal transduction proteins including a PH domain and a separate phosphatidylinositol-*bis*-phosphate binding domain. These proteins appear to be controlled by a variety of regulatory molecules [28]. The potential roles of the enzymes in cell signaling [29] and membrane traffic [30] are discussed in detail in Chapter 12.

Phospholipases C and D that act on glycosylphosphatidylinositol-anchored proteins on the cell surface are of growing importance and are discussed in Chapter 2.

3. Future directions

Since the last edition of 'Biochemistry of Lipids, Lipoproteins and Membranes' there has been remarkable progress in our knowledge of phospholipases. This progress

has included the crystal structures of many more phospholipases and the use of the techniques of molecular genetics in order to define the precise physiological function of individual enzymes. However, at this time the rate of elucidation of the structure and function of known enzymes is being matched by the discovery of new phospholipases in a very diverse array of organisms. This is particularly the case with the secreted PLA₂s and a major challenge is to understand the role or roles of many of these enzymes in their respective organisms. A major factor that has evolved is the important role of binding to the phospholipid interface in determining and regulating physiological function. Our knowledge of the molecular interactions at membrane interfaces is limited and will require advances in both X-ray crystallography and NMR to elucidate such details. There are exciting times ahead!

Abbreviations

cmc	critical micellar concentration
PLs	phospholipases
PLA ₁	phospholipase A ₁
PLA ₂	phospholipase A ₂
PLC	phospholipase C
PLD	phospholipase D

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Glycerolipids in signal transduction

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1. Introduction

Glycerolipids have emerged as essential molecules for the regulation of cell functions by hormones, neurotransmitters, growth factors, and inflammatory cytokines. Membrane phospholipids are acted upon by a host of phospholipases, lipid kinases and phosphatases to generate signaling lipids. Fig. 1 illustrates the major enzymes involved in the production of phospholipid-derived second messengers. In general, formation of signaling lipids is initiated by ligand binding to a specific cell-surface receptor, which leads to the activation of phospholipases and/or lipid kinases. The lipid-derived second messengers generated then act on specific target proteins to influence cell function. Signaling lipids and inositol phosphates, like most second messengers, are quickly metabolized to limit the response or to generate another signaling molecule.

The first glycerolipid pathway known to be associated with cell signaling was the turnover of phosphatidylinositol (PI), which leads to changes in the levels of

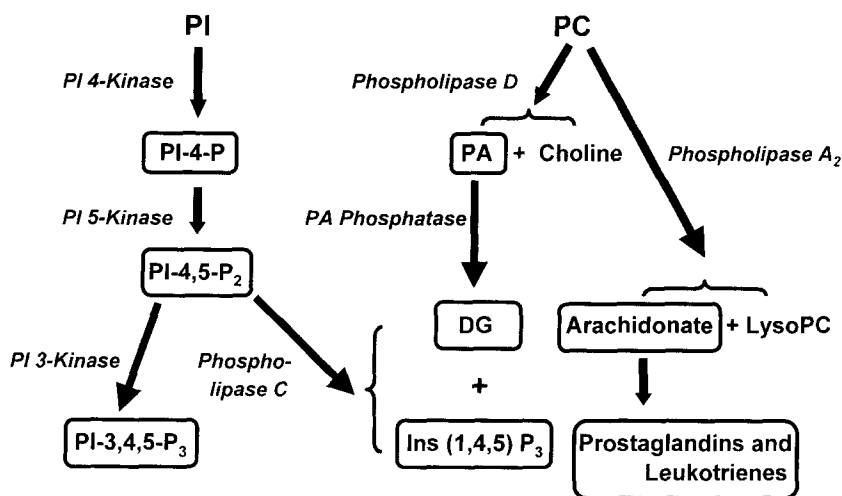


Fig. 1. Overview of phospholipase and lipid kinases that participate in signal transduction mechanisms. Abbreviations: PI, phosphatidylinositol; PI-4-P, PI-4-phosphate; PI-4,5-P₂, PI-4,5-bisphosphate; PI-3,4,5-P₃, PI-3,4,5-trisphosphate; PC, phosphatidylcholine; PA, phosphatidic acid; DG, diacylglycerol; Ins(1,4,5)P₃, inositol trisphosphate.

various phosphoinositides and the generation of diacylglycerol (DG) and inositol-1,4,5-trisphosphate (Ins-1,4,5- P_3). Both of these products are important second messengers. The turnover of phosphatidylcholine (PC) also generates second messengers: phosphatidic acid (PA), by the action of phospholipase D (PLD), and arachidonic acid and lyso-PC (LPC), by the action of phospholipase A_2 . PA has an emerging role as a second messenger and also can be converted to lyso-PA (LPA) by certain phospholipases A. Arachidonic acid also has direct intracellular targets as a second messenger, but primarily is metabolized to prostaglandins and leukotrienes (Chapter 13). Alkyl species of lyso-PC are converted to platelet-activating factor (Chapter 9). PA and DG are interconvertible by the action of the enzymes PA phosphatase and DG kinase. DG, PA, Ins-1,4,5- P_3 and the phosphoinositides primarily have intracellular sites of action. In contrast, prostaglandins, leukotrienes, platelet-activating factor, LPC and LPA exit the cell and act as autocrine, paracrine, or circulating hormones by binding to specific cell-surface heptahelical receptors on the same, neighboring, or distant target cells.

This chapter describes the mechanisms by which lipid-derived second messengers regulate various cell functions. Current information about the pathways that regulate the formation and metabolism of signaling lipids and inositol phosphates, as well as the progress in identifying protein targets for the second messengers, will be discussed. Excluded is coverage of arachidonic acid-derived signaling lipids and platelet-activating factor, as they receive extensive discussion in other chapters (Chapters 9 and 13).

2. Inositol phosphates

2.1. Mechanisms of generation and metabolism

2.1.1. Phosphoinositide-hydrolyzing phospholipase C

Mabel and Lowell Hokin were the first to observe (in 1953) that the turnover of PI increased greatly in response to cholinergic stimulation of pigeon pancreatic slices [1]. This began the study and elucidation of the PI cycle, illustrated in Fig. 2. A series of two lipid kinases (PI 4-kinase and PI 5-kinase) sequentially add phosphates to positions 4 and 5 of the inositol ring to form PI-4,5-bisphosphate (PI-4,5- P_2). PI-4,5- P_2 is the major substrate for the phosphoinositide-hydrolyzing phospholipase C (PLC) family of enzymes, which are activated in response to hormone or other agonist stimulation. Hydrolysis of PI-4,5- P_2 by PLC produces the second messengers DG and Ins-1,4,5- P_3 . The function of these messengers is discussed below (Sections 2.2, 3.2, 3.3). Both products are metabolized further and used for the resynthesis of PI, as shown in Fig. 2. The situation is more complex than shown, since the action of additional kinases and phosphatases results in the formation of inositol phosphate species phosphorylated on various combinations of up to all six hydroxyls of the inositol ring. Many of the over 60 potential species have not been isolated from biological sources, but several inositol tetrakisphosphates, Ins-1,3,4,5,6- P_5 and Ins-1,2,3,4,5,6- P_6 , are present in animal cells [2]. Known and potential functions of these molecules are described in Section 2.2.

The mammalian PLC family consists of 11 isozymes divided into four subgroups (β , γ , δ , ϵ), all of which catalyze the hydrolysis of PI-4,5- P_2 to Ins-1,4,5- P_3 and DG in

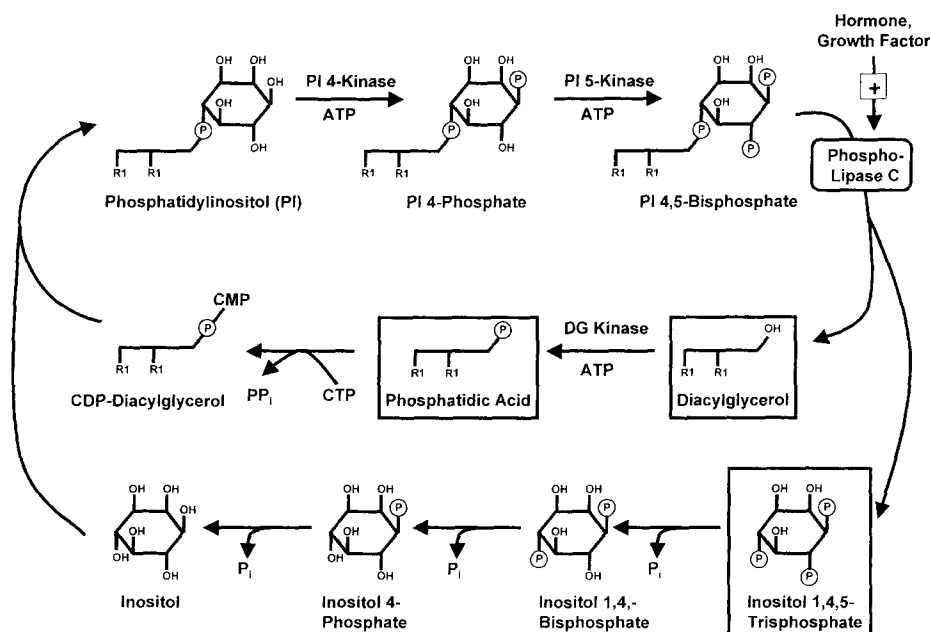


Fig. 2. Phosphatidylinositol cycle. See text for details. Shown in bold boxes are the signaling molecules derived from the PI cycle. DG kinase, diacylglycerol kinase; PI, phosphatidylinositol.

response to agonist binding to over 100 cell-surface receptors [3]. The four subgroups all contain similar catalytic X and Y domains, but differ in size and in their regulatory regions (Fig. 3). The divergent regulatory regions are responsible for differences in the types of receptors and the mechanisms by which members in each subgroup are activated. Three of the subgroups (δ , β , γ) contain three regulatory regions in common:

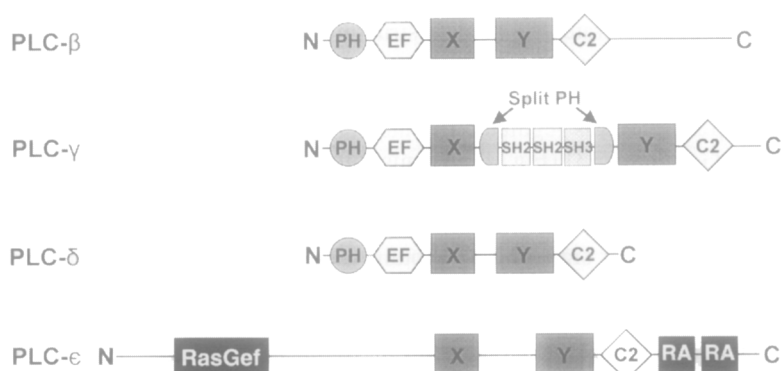


Fig. 3. Domain structures of the four isoforms of phosphatidylinositol-specific phospholipase C (PLC). The X and Y catalytic regions are conserved among all the isoforms. PH, pleckstrin homology domain; EF, EF hands; SH2 and SH3, Src homology domains 2 and 3; C2, C2 domain; RasGEF, Ras guanine nucleotide exchange factor-like domain; RA, Ras-binding domain. See text for discussion.

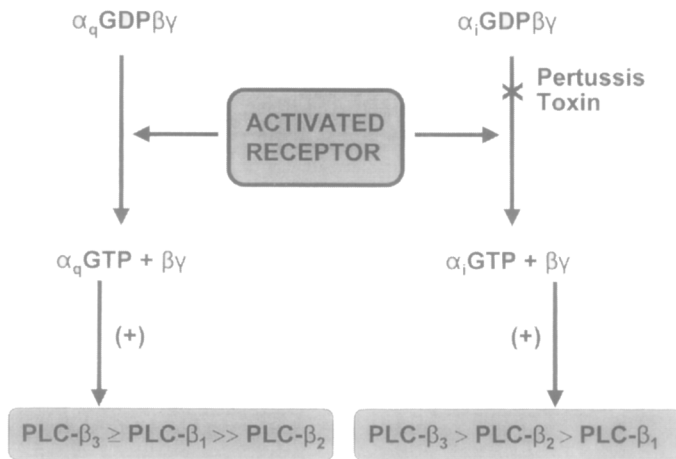


Fig. 4. Scheme for activation of phospholipase C- β (PLC- β) isoforms by G protein α and $\beta\gamma$ subunits. The left side of the figure shows the pertussis toxin-insensitive activation of PLC- β isoforms by $G_q\alpha$ subunits. The right side of the figure shows the pertussis toxin-sensitive activation of PLC- β isoforms by $\beta\gamma$ subunits derived from a G_i type of G protein.

an N-terminal pleckstrin homology (PH) domain, an EF hand domain and a C-terminal C2 domain. PH domains bind certain phosphoinositides and/or proteins (see Section 5.2.1), EF hand domains usually bind calcium [4], and C2 domains usually bind anionic phospholipids in the presence of calcium (see Section 3.2) [5]. The PLC- γ subgroup contains a second PH domain that is split by two Src homology 2 (SH2) domains and a Src homology 3 (SH3) domain. SH2 regions usually interact with phosphorylated tyrosine residues in other proteins and SH3 domains bind specific polyproline motifs in target proteins. The PLC- ϵ subgroup contains a C2 domain, as well as an N-terminal Ras guanine nucleotide exchange factor-like domain and one or two C-terminal Ras binding domains. The crystal structure of PLC- δ (minus the PH domain) has been solved (Chapter 11), which clearly illustrates the domain structure of the enzyme.

Each PLC subgroup is regulated by different mechanisms, triggered by agonist binding to specific receptors and involving recruitment of the soluble PLC to the plasma membrane. The PLC- β subgroup is regulated by interaction with α and $\beta\gamma$ subunits of certain heterotrimeric G proteins [3] (Fig. 4). Agonist binding to heptahelical receptors triggers the exchange of GTP for GDP on the α subunit of heterotrimeric G proteins, causing the α and $\beta\gamma$ subunits to dissociate. Both free subunits then interact with effector enzymes to regulate intracellular signaling. G protein α subunits hydrolyze the bound GTP back to GDP and the subunits reassociate to terminate the signaling process. There are four subfamilies of G protein α subunits, and one of these (G_{α_q}) interacts with the C2 domain and the extended C-terminal region of PLC- β to aid in membrane recruitment and induce enzyme activation. The $\beta\gamma$ subunit released by activation of the G_i subfamily also can recruit and activate PLC- β by interaction with the PH domain. Treatment of cells with pertussis toxin can determine which G protein family is responsible for PLC activation. Pertussis toxin inhibits the activation of the G_i family of G proteins, but has

no effect on the G_q family. Different isozymes in the PLC- β subfamily demonstrate differing abilities to respond to G protein α and $\beta\gamma$ subunits. For example, PLC- $\beta 1$ is quite sensitive to $G\alpha_q$ and poorly responsive to $G\beta\gamma$. Finally, PI-3-P, generated by a PI 3-kinase, binds the PH domain of PLC- $\beta 1$ and aids in recruitment of the enzyme to the membrane.

PLC- γ is activated by a large group of cell-surface receptors, including receptor protein tyrosine kinases, antigen and immunoglobulin receptors, and heptahelical receptors [3]. The basic mechanism used by all receptors involves tyrosine phosphorylation of PLC- γ and targeting of the enzyme to the plasma membrane. Receptors with intrinsic tyrosine kinase activity (e.g. growth factor receptors) first recruit PLC- γ to phosphorylated tyrosine residues on the receptor by interaction with the N-terminal SH2 domain. This allows the receptor tyrosine kinase to phosphorylate PLC- γ , inducing enzyme activation. Recruitment of PLC- γ to the membrane also involves the interactions of PI-3,4,5- P_3 with the C-terminal SH2 domain and the PH domain. Regulatory roles for the EF hand, SH3 and C2 domains of PLC- γ are not known. Activation of PLC- γ by antigen involves intracellular non-receptor protein tyrosine kinases that are either recruited to the receptor or activated by other pathways. These tyrosine kinases phosphorylate PLC- γ and, along with PI-3,4,5- P_3 , induce enzyme activation. With heptahelical receptors, a non-receptor tyrosine kinase phosphorylates and activates a growth factor receptor, which then recruits and phosphorylates PLC- γ , as described above.

The pathways used by membrane receptors to activate the PLC- δ isozymes are unclear [3]. PLC- δ is more sensitive to Ca^{2+} than the other subgroups, so an increase in intracellular Ca^{2+} levels induced by receptor-mediated activation of one of the other PLC subtypes may be responsible for PLC- δ activation. Also, a newly discovered type of GTP-binding protein, G_h (high-molecular-weight G protein) may regulate PLC- δ after stimulation of cells through α_1 -adrenergic, oxytocin, or the α subtype of thromboxane A_2 receptors.

The newly discovered PLC- ϵ appears to be regulated by two mechanisms [3]. Growth factor receptors lead to activation of the small GTPase Ras, which binds to PLC- ϵ via the C-terminal Ras binding domain and activates the enzyme. The N-terminal Ras guanine nucleotide exchange factor-like domain may prolong PLC- ϵ activation by reactivating exchange of GTP for GDP on Ras. PLC- ϵ is also activated by the α subunit of the heterotrimeric G protein G_{12} , which couples to LPA receptors, which are members of the large heptahelical (G protein-coupled) receptor family.

2.1.2. Inositol phosphate kinases and phosphatases

The initial product of PLC, Ins-1,4,5- P_3 , acts as a second messenger for the release of Ca^{2+} from intracellular stores (Section 2.2.1). However, Ins-1,4,5- P_3 is rapidly metabolized by sequential phosphatase action to free inositol or by Ins-1,4,5- P_3 3-kinase to inositol-1,3,4,5-tetrakisphosphate (Ins-1,3,4,5- P_4). Inositol polyphosphate phosphatases comprise a family of enzymes, which use both inositol phosphates and the phosphoinositides as substrates to varying degrees [6]. Group I inositol polyphosphate 5-phosphatases act on Ins-1,4,5- P_3 and Ins-1,3,4,5- P_4 and appear to be key enzymes for dampening Ca^{2+} signaling. Inositol polyphosphate 1-phosphatase hydrolyzes Ins-1,4- P_2 and Ins-1,4,5- P_3 and may be the target of lithium inhibition during therapy for manic depression.

Ins-1,4,5- P_3 3-kinases, which also remove Ins-1,4,5- P_3 , are activated by the Ca^{2+} -binding protein calmodulin as a feedback mechanism when Ins-1,4,5- P_3 causes a rise in Ca^{2+} levels [2]. Other members of the inositol phosphate kinase multi-gene family include inositol phosphate multikinases, which can phosphorylate Ins-1,4,5- P_3 to form Ins-1,3,4,5,6- P_5 , and Ins P_6 kinases, which can form Ins-1,2,3,4,5,6- P_6 from multiple inositol phosphate substrates [2].

2.2. Cellular targets

2.2.1. Control of intracellular calcium levels by inositol-1,4,5-trisphosphate

Ca^{2+} is an important intracellular signaling molecule, which regulates a host of cell functions [7]. These include contraction, proliferation, fertilization, secretion, vesicle trafficking, apoptosis, and intermediary metabolism. In keeping with its signaling role, the cytosolic level of calcium is very low (~ 100 nM), but can rise to ~ 1000 nM when cells are activated by agonist–receptor interaction. Receptor activation triggers pathways leading to the opening of Ca^{2+} channels in the endoplasmic reticulum and the plasma membrane, causing the marked elevation in intracellular Ca^{2+} concentration. The increase is transient because the second messengers acting to open Ca^{2+} channels are rapidly metabolized and Ca^{2+} pumps and exchangers remove Ca^{2+} from the cytosol to restore the resting state. A major pathway for increasing Ca^{2+} levels is the formation of Ins-1,4,5- P_3 by activation of PLC (Section 2.1.1). Ins-1,4,5- P_3 binds to its receptor, which is a Ca^{2+} channel in the endoplasmic reticulum. The role of Ins-1,4,5- P_3 may be to increase the sensitivity of the receptor to stimulation by cytosolic Ca^{2+} , rather than directly causing the channel to open.

2.2.2. Targets for other inositol phosphates

More highly phosphorylated inositol phosphates also may regulate cell function [2]. Ins-1,3,4,5- P_4 has a higher affinity than Ins-1,4,5- P_3 for hydrolysis by inositol polyphosphate 5-phosphatase and can prolong the lifetime of Ins-1,4,5- P_3 , thus enhancing increases in Ca^{2+} concentration. Ins-1,3,4,5- P_4 directly activates Ca^{2+} channels in the plasma membrane in some cells, also leading to enhanced Ca^{2+} levels. Ins-3,4,5,6- P_4 appears to inhibit plasma membrane Ca^{2+} -regulated chloride channels. Inositol hexakisphosphate (Ins P_6), also known as phytic acid, has several proposed functional targets. One is a protein kinase that phosphorylates pacsin/syndapin I, a protein involved in vesicle trafficking. A second is a DNA-dependent protein kinase that regulates DNA end-joining. Ins P_6 also may regulate mRNA transport out of the nucleus.

3. Diacylglycerols

3.1. Mechanisms of generation and metabolism

3.1.1. Hydrolysis of phospholipids

The initial step for DG generation in response to agonist–receptor engagement is the hydrolysis of PI-4,5- P_2 by PLC (Section 2.1.1). A PC-hydrolyzing PLC activity has also

phosphatase (EC 3.1.3.2) is a cytosolic enzyme absolutely dependent on Mg^{2+} and is likely involved in lipid biosynthetic reactions. It has not been characterized at the molecular level. Type II PA phosphatase does not require divalent cations and is an integral membrane protein. It is more properly termed a lipid phosphate phosphatase, since it hydrolyzes a variety of signaling lipid phosphates. Substrates include PA, LPA, ceramide 1-phosphate, sphingosine 1-phosphate, and diacylglycerol pyrophosphate. The cDNAs encoding several Type II lipid phosphate phosphatases have been cloned and these enzymes belong to a phosphatase superfamily. Superfamily members include bacterial non-specific acid phosphatases, yeast lipid phosphate phosphatases, fungal haloperoxidases, mammalian glucose-6-phosphatase, and several others. All family members share three highly conserved domains that likely comprise the active site. The mechanisms that regulate the lipid phosphate phosphatases are not clear, but several of the enzymes contain putative phosphorylation sites. Additional studies are needed to determine the functional roles of the lipid phosphate phosphatases, but they clearly can degrade a number of lipid second messengers. Thus, their primary function may be to attenuate cellular responses to agonist–receptor interaction.

3.1.2. Diacylglycerol kinases

DG is metabolized by DG kinases, which phosphorylate DG to form PA [12]. This action likely serves to attenuate DG-mediated responses, but the generated PA may be another second messenger (Section 4.3). Mammalian DG kinases comprise a family of nine isoenzymes (α , β , γ , δ , η , ϵ , ζ , ι , θ), differing in their primary structure (Fig. 6), substrate specificity and tissue distribution. All contain a conserved catalytic domain and two cysteine-rich (C1) domains. C1 domains in PKC and other proteins bind DG (Sections 3.2 and 3.3); however, the role of C1 domains in DG kinase isoforms is not known. The presence of other protein:protein and protein:lipid interaction domains (e.g. PH, EF hand, proline-rich region, ankyrin repeats) divides the isoenzymes into five classes

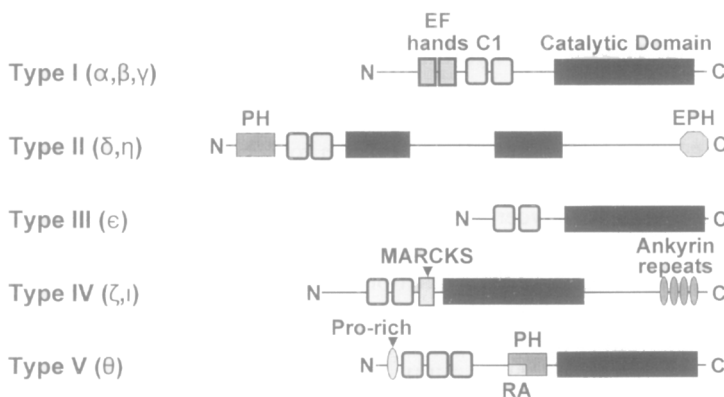


Fig. 6. Domain structures of mammalian diacylglycerol (DG) kinases. Domains are labeled from top of figure. PH, pleckstrin homology; EPH, ephrin C-terminal tail homology; MARCKS, sequence homologous to the myristoylated alanine-rich C-kinase substrate (MARCKS) phosphorylation site domain; Pro-rich, proline-rich domain; RA, Ras-associating domain.

(Fig. 6), suggesting that each class may participate in different signaling pathways. Many of the isoenzymes also have putative phosphorylation sites. Thus, regulation of DG kinases involves phosphorylation reactions and specific protein:protein and protein:lipid interactions, resulting in alterations in localization and activity of the enzymes. The diversity and complexity of the DG kinase family suggests that, like PKC (Section 3.2) and PLC (Section 2.1.1), these enzymes are crucial participants in lipid-mediated signaling processes. However, specific roles for DG kinases in the regulation of cell functions await elucidation.

3.2. Protein kinase C is an important target for diacylglycerol

PKC was the first lipid-regulated protein kinase to be described and its discovery by Nishizuka and colleagues [13,14] initiated a new era of lipid-mediated signal transduction. DG and Ca^{2+} were quickly identified as the physiological activators of PKC, placing PKC as a primary downstream participant in the hormonally activated PI pathway described by the Hokins [1] (Section 2.1.1). PKC regulates almost all cell functions, including cell differentiation, proliferation, metabolism, and apoptosis. The discovery that PKC is the target of the tumor-promoting phorbol diesters illustrated its importance for the regulation of carcinogenesis [15].

PKC is now known to be a family of twelve mammalian isoforms, divided into four classes based on structural differences and cofactor dependence [15,16] (Fig. 7). All PKCs are a single polypeptide chain, consisting of an N-terminal regulatory region and a C-terminal catalytic domain connected by a protease-sensitive hinge region. The conserved catalytic domain consists of the C3 ATP-binding region and the C4 substrate-binding domain. The conventional isoforms (α , β I, β II, γ) are activated by

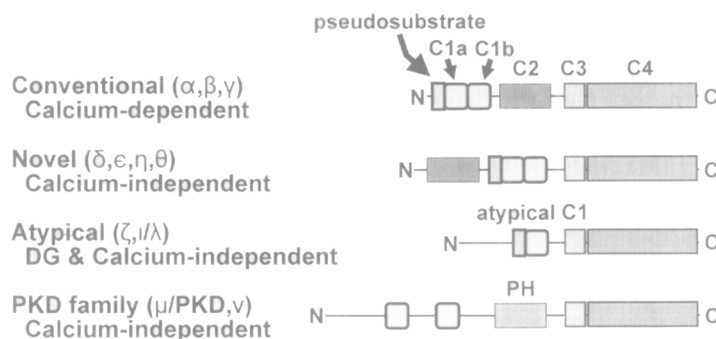


Fig. 7. Domain structure of protein kinase C (PKC) isoforms. The N-terminal half of the protein contains regulatory regions (C1, C2), while the C-terminal half contains the catalytic regions (C3, C4). The conventional isoforms contain tandem C1 domains and one C2 domain, which bind diacylglycerol (C1), Ca^{2+} (C2) and anionic phospholipids (C2). Adjacent to the C1 domains is a pseudosubstrate sequence involved in maintaining the auto-inhibitory conformation of the PKCs. The novel isoforms contain an altered C2 domain, which no longer binds Ca^{2+} . The atypical isoforms lack a C2 domain and have one atypical C1 domain, accounting for their lack of activation by diacylglycerol (DG) and Ca^{2+} . The PKD family is larger in size, has two non-adjacent C1 domains and a pleckstrin homology (PH) domain instead of a C2 domain. It is Ca^{2+} -independent, but is activated by DG and anionic phospholipids.

Ca^{2+} , DG, and phosphatidylserine. The C2 region binds phosphatidylserine and Ca^{2+} , while DG binds to the two C1 domains. A pseudosubstrate region, containing sequences resembling phosphorylation sites in PKC substrates but without a phosphorylatable residue, is at the N-terminal end of the first C1 domain. In the inactive state, PKC is folded such that the pseudosubstrate region interacts with the C4 substrate binding domain, blocking access of exogenous substrates to the active site. Binding of DG, phosphatidylserine and Ca^{2+} to the regulatory domain disrupts this interaction, leading to PKC activation.

The novel PKC isoforms (δ , ϵ , θ , η) have two DG-binding C1 domains and an atypical C2 region near the N-terminus, which binds phosphatidylserine, but not Ca^{2+} . These isoforms are Ca^{2+} -independent and are activated by phosphatidylserine and DG. The atypical PKCs (ζ , ι/λ) lack the C2 region and have an atypical C1 region that does not bind DG. Thus, these isoforms are Ca^{2+} - and DG-independent, but can still be activated by phosphatidylserine. Other mechanisms of activating the atypical isoforms likely exist. Members of a fourth subclass (μ , ν) are larger than the other isoforms and have two separated C1 regions and a PH domain in the extended regulatory region [17]. These isoforms are activated by phosphatidylserine and DG, but may have additional mechanisms of regulation.

It has become clear that regulation of PKC involves more than binding of activators to induce conformational changes in the enzyme. The lipid activators of PKC also target the enzyme to membranes, which regulates full activation of the enzyme, access of the enzyme to substrates, and proteolytic degradation of the enzyme [16]. Membrane targeting involves the cooperation between lipid binding to the C1 and C2 domains and probable interactions of other regions with membrane lipids. Protein:protein interactions also regulate PKC localization and function [15]. PKC isoforms interact with a wide variety of proteins, including RACKs (receptors for activated C kinase), STICKs (substrates that interact with C kinase), cytoskeletal proteins, and scaffolding proteins. These interactions function either to target specific PKC isoforms to sites of activator generation and/or substrates or to integrate PKC into other signaling pathways. Finally, all PKC isoforms require phosphorylation during maturation of the enzyme for catalytic competence. Phosphorylation within the activation loop by phosphoinositide-dependent protein kinase-1 or a related kinase is required for enzyme activity. Also, two sites in the C-terminal tail of PKC must be autophosphorylated for proper localization and catalytic competence of the enzyme. The multiplicity of PKC isoforms and the complexity of their regulation affirm that this family of enzymes plays critical roles in various signaling pathways.

3.3. Non-PKC targets of diacylglycerol

Most of the PKC isoforms are targets of DG and mediate much of the signaling by this lipid second messenger. However, other proteins contain C1 domains that may bind DG. Some of these appear to be atypical C1 domains, similar to the C1 domain in PKC- ζ [16]. These include DG kinase isoforms, the protein kinase Raf, and the Rac guanine nucleotide exchange factor Vav. In contrast, DG and phorbol diesters have been found to bind to several non-kinase proteins, including the chimaerins, which are Rac GTPase-

activating proteins, Unc-13 and related proteins, which may be scaffolding proteins, and RasGRP, a guanine nucleotide release protein for Ras [15]. The main function of DG may be to recruit these proteins to membrane sites where they regulate other enzyme activities (e.g. RasGRP activates Ras). Thus, DG may have other functional roles beyond activation of PKC.

4. Phospholipase D and the generation of phosphatidic acid

4.1. Discovery and molecular nature of phospholipase D

PLD (EC 3.1.4.4) was discovered in plants over 50 years ago as an enzyme that catalyzed the hydrolysis of PC to form PA and choline [18]. However, PLD was not of interest in the field of signal transduction until mammalian PC-hydrolyzing PLD was shown to be activated by extracellular stimuli in the late 1980's (reviewed in [18]). Mammalian PLD is activated by a wide variety of hormones, neurotransmitters, cytokines, and growth factors and plays essential roles in the regulation of numerous cellular functions [18,19]. A number of other eukaryotic PLD activities have been described, including oleate-activated PLD, phosphoinositide-specific PLD, phosphatidylserine/phosphatidylethanolamine-hydrolyzing PLDs, *N*-acyl-phosphatidylethanolamine PLD, lyso-PLD, and glycosylphosphatidylinositol-PLD. With the exception of the glycosylphosphatidylinositol-PLD, which is a serum enzyme and has no signaling role, none of these other enzymes have been identified at the molecular level. Thus, this section will focus on the molecularly characterized PC-hydrolyzing PLDs that clearly function in glycerolipid signaling.

PC-PLD enzymes are members of a diverse gene superfamily that includes cardiolipin and phospholipid synthases, endonucleases and certain viral proteins, which share a common 'HKD' motif involved in catalysis (E.V. Koonin, 1996; C.P. Ponting, 1996). PC-PLDs are present in most organisms, including bacteria, plants, flies, yeast, worms, and mammals [18]. Some organisms express only one PC-PLD gene (e.g. yeast), while others express multiple genes (plants, mammals). The domain structure of representative PC-PLDs from various organisms is shown in Fig. 8. All have four conserved domains (I–IV), which comprise the catalytic core of the enzyme. The exact functions of Domains I and III are unknown. Domains II and IV contain the short conserved sequence HxKx₄Dx₆GG/S/TxN termed the 'HKD motif' or the phosphatidyltransferase motif, present in all members of the PLD superfamily. PLD activity requires both HKD motifs and Domains I–IV may form a bilobed, 'dimer' structure, such that both HKD motifs are present in the active site (I. Leiros, 2000). The histidines are thought to attach the phosphatidyl moiety to the enzyme, forming a phosphohistidine intermediate. This is a hallmark of the 'transphosphatidylation' reaction catalyzed by PC-PLD and other members of the PLD superfamily. The phosphatidyl moiety is transferred to a H₂O molecule to form PA, or it can be transferred to an alcohol acceptor to form a phosphatidylalcohol. This property is the basis for a commonly used PLD assay, in which cells are incubated in the presence of a short-chain primary alcohol (e.g. 1-butanol, ethanol) and the formation of the phosphatidylalcohol (which is poorly metabolized) is measured.

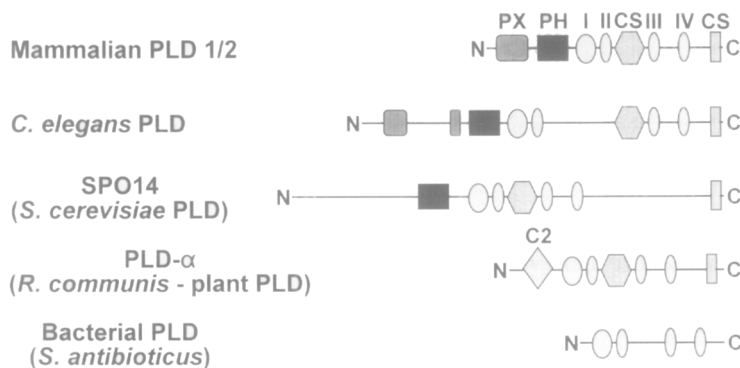


Fig. 8. Comparison of domain structures of phospholipase D (PLD) isoforms from various organisms. Domains are labeled at the top of the figure, except for the C2 domain found only in plant PLD isoforms. All isoforms contain four conserved regions involved in catalysis (I, II, III, IV). All isoforms except the bacterial enzymes contain two conserved regions (CS) of unknown function. PX, phox homology domain; PH, pleckstrin homology domain. The PX domain in *C. elegans* PLD is split.

Incubation of cells with primary alcohols can be used to reduce the production of PA by PLD, thus interfering with PLD-dependent regulation of cellular functional responses.

Most of the PC-PLDs from higher organisms are inactive in unstimulated cells and are regulated by the interaction of intracellular signaling proteins and lipids with domains present in the various proteins [18] (Fig. 8). The active site of PLD is likely blocked by the folded structure of the protein and requires a conformational change to release the autoinhibition. Mammalian and *Caenorhabditis elegans* PLDs each contain an atypical PH domain and a PX (phox homology) domain in the N-terminal half of the molecule. The *Saccharomyces cerevisiae* enzyme contains only the PH domain. The PH domain in human PLD1 binds PI-4,5-P₂ (M.N. Hodgkin, 2000), but it is controversial whether or not this binding is responsible for the PI-4,5-P₂ dependence of PLD activity. PX domains in other proteins were recently shown to target to membranes by binding selected phosphoinositides [20]. The function of the PX domain in PLDs is not known. The plant PLDs lack PH and PX domains, but have a C2 domain in the N-terminal region (Fig. 8). The C2 domain binds Ca²⁺ and anionic phospholipids and is likely responsible for the dependence of the plant enzymes on Ca²⁺ for activity. All eukaryotic PC-PLDs also contain conserved regions between catalytic Domains II and III and at the C-terminus. The functions of these regions are unknown.

4.2. Localization and regulation of phospholipase D

Numerous receptor agonists induce activation of mammalian PC-PLD in vivo [19]. Activation does not involve direct interaction of occupied receptors with PLD, but instead uses intracellular signaling intermediates. Evidence is strong that PLD is regulated in vivo by PKC, the ARF, Rho, and Ras/RalA families of small GTPases, and the intracellular levels of PI-4,5-P₂. The cDNAs for two mammalian PLD isoforms (PLD1, PLD2) have been cloned and each gene can be alternatively spliced to yield two products. In vitro assays of the recombinant PLD enzymes show that PLD1a and PLD1b

require PI-4,5-P₂ for activity and are synergistically activated by PKC- α or - β , ARF1, and RhoA (see Fig. 5). It is not known if such synergism between multiple activators takes place in intact cells. In contrast, PLD2a and PLD2b require PI-4,5-P₂, are only slightly activated by ARF1 and are insensitive to PKC and RhoA. For the most part, the functional binding sites on the PLD molecule for these regulators are unknown. The PI-4,5-P₂-binding site may be the PH domain or the conserved region between Domains II and III (Fig. 8). The ARF-binding site has not been identified, despite intensive effort. The PKC- α/β binding site appears to be in the N-terminal third of PLD1 and the RhoA-binding site is in the C-terminal third, but the specific motifs involved have not been defined.

PLD1 and PLD2 usually are differentially localized in cell types that have been examined [18]. PLD1 is associated with intracellular membrane compartments, most likely endosomal/lysosomal membranes. PLD2 is associated with the plasma membrane and has been found in caveolae. Given the *in vitro* differences in the regulation of PLD1 and PLD2 and their differential localization, it is likely that each has unique functions in cellular regulation.

4.3. Functions regulated by phospholipase D and phosphatidic acid

Studies that selectively manipulate PLD1 and PLD2 levels and activity in intact cells, in order to identify PLD isoform-dependent downstream signaling elements and cellular functions, are still in the early stages. Similarly, while PA regulates a number of proteins *in vitro*, it is not clear whether PA has the same role *in vivo* for most of the proteins. However, pharmacological approaches and PLD localization studies have strongly implicated PLD and PA as important regulators of several cell functions.

4.3.1. Membrane trafficking, secretion and cell proliferation

Both PLD1 and PLD2 have been implicated in membrane trafficking events, including budding of vesicles from the Golgi apparatus and the *trans*-Golgi network and ARF-dependent clathrin coat assembly in the endosomal/lysosomal system [18]. The production of PA by PLD may promote the recruitment of adapter and coat proteins to sites of vesicle budding. PA also can induce activation of Type I PI 4-phosphate kinases, leading to increased levels of PI-4,5-P₂ and additional protein recruitment and PLD activation. Scission of vesicles is mediated by the GTPase dynamin or its analogues, which are also regulated by anionic phospholipids. Thus, the increased levels of PA and PI-4,5-P₂ in the budding membrane are thought to mediate recruitment of the proteins necessary to complete the budding process.

Evidence also implicates PLD in the regulation of stimulated secretory and degranulation responses, such as the insulin-stimulated translocation of GLUT4 to the plasma membrane, release of granule contents from mast cells and neutrophils, exocytosis from chromaffin cells, and the release of matrix metalloproteases from cancer cells [18]. The mechanisms by which the PLD product PA mediates secretion are not known, but also may involve recruitment of particular proteins to secretory vesicle and granule membranes.

PA is thought to have a similar recruitment role in the regulation of Raf1, a key

protein kinase that regulates cell proliferation (M.A. Rizzo, 2000). Raf1 is a target of the small GTPase Ras and triggers cell proliferation through activation of the mitogen-activated protein kinase cascade. PLD activation is required for the activation of Raf1 in intact cells and a PA-binding site in Raf1 has been identified. However, PA has no direct effect on Raf1 activity. Mutation of the PA-binding site prevents recruitment of Raf1 to the membrane and inhibits activation of the downstream mitogenic kinase cascade. This suggests that the role of PA is to recruit Raf1 to the membrane where Ras is located.

4.3.2. NADPH oxidase

Pharmacological data strongly indicate that PLD activation is needed for receptor-mediated activation of the phagocyte multi-component enzyme NADPH oxidase [21]. This enzyme functions in host defense against infection and in tissue damage during inflammatory diseases. Study of the processes regulating this enzyme is providing insight into the complex mechanisms by which signaling lipids mediate functional responses. Activation of NADPH oxidase requires recruitment of four cytosolic proteins, including the small GTPase Rac, to assemble with heterodimeric flavocytochrome b_{558} in the membrane. The flavocytochrome then undergoes a presumed conformational change, allowing it to transfer electrons from NADPH to O_2 to form superoxide anion. The cytosolic proteins exist in inactive complexes and must be dissociated or undergo conformational changes to move to the membrane. The entire process involves phosphorylation of components (by lipid-activated protein kinases, such as PKC), lipid- and phosphorylation-regulated SH3 region-mediated protein : protein interactions, and direct interaction of signaling lipids (PA, DG, arachidonic acid) with various oxidase proteins. PA and arachidonic acid each cause conformational changes in a cytosolic oxidase component, $p47^{phox}$, and the membrane-bound flavocytochrome. PA (plus DG) or arachidonic acid (alone) induce phosphorylation-independent oxidase activation in purified cell-free systems containing only oxidase proteins, presumably by their direct interactions with $p47^{phox}$ and flavocytochrome b_{558} . Two of the cytosolic oxidase components, $p47^{phox}$ and $p40^{phox}$, have N-terminal PX domains, which bind phosphoinositides and likely participate in the recruitment process [20]. New data indicate that PA binds to the $p47^{phox}$ PX domain with greater affinity and efficacy than phosphoinositides (L.C. McPhail, unpublished results). Thus, PA may be the physiological ligand for this particular PX domain and may regulate NADPH oxidase activation partially through its PX domain interaction. Studies are ongoing, but it is clear that PA and other signaling lipids induce activating conformational changes in NADPH oxidase components and regulate membrane localization of oxidase proteins by both direct binding and protein kinase-dependent mechanisms.

5. Phosphoinositides

The phosphoinositides comprise a large group of inositol-containing phospholipids, with phosphates attached to positions D3, D4, or D5 (or various combinations of these positions) of the inositol ring (Figs. 9 and 10). They are present at low concentrations in cells: PI, the precursor phospholipid, is present at less than 10% of total membrane

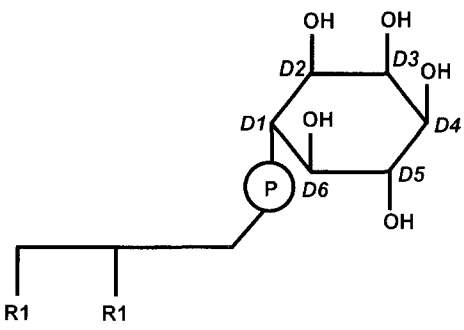


Fig. 9. Numbering of the inositol ring. Phosphates can be attached at positions D1 through D6 on the inositol ring. Shown is phosphatidylinositol, in which inositol is attached to the glycerol backbone by a phosphate in position D1.

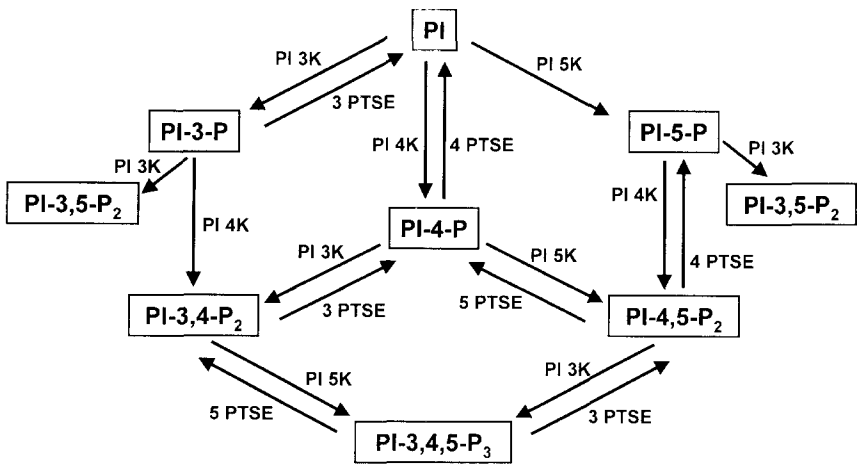


Fig. 10. Interconversion of phosphoinositides. The pathways by which phosphoinositide kinases and phosphatases synthesize and degrade various phosphoinositides are shown. Abbreviations: PI, phosphatidylinositol; PI-4-P, PI-4-phosphate; PI-4,5-P₂, PI-4,5-bisphosphate; PI-3,4,5-P₃, PI-3,4,5-trisphosphate; PI 3K, PI 3-kinase; 3 PTSE, PI 3-phosphatase.

lipid; PI-4,5-P₂, the substrate for PLC, is about 5%; phosphoinositides phosphorylated at the 3 position are less than 0.25% of the total inositol-containing lipids [22]. The phosphoinositides regulate a large number of signaling pathways and cellular functions by binding to specific sites on proteins to alter their localization or activity (Section 5.2). A complex set of PI kinases and phosphatases is involved in the interconversion (Fig. 10) (Section 5.1), and considerable progress in identifying these enzymes at the molecular level has been made. Phosphoinositide-metabolizing enzymes are regulated by receptor-triggered signaling processes, which results in highly localized changes in the levels of specific phosphoinositides to coordinate temporal and spatial regulation of cellular functions.

5.1. *Phosphoinositide kinases and phosphatases*

5.1.1. *Phosphatidylinositol 4- and 5-kinases*

The enzymes that phosphorylate positions D4 and D5 of the inositol ring fall into two major categories, i.e. those that classically act on PI-4-P and those that utilize PI [23]. The PI 4-kinases convert PI to PI-4-P and are divided into two types (II and III), based on biochemical differences (sensitivity to detergents and adenosine). The cDNA for a Type II enzyme has recently been cloned and is suggested to establish a novel family of PI 4-kinases because it lacks the phosphoinositide kinase (PIK) domain (S. Minogue, 2001). The PIK domain is found in the Type III PI kinases and PI 3-kinases (Section 5.1.2) and has an unknown function. No homology to other known proteins was found in the Type II enzyme. Several Type III enzymes have been identified molecularly in yeast and mammals and are categorized as α and β isoforms, based on sequence homology. All contain the PIK domain and a conserved catalytic region. The mammalian α isoform additionally contains ankyrin-like repeats, a SH3 domain, and a polyproline motif. Although it is likely that the PI kinases are regulated by extracellular signals, the mechanisms involved are not known.

The enzymes that act on monophosphorylated forms of PI (PIP) are divided into two families, based on their preferred site of phosphorylation [23,24]. The PIP 5-kinases (also known as Type I) phosphorylate position 5 and PIP 4-kinases (Type II) phosphorylate position 4 of the inositol ring. The PIP 5-kinases are more promiscuous in substrate specificity and phosphorylate both PI-4-P (to form PI-4,5-P₂) and PI-3-P (to form PI-3,5-P₂ or PI-3,4-P₂). In addition, they can add two phosphates to PI-3-P in vitro to form PI-3,4,5-P₃. The PIP 4-kinases prefer PI-5-P as substrate and form PI-4,5-P₂. Multiple isoforms of each have been found in plants and animals. The two groups share homology in their kinase domain, which resembles only slightly the kinase domains in other phosphoinositide and protein kinases. The Type II enzymes contain polyproline sequences, indicating possible interaction with SH3 domains. The two groups do not possess other known domains or motifs. A crystal structure of a PIP 4-kinase shows the enzyme exists as a dimer, forming an extended basic flat face that interacts with membranes containing anionic phospholipids (V.D. Rao, 1998).

Regulation of the PIP 5- and 4-kinases by receptor-mediated signals is under intensive study, but physiologically relevant mechanisms are unclear [23,24]. Some of the Type II enzymes associate with cell-surface growth factor and cytokine receptors. The Type I enzymes are markedly stimulated by PA in vitro and this also may occur in intact cells (D.R. Jones, 2000). The Type I enzymes may be regulated by GTP-binding proteins, since the non-hydrolyzable GTP analog GTP γ S increases PIP 5-kinase activity in cells.

5.1.2. *Phosphoinositide 3-kinases*

Phosphoinositide 3-kinases catalyze the addition of phosphate to the D3 position of the inositol ring of phosphoinositides. Phosphoinositide 3-kinases comprise a large family of enzymes divided into three classes (I, II, and III), with each class showing differences in structure (Fig. 11), substrate specificity, and regulation [23]. The class I enzymes are heterodimeric proteins, consisting of a 110 kDa catalytic subunit and a 50–101 kDa adaptor/regulatory subunit. The adaptor/regulatory subunit inhibits catalytic activity

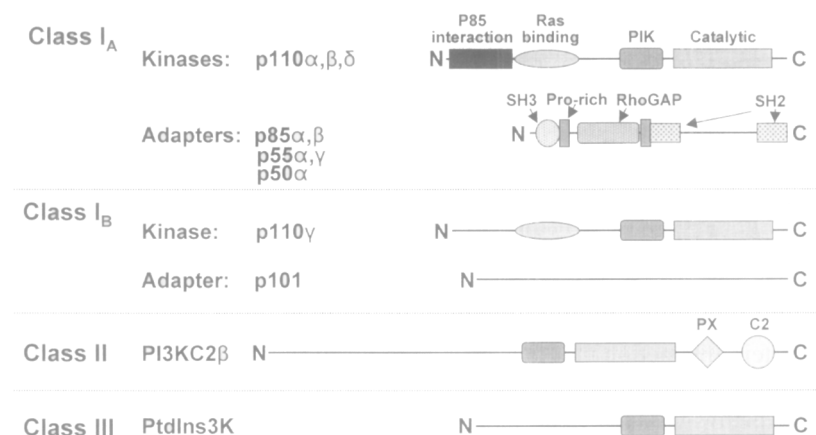


Fig. 11. Classes of mammalian phosphoinositide 3-kinases. Class I enzymes consist of a kinase subunit and an adapter (regulatory) subunit. Functional domains are labeled at the top of the figure or above each isoform. Domains include: p85 interaction domain; Ras binding domain; phosphoinositide kinase (PIK) domain; catalytic domain; Src homology 2 (SH2) and 3 (SH3) domains; proline-rich region (Pro-rich); Rho-GTPase-activating protein (RhoGAP) domain; phox homology (PX) domain; and C2 domain.

and recruits the enzyme to specific subcellular sites by regulated interaction with other proteins. In vitro, the class I enzymes use PI, PI-4-P, and PI-4,5-P₂ as substrates. However, the major products in vivo are PI-3,4-P₂ and PI-3,4,5-P₃, so intracellular regulatory mechanisms must limit substrate specificity.

The class I enzymes are further subdivided into two classes (IA and IB), based on differences in sequence and receptor regulation (Fig. 11). The class IA enzymes are regulated by receptor protein tyrosine kinases and consist of three 110 kDa catalytic subunits (α , β , δ) from different genes and at least five regulatory subunits (p85 α , p85 β , p55 α , p55 γ , p50 α), which can interact interchangeably. The p55 α and the p50 α proteins are alternatively spliced products of the p85 α gene. By convention, all of these subunits are referred to as p85. The class IB enzyme is activated by G protein-coupled receptors and is made up of one p110 catalytic subunit (γ) and a 101 kDa regulatory subunit, with no homology to the class IA adaptor/regulatory subunits. The p110 γ subunit cannot interact with the class IA regulatory subunits, because it lacks an N-terminal interaction region present in p110 α , β , and δ (Fig. 11). The p110 subunits share a number of other functional domains, including the C-terminal catalytic domain, a PIK domain shared with the Type III PI 4-kinases, and a Ras-binding domain. When activated by receptor agonists, the small GTPase Ras can bind to p110 and increase catalytic activity, indicating that the class I enzymes are Ras effectors.

The class IA p85 subunits also share a number of protein-interaction domains (Fig. 11), whose function is to recruit the holoenzyme to specific cellular locations by interaction between the domains and specific target proteins. All p85 proteins contain two SH2 domains, which bind to specific motifs containing phosphorylated tyrosine residues. Tyrosine phosphorylation occurs in many receptor complexes (e.g. receptor tyrosine kinases such as the epidermal growth factor receptor) in response to

extracellular stimuli. Binding of the SH2 domains of p85 to phosphotyrosine motifs in the receptor complex mediates the recruitment of p110 to the membrane, where it can access substrates and undergo additional activation by Ras. The longer forms of p85 also contain a SH3 domain in the N-terminal region, which binds to polyproline motifs in target proteins, and either two (p85 α) or three (p85 β) polyproline (Pro-rich) motifs. The shorter p85 isoforms contain only one polyproline motif. Only a few binding partners (e.g. the GTPase dynamin for the SH3 region, the Src family of protein tyrosine kinases for the polyproline motifs) for the N-terminal region are known. The longer forms of p85 contain a Rho-GTPase-activating protein homology domain, which mediates interaction of the Rho family of GTPases (e.g. Rac, Cdc42) with p85. Whether these different protein-interaction motifs cooperate to localize and activate the enzyme, antagonize each other under certain conditions, or allow the enzyme to participate in different signaling pathways is not clear.

The class IB catalytic subunit (p110 γ) is activated by interaction with $\beta\gamma$ subunits of heterotrimeric G proteins. It shares the catalytic, PIK, and Ras-binding domains with the class IA p110 subunits (Fig. 11). The function of the p101 adaptor/regulatory subunit is not clear, since it has no homology to the p85 subunits. However, p101 appears to be necessary for the activation of p110 γ by $\beta\gamma$ subunits.

The class II phosphoinositide kinases (α , β) are single polypeptides, sharing the catalytic and PIK domains with the class I p110 subunits. These kinases also contain an atypical C2 domain, which may mediate the binding of the protein to anionic phospholipids in membranes. The C2 domain lacks residues that are important for Ca^{2+} binding, so Ca^{2+} most likely does not play a role in phospholipid binding. The two mammalian isoforms both have a C-terminal PX domain, which may bind PI-4,5- P_2 (X. Song, 2001). Both the C2 and the PX domains are likely to regulate membrane recruitment and activation of the class II enzymes. In vitro, the class II enzymes prefer PI and PI-4-P as substrates, thus forming PI-3-P and PI-3,4- P_2 . Their specificity in vivo is not known.

The class III phosphoinositide 3-kinases share only the catalytic and PIC domains with the enzymes in the other two classes. The first class III enzyme was identified in yeast, where it is essential for transport of proteins. These enzymes phosphorylate only PI and are thought to be the major contributors to the synthesis of PI-3-P. A putative adaptor/regulator for the class III enzymes has been identified, which is a protein serine/threonine kinase. It was thought that the class III enzymes are not subject to receptor-mediated regulation, but recent work suggests that they regulate the process of phagosome maturation in professional phagocytic cells (O.V. Vieira, 2001). Since several protein domains have been shown to bind PI-3-P (Section 5.2.2), it is likely that the class III enzymes have important signaling roles.

5.1.3. PTEN and other phosphoinositide phosphatases

Inositol phosphatases remove phosphates from the inositol ring and many of these enzymes can use both soluble inositol phosphates and lipid phosphoinositides as substrates. The focus in this section is on the lipid phosphatases that hydrolyze phosphate from the D3, D4, or D5 position of the phosphoinositides. The overall function of these phosphatases is to down-regulate phosphoinositide-dependent signaling reactions.

Type II 5-Phosphatases



3-Phosphatases



Fig. 12. Domain structures of representative phosphoinositide phosphatases. Shown are selected members of the Type II 5-phosphatases and the 3-phosphatases. Domains include: catalytic (different for the 5- and the 3-phosphatases); Rho-GTPase-activating protein (RhoGAP) domain; Sac homology domain, a region with lipid phosphatase activity; polyproline (Pro-rich) region; Src homology 2 (SH2) domain; C2 domain; a phosphatidylinositol-4,5-bisphosphate binding motif (PIP₂-binding) and a PDZ motif.

Mutations in genes encoding some of these enzymes are responsible for certain dys- trophies and myopathies and also are common in various tumors. These phosphatases are classified into three families, based on selectivity for the different positions on the inositol ring: the 5-phosphatases, the 3-phosphatases, and the 4-phosphatases [25]. Domain structures of representatives from the 5- and 3-phosphatase families are shown in Fig. 12.

The largest family is the 5-phosphatases, which is divided into two major groups, based on substrate preference and structural differences [6,25]. They all share a homologous phosphatase domain, defined by two signature motifs involved in catalysis and divalent metal ion binding (Y. Tsujishita, 2001). Type I enzymes hydrolyze only water-soluble substrates, as discussed in Section 2.1.2. Type II enzymes prefer PI-4,5-P₂ and/or PI-3,4,5-P₃ as substrates. Although they share the 5-phosphatase catalytic domain, Type II members differ with respect to other regions. One subclass is typified by OCRL-1 that, when mutated, causes Lowe syndrome or oculocerebrorenal dystrophy. Characteristics of the disease include cataracts, mental retardation and the development of renal failure. OCRL-1 contains a C-terminal Rho-GTPase-activating protein domain of unknown function, which could be involved in localization of the enzyme.

A second Type II subclass consists of the synptojanins, which participate in synaptic vesicle trafficking. These proteins have a C-terminal polyproline region, presumably mediating interactions with specific SH3 domain-containing proteins. Synptojanins also contain an N-terminal Sac homology domain, which possesses a separate lipid phosphatase activity (W.E. Hughes, 2000). The Sac domain shows selectivity for monophosphoinositides or phosphoinositides without adjacent phosphates (i.e. PI-3,5-P₂). Thus, synptojanins have two phosphatase domains, which allows hydrolysis of PI-4,5-P₂ to PI.

A third Type II sub-class consists of SHIP1 and SHIP2, which show specificity for 3-phosphate-containing substrates (e.g. PI-3,4,5-P₃). These enzymes contain a SH2

domain at the N-terminus, which is responsible for their acronym (*SH2*-containing inositol phosphatase). SHIP1 is restricted to hematopoietic cells, while SHIP2 is more widely distributed. The SHIPs also can be phosphorylated on C-terminal region Tyr residues, which allows interaction with SH2 domain-containing proteins (e.g. Shc). SHIPs additionally contain one or more polyproline motifs in the C-terminal half of the protein, which mediates binding to SH3 domain-containing proteins. The SH2 domains and the other motifs mediate assembly of SHIPs into growth factor and cytokine receptor signaling complexes, where the enzymes can hydrolyze newly formed PI-3,4,5-P₃. This limits the duration of PI-3,4,5-P₃-mediated signaling.

The fourth Type II subclass of 5-phosphatases (not shown in Fig. 12), like the SHIPs, shows selectivity for PI-3,4,5-P₃. These enzymes lack SH2 domains, but contain numerous N-terminal polyproline motifs, as well as an immunoreceptor tyrosine-based activation motif, usually found only in receptor molecules (M.V. Kisseleva, 2000). Binding partners for these motifs, which presumably regulate localization and function of the enzyme, are not identified.

The 3-phosphatases are a novel family of proteins, which contain a protein tyrosine phosphatase active site motif [26]. PTEN/MMAC1/TEP1 (acronyms for phosphatase and *tensin* homolog, *mutated in multiple advanced cancers*, and *TGFβ*-regulated and *epithelial cell-enriched phosphatase*) was the first member identified and it was originally thought to be a dual specificity protein tyrosine kinase involved in tumor suppression. Mutations in the PTEN gene cause tumorigenesis and contribute to specific syndromes with increased risk for breast and thyroid cancers. The known functions of PTEN are to switch off cell proliferation and to permit programmed cell death (apoptosis), which clearly are tumor suppressive activities. Maehama and Dixon (Maehama, 1998) first reported that PTEN had lipid phosphatase activity, hydrolyzing the D3 phosphate from PI-3,4,5-P₃ with high efficiency. Hydrolysis of PI-3,4,5-P₃ seems to be the primary function of the enzyme in vivo, based on studies using mutations that eliminate lipid, but not protein, phosphatase activity. Regulation of PTEN function is most likely by membrane translocation, mediated by a N-terminal PI-4,5-P₂ binding domain, a C2 domain in the C-terminal region, and a C-terminal PDZ-binding site. Both the PI-4,5-P₂ binding domain and the C2 domain mediate binding to membrane phospholipids. Interaction with PDZ proteins, which are scaffolding proteins, may aid in assembly of PTEN with signaling complexes.

A second group of enzymes, the myotubularins, also hydrolyze the D3 phosphate with preference for PI-3-P [26]. Mutations in two myotubularin family members, MTM1 and MTMR2, cause two human diseases: X-linked myotubular myopathy and Type 4B Charcot-Marie-Tooth syndrome, respectively. The mutations usually affect catalytic activity, suggesting the requirement for phosphatase activity in the normal functioning of the proteins. However, the physiological functions controlled by the myotubularins are not understood. These proteins contain several regulatory domains, including a PDZ-binding motif. Some family members also contain lipid-binding domains (i.e. PH or FYVE domains), which may help to localize the enzyme near its substrate.

The 4-phosphatases are also involved in regulation of 3-phosphoinositide signaling. These enzymes hydrolyze the D4 phosphate of PI-3,4-P₂ to yield PI-3-P [6]. This action will limit the duration of receptor-mediated increases in the levels of PI-3,4-P₂ (see

Section 5.2.1). The 4-phosphatases are Mg^{2+} -independent and may be down-regulated by receptor-triggered proteolysis. Little is known of their physiological functions.

5.2. Cellular targets

Phosphoinositides exert their effects by binding to specific domains in target proteins. Lipid binding may cause a conformational change in the target protein, thereby altering enzyme activity, or it may affect the localization of the target protein and access to enzymatic substrates. Multiple domains and motifs have been shown to bind phosphoinositides *in vitro* [25,27]. PH domains generally show specificity for PI-4,5- P_2 , PI-3,4- P_2 or PI-3,4,5- P_3 . PI-4,5- P_2 also binds to the ENTH (acronym for epsin N-terminal homology) domain and to certain Lys/Arg-rich motifs in proteins. Some SH2 and PTB domains bind PI-3,4,5- P_3 *in vitro*, but the relevance of this interaction *in vivo* is unclear. FYVE domains [named after the first four proteins shown to contain it: *Fab1p*, *YOTB*, *Vac1p*, and *early endosome antigen 1* (EEA1)] and certain PX domains selectively bind PI-3-P. Other PX domains show greater specificity for other lipids, as discussed in Section 5.2.2.

5.2.1. Domains that bind polyphosphoinositides

PH domains contain about 120 amino acids and were first identified as targets for PI-4,5- P_2 . These domains are found in over 100 proteins, including protein kinases, phospholipases, and regulatory proteins for small GTPases (exchange factors and GTPase-activating proteins). PH domains are divided into four groups, based on their selectivity for binding various phosphoinositides [27]. Group I, typified by PH domains in Bruton's tyrosine kinase, exchange factors for the small GTPase Arf (ARNO, GRP1), and GTPase-activating proteins for Arfs (centaurin α) and Ras (GAP1), show high affinity binding for PI-3,4,5- P_3 . Group II PH domains selectively bind PI-4,5- P_2 and are found in PLC δ , oxysterol-binding protein, β -adrenergic receptor kinase, and β -spectrin. Group III PH domains are found in the protein kinases Akt/protein kinase B (PKB) and phosphoinositide-dependent kinase 1 (PDK1) and bind PI-3,4- P_2 and PI-3,4,5- P_3 with equal affinities. Group IV PH domains bind phosphoinositides with relatively low affinity, typified by the PH domain in the endocytic GTPase dynamin.

The Group III PH domain-containing proteins are important cellular targets for PI-3,4,5- P_3 and PI-3,4- P_2 , because of their central roles in the regulation of cell survival and proliferation. Akt/PKB is a 57 kDa Ser/Thr protein kinase identified as the cellular homolog of the viral oncoprotein v-Akt. It was also shown to have high homology to protein kinases A and C (hence the name PKB). Akt/PKB is activated by the binding of PI-3,4,5- P_3 or PI-3,4- P_2 , which localizes the enzyme to the membrane and induces conformational changes allowing phosphorylation of a critical Thr residue by PDK1. PDK1 is a 63 kDa Ser/Thr kinase, which binds to membranes containing polyphosphoinositides, placing the enzyme in close proximity to Akt/PKB. After interaction with another protein kinase (PKC-related kinase 1), PDK1 is then able to phosphorylate a C-terminal Ser residue in Akt/PKB, rendering the enzyme fully active. Active Akt/PKB phosphorylates a variety of substrates involved in the regulation of apoptosis, cell metabolism, and gene expression. The overall effect of

activated Akt/PKB is to promote cell survival, increase cell size, and enhance cell proliferation.

PI-4,5-P₂ has additional binding sites in proteins, i.e. stretches of Lys/Arg-rich sequences, found primarily in cytoskeletal proteins and in PTEN, and the ENTH domain, a region of about 140 residues found in several proteins involved with endocytosis [27]. Cytoskeletal protein targets include the ezrin/radixin/moesin family of proteins and vinculin, which help to cross-link actin to the plasma membrane, and proteins that regulate actin polymerization, such as gelsolin, cofilin, and profilin (M.J. Bottomley, 1998). ENTH domain-containing proteins include epsin and other proteins involved in assembly of the clathrin lattice. PI-4,5-P₂ also accumulates in the early phagosome, where it recruits PLC. Thus, PI-4,5-P₂ plays major roles in the regulation of the cytoskeleton, endocytosis, and phagocytosis (T. Nebl, 2000; A. Simonsen, 2001).

5.2.2. Domains that bind phosphatidylinositol-3-phosphate

A role for phosphoinositides in endosomal trafficking was first discovered in yeast, where PI-3-P was shown to be the major phosphoinositide involved. Recently, a specific binding domain for PI-3-P, the FYVE domain, was discovered (H. Stenmark, 1996). This domain consists of about 60–80 residues and coordinates two Zn²⁺ atoms [27]. Binding of PI-3-P involves initial contact between a hydrophobic loop of the domain with the membrane, followed by interaction with the headgroup of PI-3-P. Use of the isolated domain coupled to green fluorescent protein demonstrated that PI-3-P is selectively enriched in early endosomes. Many FYVE domain-containing proteins target to endosomes, where they regulate endocytosis, endosomal fusion, and vacuolar transport. About 60 proteins are known to contain FYVE domains, including early endosomal antigen 1, which is essential for endosomal fusion, and PI-3-P-metabolizing enzymes (some of the myotubularin-related 3-phosphatases and the PI 5-kinase PIKfyve).

Another binding target for PI-3-P is the PX domain, first identified as a conserved motif of about 120 residues in the N-terminal regions of the NADPH oxidase proteins p47^{phox} and p40^{phox}. It is present in over 70 human and yeast proteins, and PX domain-containing proteins are also found in plants, flies, and worms [27]. The proteins have diverse functions, although a large group includes the sorting nexins, which are involved in targeting of proteins to lysosomes. Recent studies established that several PX domains bind selectively to PI-3-P, including those in several sorting nexins, the yeast SNARE protein Vam7p, and the NADPH oxidase component p40^{phox}. However, PX domains from several other proteins (p47^{phox}, class II PI 3-kinase, cytokine-independent survival kinase) preferentially bind polyphosphoinositides (PI-4,5-P₂, PI-3,4-P₂, PI-3,4,5-P₃). The PX domain of p47^{phox} also binds PA with higher affinity than the polyphosphoinositides (L.C. McPhail, unpublished results). The lipid-binding specificity for numerous other PX domains is not known. The diversity in lipid-binding and in the types of proteins containing PX domains suggest that PX domains have multiple functions in lipid-signaling pathways.

6. *Lysophosphatidic acid and other lysophospholipids*

For many years it was believed that lysophospholipids were simply intermediates in the metabolism of glycerolipids. However, it has now been clearly established that LPA, LPC and the sphingolipids sphingosylphosphorylcholine and sphingosine-1-phosphate act as extracellular signaling molecules [28]. These lipids bind to a family of related cell-surface heptahelical receptors and are implicated in tumorigenesis, angiogenesis, immunity, atherosclerosis, and neuronal survival. Because sphingolipids are discussed in a separate chapter, the focus here will be on LPA and lysophosphatidylcholine.

6.1. *Sources of lysophospholipids*

A number of cell types have been shown to produce lysophospholipids, including platelets, macrophages, other leukocytes, some epithelial cells, and some tumor cells [29]. The regulation of the synthesis and metabolism of extracellular lysophospholipids is not well understood. Phosphatidylcholine is likely to be the major source for both LPC and LPA and several pathways are possible. The action of a phospholipase A directly yields LPC. LPA can be formed by the sequential actions of PLD and a phospholipase A or of a phospholipase A and a lyso-PLD. The nature of the phospholipases A₂ or A₁ involved may vary, depending on cellular source and stimulus, and is not clearly delineated. Although lyso-PLD activity has been detected in various cell types, the cDNA encoding the enzyme has not been cloned. It is not known how lysophospholipids exit the cell, once they are synthesized. Albumin is probably the main LPA-binding protein in plasma, although plasma gelsolin also binds LPA with higher affinity than albumin. LPC is present in oxidized low density lipoproteins. Metabolism of lysophospholipids is by reacylation and/or, for LPA, dephosphorylation. The enzyme LPA acyltransferase can convert LPA to PA. Lipid phosphate phosphatase-1 is positioned in the plasma membrane of cells such that it can dephosphorylate extracellular LPA to form monoacylglycerol.

6.2. *Lysophospholipids are extracellular signaling molecules*

LPA induces a number of cellular responses, including proliferation, decreased apoptosis, platelet aggregation, smooth muscle contraction, chemotaxis, and tumor cell invasion. LPC has inflammatory effects in the body, including up-regulation of endothelial cell adhesion molecules and growth factors, and activation of macrophages. Both LPA and LPC may be elevated in certain pathophysiological conditions, such as ovarian cancer and atherosclerosis. LPA and LPC are ligands for members of a family of heptahelical receptors, formerly known as the EDG receptors and now termed the LP receptors [28]. Family members number 12 so far and can be divided into three distinct groups. The first group consists of three receptors (termed LPA₁₋₃) showing high selectivity for LPA. The second group (5 receptors, termed S1P₁₋₅) preferentially binds sphingosine-1-phosphate. The third group (SPC₁, LPC₁, PSY, GPR4) is more divergent and shows greater specificity for LPC and sphingosylphosphorylcholine. As

a group, the LP receptors couple to several G proteins, including G_i , G_q , and $G_{12/13}$. Thus, the lysophospholipids exert their effects on cell functions by triggering a variety of G protein-linked second messenger pathways, including activation of PLC, PLD, phosphoinositide 3-kinases, PKC, MAP kinases, and small GTPases. LPA₁ receptor (also called EDG-2) null mice have a complex phenotype, with 50% neonatal lethality and neuronal and other abnormalities (J.J. Contos, 2000). This suggests a critical role for LPA₁ in multiple aspects of development. LPC₁ null mice show T cell abnormalities and develop a late-onset autoimmune disease, similar to systemic lupus erythematosus (L.Q. Li, 2001). Creation of null mice deficient in other receptors in the LP family is needed to help elucidate the spectrum of biological functions of these lysophospholipids.

7. Future directions

Much remains to be learned about glycerolipid signaling. It is becoming clear that the signaling intermediates and protein targets involved in the regulation of specific cell functions assemble and disassemble in dynamic complexes at specific intracellular locations. Movement of proteins in and out of these complexes is likely orchestrated by the transient presence of specific signaling lipids, along with the cytoskeleton. It will be a challenge to design tools to visualize these processes at high resolution within living cells without disrupting the signaling pathways or the functions being regulated. The engineering of fluorescently tagged protein domains that recognize specific lipids is providing some insight [20]. With the dawn of the genomic age, it should soon be possible to identify all of the enzymes involved in the synthesis and metabolism of the signaling lipids, based on homologies with known proteins. Similarly, additional targets for the lipids will be found. However, discerning the biological functions of the numerous enzymes and target proteins involved in lipid signaling will continue to require extensive research using biochemical, cell biological, and molecular tools. Such a global and integrated approach has been established for understanding G protein-dependent cell signaling (The Alliance for Cellular Signaling, <http://cellularsignaling.org/>), which, of course, encompasses glycerolipid signaling.

Abbreviations

DG	Diacylglycerol
Ins-1,4,5-P ₃	Inositol-1,4,5-trisphosphate
InsP ₆	Inositol hexakisphosphate
LPA	Lysophosphatidic acid
LPC	Lysophosphatidylcholine
PA	Phosphatidic acid
PC	Phosphatidylcholine
PH	Pleckstrin homology
PKB	Protein kinase B
PKC	protein kinase C

PLC	Phospholipase C
PLD	Phospholipase D
PI	Phosphatidylinositol
PIP	Phosphatidylinositol monophosphate
PI-4,5-P ₂	Phosphatidylinositol-4,5-bisphosphate
PX	Phox homology
SH2	Src homology 2
SH3	Src homology 3

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The eicosanoids: cyclooxygenase, lipoxygenase, and epoxigenase pathways

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1. Introduction

1.1. Terminology, structures, and nomenclature

The term 'eicosanoids' is used to denote a group of oxygenated, twenty carbon fatty acids (Fig. 1) [1]. The major precursor of these compounds is arachidonic acid (all *cis* 5,8,11,14-eicosatetraenoic acid), and the pathways leading to the eicosanoids are known collectively as the 'arachidonate cascade'. There are three major pathways within the cascade, including the cyclooxygenase, lipoxygenase, and epoxigenase pathways. In each case, these pathways are named after the enzyme(s) that catalyzes the first committed step. The prostanoids, which include the prostaglandins and thromboxanes, are formed via the cyclooxygenase pathway. The first part of our discussion will focus on the prostanoids. Later in this chapter, we will describe the lipoxygenase and epoxigenase pathways.

The structures and biosynthetic interrelationships of the most important prostanoids are shown in Fig. 2 [1]. PG is the abbreviation for prostaglandin, and TX is the abbreviation for thromboxane. Naturally occurring prostaglandins contain a cyclopentane ring, a *trans* double bond between C-13 and C-14, and an hydroxyl group at C-15. The letters following the abbreviation PG indicate the nature and location of the oxygen-containing substituents present in the cyclopentane ring. Letters are also used to label thromboxane derivatives (e.g., TXA and TXB). The numerical subscripts indicate the number of carbon-carbon double bonds in the side chains emanating from the cyclopentane ring (e.g., PGE₁ vs. PGE₂). In general, those prostanoids with the '2' subscript are derived from arachidonate; the '1' series prostanoids are formed from 8,11,14-eicosatetraenoate, and the '3' series compounds are derived from 5,8,11,14,17-eicosapentaenoate. Greek subscripts are used to denote the orientation of ring hydroxyl groups (e.g., PGF_α).

Prostanoids formed by the action of cyclooxygenases have their aliphatic side chains emanating from C-8 and C-12 of the cyclopentane ring in the orientations shown in Fig. 2. Prostanoids known as isoprostanes have their aliphatic groups in various other orientations [2]. Isoprostanes are formed from arachidonic acid by nonenzymatic autooxidation, and, somewhat surprisingly, isoprostanes and their metabolites are found in greater quantities in urine than metabolites of prostanoids formed enzymatically via

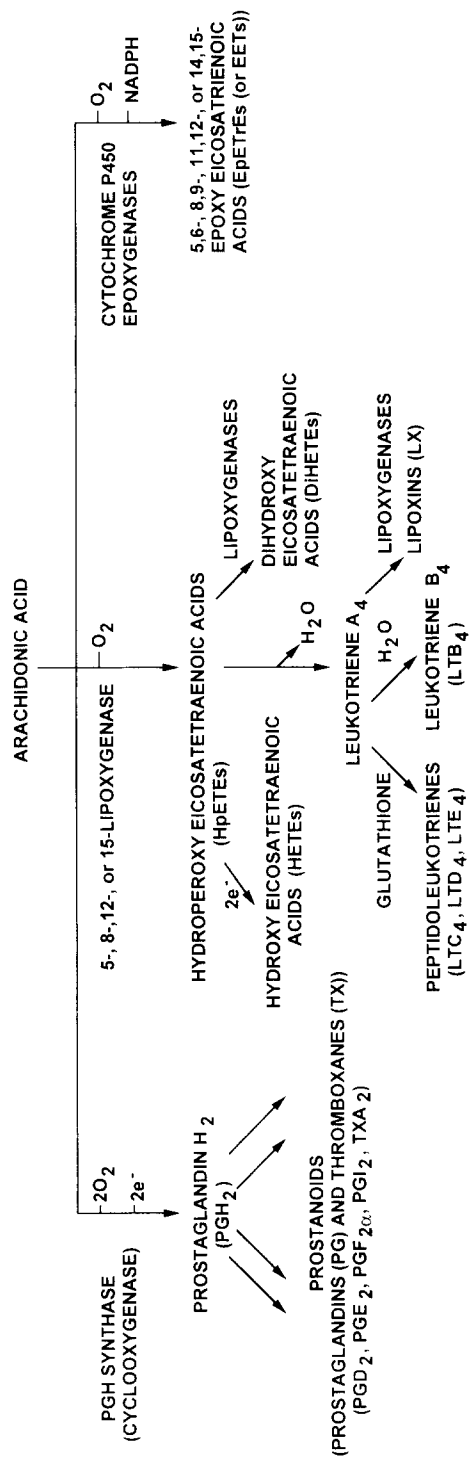


Fig. 1. Cyclooxygenase, lipoxygenase, and epoxygenase pathways leading to the formation of eicosanoids from arachidonic acid.

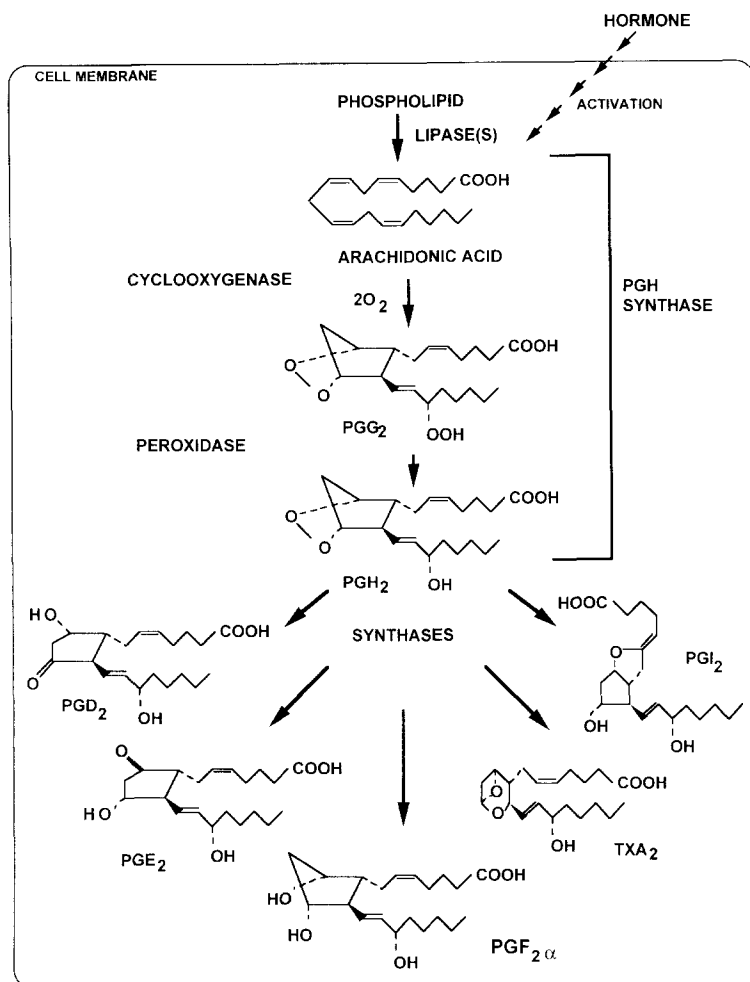


Fig. 2. Structures and biosynthetic relationships among the most common prostanoide.

the cyclooxygenase. Particularly in pathological conditions which support autooxidation (e.g., CCl_4 toxicity), isoprostanes are produced in abundance [2].

1.2. Prostanoid chemistry

Prostaglandins are soluble in lipid solvents below pH 3.0 and are typically extracted from acidified aqueous solutions with ether, chloroform/methanol, or ethyl acetate. PGE, PGF, and PGD derivatives are relatively stable in aqueous solution at pH 4–9; above pH 10, both PGE and PGD are subject to dehydration. PGI_2 , which is also known as prostacyclin, contains a vinyl ether group that is very sensitive to acid-catalyzed hydrolysis; PGI_2 is unstable below pH 8.0. The stable hydrolysis product

of PGI_2 is 6-keto- $\text{PGF}_{1\alpha}$. PGI_2 formation is usually monitored by measuring 6-keto- $\text{PGF}_{1\alpha}$ formation. TXA_2 , which contains an oxane–oxetane grouping in place of the cyclopentane ring, is hydrolyzed rapidly ($t_{1/2} = 30$ s at 37°C in neutral aqueous solution) to TXB_2 ; TXA_2 formation is assayed by quantifying TXB_2 . Prostaglandin derivatives are commonly quantified with immunoassays or by mass spectrometry using deuterium-labeled internal standards.

2. Prostanoid biosynthesis

Eicosanoids are not stored by cells, but rather are synthesized and released rapidly (5–60 s) in response to extracellular hormonal stimuli. The pathway for stimulus-induced prostanoid formation as it might occur in a model cell is illustrated in Fig. 2 [1]. Prostanoid formation occurs in three stages: (a) mobilization of free arachidonic acid (or 2-arachidonyl-glycerol (2-AG); see below) from membrane phospholipids; (b) conversion of arachidonate (or 2-AG) to the prostaglandin endoperoxide PGH_2 (or 2- PGH_2 -glycerol); and (c) cell-specific conversion of PGH_2 (or 2- PGH_2 -glycerol) to one of the major prostanoids.

2.1. Mobilization of arachidonate

Prostaglandin synthesis is initiated by the interaction of various hormones (e.g., bradykinin, angiotensin II, thrombin) with their cognate cell surface receptors (Figs. 2 and 3) which, in turn, causes the activation of one or more cellular lipases. Although in principle, there are a variety of lipases and phospholipases that could participate in this arachidonate mobilizing phase, the high molecular weight cytosolic phospholipase A_2 (PLA_2) and certain of the nonpancreatic, secretory PLA_2 s appear to be the relevant lipases (Chapter 11). The current consensus regarding the roles of PLA_2 s in prostanoid synthesis is that typically cytosolic PLA_2 is involved directly in mobilizing arachidonic acid for the constitutive, prostaglandin endoperoxide H synthase-1 (PGHS-1) whereas cytosolic PLA_2 is indirectly, and secretory PLA_2 is directly, involved in mobilizing arachidonate for the inducible PGHS-2 (Fig. 3); [3–5]. This is discussed in more detail below in describing the functions of the PGHS isoforms.

2.2. Cytosolic and secreted phospholipase A_2 s

Cytosolic PLA_2 is found in the cytosol of resting cells, but as illustrated in Fig. 3, hormone-induced mobilization of intracellular Ca^{2+} leads to the translocation of cytosolic PLA_2 to the ER and nuclear envelope. There, cytosolic PLA_2 cleaves arachidonate from the *sn*2 position of phospholipids on the cytosolic surface of the membranes. The arachidonate then traverses the membrane where it acts as a substrate for PGHSs which are located on the luminal surfaces of the ER and the associated inner and outer membranes of the nuclear envelope [1]. The activity of cytosolic PLA_2 is also augmented by phosphorylation by a variety of kinases [3]. The translocation of cytosolic PLA_2 involves the binding of Ca^{2+} to an N-terminal CalB domain and then the binding of the

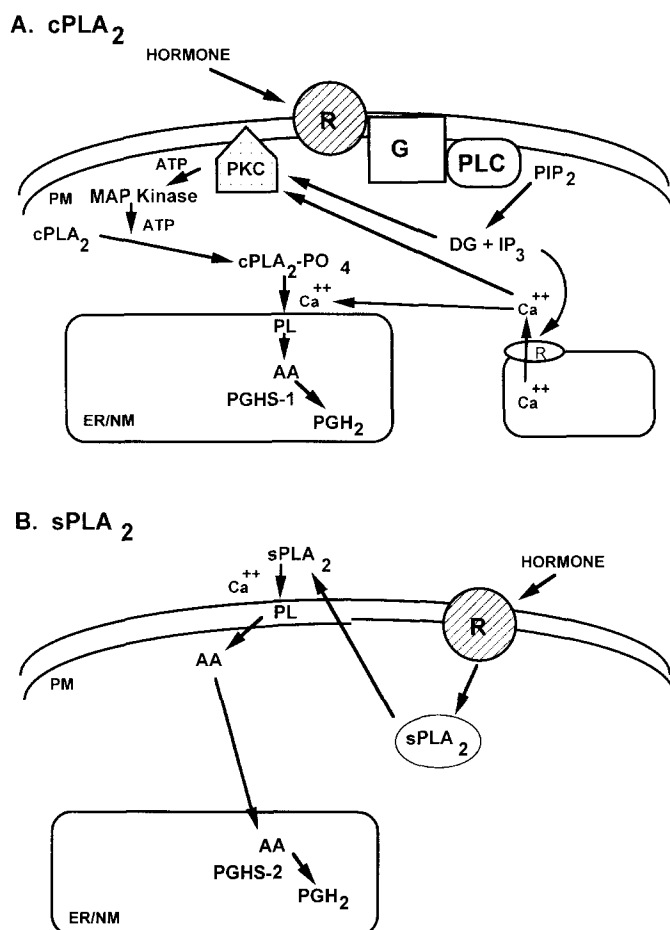


Fig. 3. Activation of cytosolic (c) PLA₂ (panel A) and Type II nonpancreatic secretory (s) PLA₂ (panel B) involved in mobilizing arachidonic acid from phospholipid. Abbreviations include: PL, phospholipid; ER/NM, endoplasmic reticulum/nuclear membrane; PM, plasma membrane; AA, arachidonic acid; PGHS, prostaglandin endoperoxide H synthase; R, receptor; G, G protein; PLC, phospholipase C; PIP₂, phosphatidylinositol-4,5-bis-phosphate; PKC, protein kinase C. The activation of secretory PLA₂ requires participation of cytosolic PLA₂ but the mechanism has not yet been determined.

Ca²⁺/CalB domain to intracellular membranes [6]; Ca²⁺ is involved in translocation of the enzyme but not in the catalytic mechanism of cytosolic PLA₂ (see Dessen et al., 1999).

Stimulus-dependent arachidonate mobilization by secretory PLA₂s depends on the ability of these enzymes to be released from cells and to rebind to the cell surface through heparin sulfate containing proteoglycans (Fig. 3). Once bound, these enzymes appear to act on phosphatidylcholine present on the extracellular face of the plasma membrane [7]. Secretory PLA₂ is shown in Fig. 3 in a vesicle that moves to the surface of the cell in response to an agonist, is secreted into the medium and then binds to

a proteoglycan on the cell surface. Presumably, free arachidonate released through the actions of secretory PLA₂ can enter the cell and make its way to the ER, where it is acted upon by PGHSs. There are many different nonpancreatic secretory PLA₂s but to date, only secretory PLA₂s IIA and V have been shown to be involved in releasing arachidonate for prostanoid synthesis [7].

The crystal structure of secretory PLA₂ IIA from human synovial fluid has been determined. Secretory PLA₂s require high concentrations of Ca²⁺ (~1 mM) such as those found extracellularly for maximal activity. Ca²⁺ is involved in phospholipid substrate binding and catalysis by secretory PLA₂s. Unlike cytosolic PLA₂, secretory PLA₂ shows no specificity toward either the phospholipid head group or the acyl group at the *sn*2 position. The levels of secretory PLA₂s are regulated transcriptionally in response to cell activation.

Cytosolic PLA₂ is directly involved in the immediate arachidonate release that occurs when a cell is challenged with a circulating hormone or protease. For example, thrombin acting through its cell surface receptor activates cytosolic PLA₂ in platelet cells to cause arachidonate release that results in TXA₂ formation. This entire process occurs in seconds. secretory PLA₂, on the other hand, plays a prominent role in 'late-phase' prostaglandin formation which occurs 2–3 h after cells have been exposed to a mediator of inflammation (e.g., endotoxin or interleukin-1) or a growth factor such as platelet-derived growth factor.

2.3. Mobilization of 2-arachidonyl-glycerol (2-AG)

In 2000, Kozak et al. made the insightful discovery that 2-AG is an efficient substrate for PGHS-2 but not PGHS-1 [8]. PGHS-2 converts 2-AG to 2-PGH₂-glycerol, and this intermediate is converted to the 2-prostanoyl-glycerol derivatives but not 2-thromboxane-glycerol. At the time of this writing, it is not clear under what conditions or to what degree alternative substrates such as 2-AG are used in vivo to form products like the 2-prostanoyl-glycerol derivatives. However, one can imagine that these products have a unique set of biological roles that distinguish them from the classical prostaglandins derived from arachidonic acid itself. Although the pathway leading to the formation of 2-AG itself has not been defined in the context of prostanoid metabolism, 2-AG could be formed from phosphatidylcholine through the sequential actions of phospholipase C and acylglycerol lipase [8].

2.4. Prostaglandin endoperoxide H₂ (PGH₂) formation

Once arachidonate is released, it can be acted upon by PGHS [9]. There are two PGHS isozymes, PGHS-1 and PGHS-2. PGHS-1 appears to use fatty acids such as arachidonate exclusively as substrates. In contrast, PGHS-2 utilizes both fatty acids and 2-AG about equally well [8]. The PGHSs exhibit two different but complementary enzymatic activities (Fig. 2): (a) a cyclooxygenase (*bis*-oxygenase) which catalyzes the formation of PGG₂ (or 2-PGG₂-glycerol) from arachidonate (or 2-AG) and two molecules of O₂; and (b) a peroxidase which facilitates the two-electron reduction of the 15-hydroperoxyl group of PGG₂ (or 2-PGG₂-glycerol) to PGH₂ (or 2-PGH₂-glycerol)

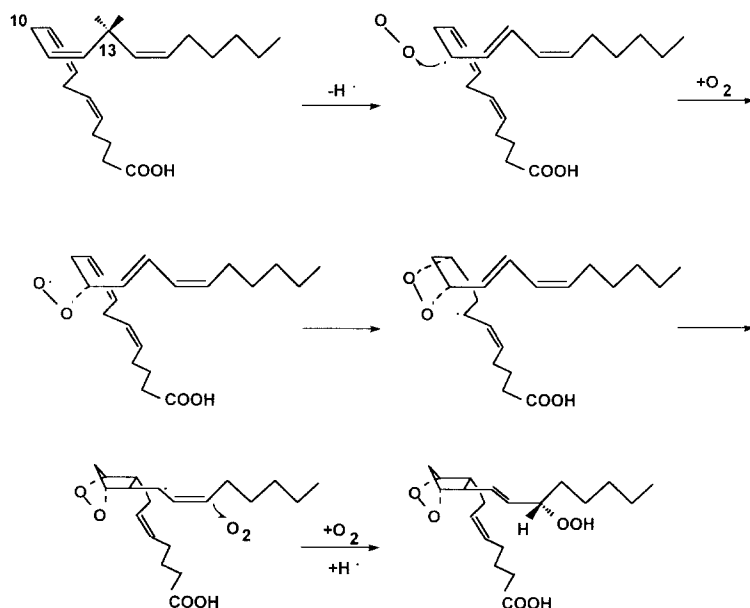


Fig. 4. Mechanism for the cyclooxygenase reaction showing the conversion of arachidonic acid and two molecules of oxygen to PGG_2 .

(Fig. 2). The oxygenase and peroxidase activities occur at distinct but interactive sites within the protein.

The initial step in the cyclooxygenase reaction is the stereospecific removal of the 13-pro-*S* hydrogen from arachidonate. As depicted in Fig. 4 [9], an arachidonate molecule becomes oriented in the cyclooxygenase active site with a kink in the carbon chain at C-9. Abstraction of the 13-pro-*S* hydrogen and subsequent isomerization leads to a carbon-centered radical at C-11 and attack of molecular oxygen at C-11 from the side opposite that of hydrogen abstraction. The resulting 11-hydroperoxyl radical adds to the double bond at C-9, leading to intramolecular rearrangement and formation of another carbon-centered radical at C-15. This radical then reacts with another molecule of oxygen. The 15-hydroperoxyl group of PGG_2 can undergo a two-electron reduction to an alcohol yielding PGH_2 in a reaction catalyzed by the peroxidase activity of PGHSs.

2.5. PGHS active site

Depicted in Fig. 5 is a model of the cyclooxygenase and peroxidase active sites of ovine PGHS-1 [1]. The cyclooxygenase is an unusual activity that exhibits a requirement for hydroperoxide and undergoes a suicide inactivation [1]. The reason for the hydroperoxide activating requirement is that in order for the cyclooxygenase to function, a hydroperoxide must oxidize the heme prosthetic group located at the peroxidase active site to an oxo-ferryl heme radical cation. This oxidized heme intermediate abstracts an electron from Tyr385. Finally, the resulting Tyr385 tyrosyl

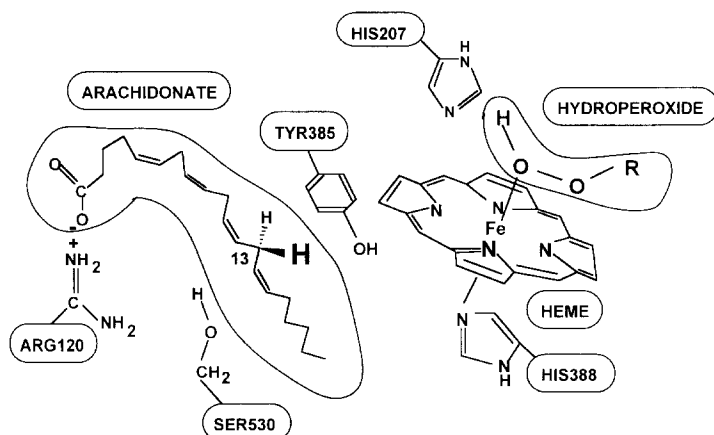


Fig. 5. Model of the cyclooxygenase and peroxidase active sites of the ovine PGHS-1.

radical abstracts the 13-pro-*S* hydrogen from arachidonate, initiating the cyclooxygenase reaction. Once the cyclooxygenase reaction begins, newly formed PGG_2 can serve as the source of the activating hydroperoxide; prior to PGG_2 formation, ambient cellular hydroperoxides apparently serve to initiate heme oxidation and cyclooxygenase catalysis. Ser530, the site of acetylation of PGHS-1 by aspirin (Section 2.7), is shown within the cyclooxygenase active site in Fig. 5. Also shown is Arg120. The guanidino group of this residue serves as the counterion for the carboxylate group of arachidonate.

2.6. Physico-chemical properties of PGHSs

PGHS-1 was first purified from ovine vesicular gland, and most biochemical studies have been performed using this protein. PGHS-1 is associated with the luminal surfaces of the ER and the inner and outer membranes of the nuclear envelope. Detergent-solubilized ovine PGHS-1 is a dimer with a subunit molecular mass of 72 kDa. The reason for the existence of the dimer is unknown, but separation of the enzyme into monomers eliminates enzyme activity. The protein is *N*-glycosylated and is a hemoprotein containing one protoporphyrin IX per monomer [1]. The sequences of cDNA clones for PGHS-1 from many mammals indicate that initially the protein has a signal peptide of 24–26 amino acids, that is cleaved to yield a mature protein of 574 amino acids.

PGHS-2 was discovered in 1991 as an immediate early gene product in phorbol ester-activated murine 3T3 cells and in *v-src*-transformed chicken fibroblasts [1,10]. PGHS-1 and PGHS-2 from the same species have amino acid sequences that are 60% identical. The major sequence differences are in the signal peptides and the membrane binding domains (residues 70–120 of PGHS-1); in addition, PGHS-2 contains a unique 18 amino acid insert near its carboxyl terminus. The role of this 18 amino acid cassette in PGHS-2 is unknown.

The crystal structures of PGHS-1 and PGHS-2 have been determined. As noted earlier, two subunits of the enzymes form homodimers. Each monomer contains three

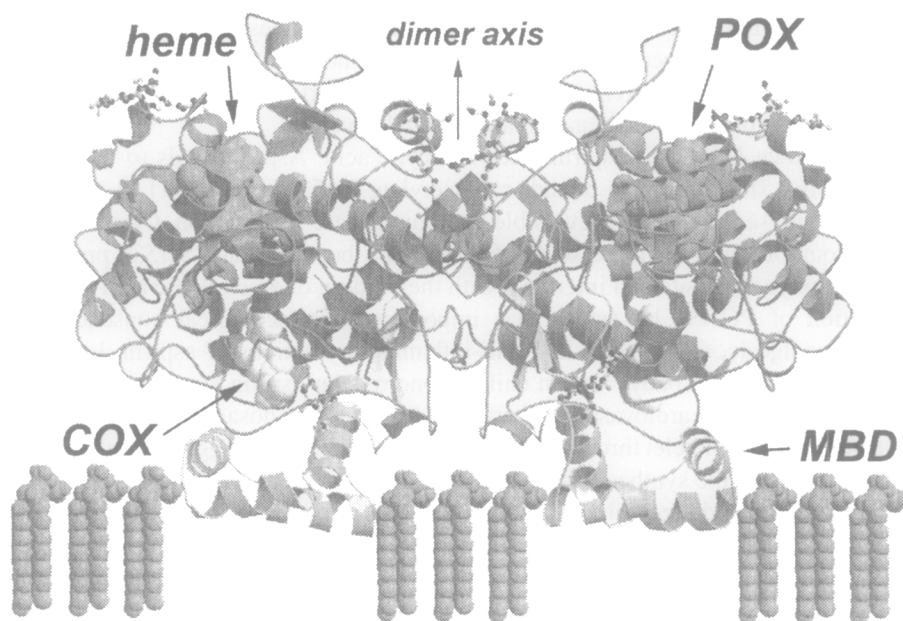


Fig. 6. Ribbon diagram of the structure of ovine PGHS-1 homodimer interdigitated via its membrane binding domain (MBD) into the luminal surface of the endoplasmic reticulum. The arrow denotes the dimer interface. Abbreviations include: POX, peroxidase; COX, cyclooxygenase.

sequential folding domains: an N-terminal epidermal growth factor-like domain of about 50 amino acids, an adjoining region containing about 70 amino acids that serves as the membrane binding domain, and a C-terminal globular catalytic domain (Fig. 6).

PGHS-1 and PGHS-2 are integral membrane proteins. However, their interactions with membranes do not involve typical transmembrane helices. Instead, analysis of the crystal structures and membrane domain labeling studies have established that PGHSs interact *monotopically* with only one surface of the membrane bilayer [1]. The interaction involves four short amphipathic α -helices present in the membrane binding domain noted above. The side chains of hydrophobic residues located on one surface of these helices interdigitate into and anchor PGHSs to the luminal surface of the ER and the inner and outer membranes of the nuclear envelope (Fig. 6).

2.7. PGHSs and nonsteroidal anti-inflammatory drugs

Prostaglandin synthesis can be inhibited by both nonsteroidal anti-inflammatory drugs (NSAIDs) and anti-inflammatory steroids. Both PGHS isozymes are pharmacological targets of common NSAIDs (e.g., aspirin, ibuprofen, naproxen). However, only prostaglandin synthesis mediated by PGHS-2 is inhibited by anti-inflammatory steroids, which block the synthesis of PGHS-2, at least in part, at the level of transcription [1]. Additionally, PGHS-2 is inhibited by 'COX-2 inhibitors' including rofecoxib and celecoxib. These drugs belong to a special class of NSAIDs specific for this isoform

and are often referred to as COX-2 drugs. The selectivity of these latter drugs depends on subtle structural differences between the cyclooxygenase active sites of PGHS-1 and PGHS-2 [1]. These same differences account, at least in part, for the ability of PGHS-2 but not PGHS-1 to bind and oxygenate 2-arachidonyl-glycerol.

The best known NSAID is aspirin, acetylsalicylic acid. Aspirin binds to the cyclooxygenase active site, and, once bound, can acetylate Ser530 (Fig. 5) [1]. Acetylation of this active site serine causes irreversible cyclooxygenase inactivation. Curiously, the hydroxyl group of Ser530 is not essential for catalysis, but acetylated Ser530 protrudes into the cyclooxygenase site and interferes with the binding of arachidonic acid [1].

Acetylation of PGHSs by aspirin has important pharmacological consequences. Besides the analgesic, anti-pyretic, and anti-inflammatory actions of aspirin, low-dose aspirin treatment — one ‘baby’ aspirin daily or one regular aspirin every three days — is a useful anti-platelet cardiovascular therapy [11]. This low-dosage regimen leads to selective inhibition of platelet thromboxane formation (and platelet aggregation) without appreciably affecting the synthesis of other prostanoids in other cells. Circulating blood platelets lack nuclei and are unable to synthesize new protein. Exposure of the PGHS-1 of platelets to circulating aspirin causes irreversible inactivation of the platelet enzyme. Of course, PGHS-1 (and PGHS-2) inactivation also occurs in other cell types, but cell types other than platelets can resynthesize PGHSs relatively quickly. For new PGHS-1 activity to appear in platelets, new platelets must be formed. Because the replacement time for platelets is five to ten days, it takes time for the circulating platelet pool to regain its original complement of active PGHS-1.

There are many NSAIDs other than aspirin [1,12]. In fact, this is one of the largest niches in the pharmaceutical market, currently accounting for about five billion dollars in annual sales. Like aspirin, other NSAIDs act by inhibiting the cyclooxygenase activity of PGHS [1,11,12]. However, unlike aspirin, most of these drugs cause reversible enzyme inhibition simply by competing with arachidonate for binding. A well-known example of a reversible nonsteroidal anti-inflammatory drug is ibuprofen. All currently available NSAIDs inhibit both PGHS-1 and PGHS-2 [1,12]. However, inhibition of PGHS-2 appears to be primarily responsible for both the anti-inflammatory and analgesic actions of NSAIDs [11]. Dual inhibition of PGHS-1 and PGHS-2 with common NSAIDs causes unwanted ulcerogenic side-effects [1,5,12]. Indeed, the newly developed COX-2 drugs rofecoxib and celecoxib exhibit the anti-inflammatory and analgesic actions of classical NSAIDs but have lesser side-effects, particularly gastro-intestinal side-effects [11].

Recent attention has been focussed on COX-2 inhibitors as prophylactic agents in the prevention of colon cancer. About 85% of tumors of the colon express elevated levels of PGHS-2 and classical NSAIDs and COX-2 inhibitors reduce mortality due to colon cancer (Takuku et al., 1998; Dubois, 2001).

2.8. Regulation of PGHS-1 and PGHS-2 gene expression

PGHS-1 and PGHS-2 are encoded by separate genes [1]. Apart from the first two exons, the intron/exon arrangements are similar. However, the PGHS-2 gene (~8 kb) is considerably smaller than the PGHS-1 gene (~22 kb). The PGHS-1 gene is on human chromosome 9, while the PGHS-2 gene is located on human chromosome 1.

The expressions of the PGHS-1 and PGHS-2 genes are regulated in quite different ways. PGHS-1 is expressed more or less constitutively in almost all tissues, whereas PGHS-2 is absent from cells unless induced in response to cytokines, tumor promoters, or growth factors [1]. Cells use PGHS-1 to produce prostaglandins needed to regulate 'housekeeping activities' typically involving rapid responses to circulating hormones (Fig. 2). PGHS-2 produces prostanoids which function during specific stages of cell differentiation or replication; it is not yet clear what these latter functions are nor whether they involve products formed from arachidonate or 2-AG or both. There is some indirect evidence suggesting that at least some of the products formed via PGHS-2 operate at the level of the nucleus through peroxisomal proliferator-activated receptors to modulate transcription of specific genes (Chapters 10 and 16) [1,5].

Relatively little is known concerning the regulation of expression of PGHS-1, although the enzyme must be under developmental control. The regulation of expression of PGHS-2 continues to be an area of intensive investigation. Much of what is known about PGHS-2 comes from studies with cultured fibroblasts, endothelial cells and macrophages [1]. Typically, PGHS-2 is induced rapidly (1–3 h) and dramatically (20- to 80-fold). Platelet-derived growth factor, phorbol ester, and interleukin-1 β induce PGHS-2 expression in fibroblasts and endothelial cells. Bacterial lipopolysaccharide, interleukin-1 β , and tumor necrosis factor α stimulate PGHS-2 in monocytes and macrophages. While only a limited number of tissues and cell types have been examined, it is likely that PGHS-2 can be induced in almost any cell or tissue with the appropriate stimuli. Importantly, as noted earlier, PGHS-2 expression, but not PGHS-1 expression, can be completely inhibited by anti-inflammatory glucocorticoids such as dexamethasone [1,12].

The promoters of the two PGHS genes are indicative of their mode of regulation. PGHS-1 has a TATA-less promoter, a feature common to housekeeping genes. Reporter plasmids constructed with the 5'-upstream region of the PGHS-1 gene have failed to show any significant inducible transcription from this promoter, supporting the concept that regulation of PGHS-1 occurs only developmentally. The PGHS-2 promoter, on the other hand, contains a TATA box, and experiments with reporter plasmids containing the PGHS-2 promoter and upstream 5'-flanking sequence have demonstrated that PGHS-2 is highly regulatable. Transcriptional activation of the PGHS-2 gene appears to be one important mechanism for increasing PGHS-2 expression. Transcription of PGHS-2 can be controlled by multiple signaling pathways including the cAMP pathway, the protein kinase C pathway (phorbol esters), viral transformation (*src*), and other pleiotropic pathways such as those activated by growth factors, endotoxin, and inflammatory cytokines. These latter agents (e.g., platelet-derived growth factor, lipopolysaccharide, interleukin-1 β , tumor necrosis factor α) likely share convergent pathways involving nuclear factor κ B (NF κ B) and the CAAT enhancer binding protein (C/EBP), two transcription factors common to inflammatory responses, and one or more of the established mitogen activated protein kinase cascades: ERK1/2, JNK/SAPK, and p38/RK/Mpk2.

The nucleotide sequence of the PGHS-2 gene promoter is known but identification of *cis*-elements responsible for the regulation of this gene is incomplete. Regulatory elements in the 5'-flanking regions of the PGHS-2 gene that are known to regulate

transcription are the overlapping E-box and CRE-1 sequence most proximal to the TATA box, a C/EBP-2 sequence, a NF κ B binding site and a downstream CRE-2. In fibroblasts and endothelial cells, the most critical of the regulatory sequences is the CRE-1. This type of response element typically is activated by hetero- and homo-dimers of the c-Fos, c-Jun and ATF families of bZIP proteins (AP-1) and the cAMP regulatory binding protein (CREB). Activation of c-Jun is required for transcriptional activation of the PGHS-2 promoter in response to growth factors. It is not known if c-Jun forms homodimers, or interacts with other bZIP proteins.

In macrophages bacterial endotoxin stimulates PGHS-2 expression through cooperative activation via the CRE-1, C/EBP-2, NF κ B and CRE-2 regulatory elements (Wingerd, 2002). Tumor necrosis factor α -stimulated expression is dependent on a NF κ B site and the C/EBP-1 site. The C/EBP β and C/EBP δ transcription factors are commonly involved in the regulation of inflammatory responses, and C/EBP regulatory elements are frequently found in promoters of so called acute phase genes. This family of transcription factors is activated by most of the inflammatory stimuli which induce PGHS-2 expression. C/EBPs bind to PGHS-2 promoters and function in conjunction with USF-1, NF κ B, and c-Jun *cis*-regulatory proteins to activate transcription.

2.9. PGH₂ metabolism

Although all the major prostanoids are depicted in Fig. 2 as being formed by a single cell, prostanoid synthesis appears to be cell-specific [1]. For example, platelets form mainly TXA₂, endothelial cells form PGI₂ as their major prostanoid, and PGE₂ is the major prostanoid produced by renal collecting tubule cells. The syntheses of PGE₂, PGD₂, PGF_{2 α} , PGI₂, and TXA₂ from PGH₂ are catalyzed by PGE synthase, PGD synthase, PGF α synthase, PGI synthase, and TXA synthase, respectively [1]. Formation of PGF_{2 α} involves a two-electron reduction of PGH₂, and a PGF α synthase utilizing NADPH can catalyze this reaction. All other prostanoids are formed via isomerization reactions involving no net change in oxidation state from PGH₂.

PGI synthase and TXA synthase are hemoproteins with molecular weights of 50–55,000. Both of these proteins are cytochrome P-450s. Both enzymes, like PGHSs, undergo suicide inactivation during catalysis. TXA synthase is found in abundance in platelets and lung. PGI synthase is localized to endothelial cells, as well as both vascular and nonvascular smooth muscle [13]. Both TXA and PGI synthases are found on the cytoplasmic face of the ER. PGH₂ formed in the lumen of the ER via PGHSs diffuses across the membrane and is converted to a prostanoid end product on the cytoplasmic side of the membrane.

At least four different proteins have been shown to have PGE synthase activity including a cytosolic PGE synthase, an inducible, membrane-associated PGE synthase and two cytosolic glutathione-S-transferase isozymes [13]. Cytosolic PGE synthase appears to be more tightly coupled to PGHS-1. Microsomal PGE synthase has been reported to be more tightly coupled to PGHS-2 and interestingly, its expression like that of PGHS-2 is inhibited by anti-inflammatory steroids such as dexamethasone. All PGE synthases require reduced glutathione as a cofactor. Glutathione facilitates cleavage of the endoperoxide group and formation of the 9-keto group [14]. PGF α synthase

activity has been partially purified from lung. Structurally, the enzyme is a member of the aldose reductase family of proteins. Glutathione-dependent and -independent PGD synthases have been isolated. The glutathione-dependent forms also exhibit glutathione-S-transferase activity. A glutathione-independent form of PGD synthase has been purified from brain [13].

3. Prostanoid catabolism and mechanisms of action

3.1. Prostanoid catabolism

Once a prostanoid is formed on the cytoplasmic surface of the ER, it diffuses to the cell membrane and exits the cell probably via carrier-mediated transport [14]. Prostanoids are local hormones that act very near their sites of synthesis. Unlike typical circulating hormones that are released from one major endocrine site, prostanoids are synthesized and released by virtually all organs. In addition, all prostanoids are inactivated rapidly in the circulation. The initial step of inactivation of PGE₂ is oxidation to a 15-keto compound in a reaction catalyzed by a family of 15-hydroxyprostaglandin dehydrogenases. Further catabolism involves reduction of the double bond between C-13 and C-14, ω -oxidation, and β -oxidation.

3.2. Physiological actions of prostanoids

Prostanoids act both in an autocrine fashion on the parent cell and in a paracrine fashion on neighboring cells [15,16]. Typically, the role of a prostanoid is to coordinate the responses of the parent cells and neighboring cells to the biosynthetic stimulus, a circulating hormone. The actions of prostanoids are mediated by G-protein-linked prostanoid receptors of the seven *trans* membrane domain receptor superfamily [16].

Those examples which have been studied in the most detail are the renal collecting tubule-thick limb interactions involving PGE₂ synthesized by the collecting tubule and the platelet-vessel wall interactions involving PGI₂ and TXA₂ [14]. For example, in the case of platelets TXA₂ is synthesized by platelets when they bind to subendothelial collagen that is exposed by microinjury to the vascular endothelium. Newly synthesized TXA₂ promotes subsequent adherence and aggregation of circulating platelets to the subendothelium. In addition, TXA₂ produced by platelets causes constriction of vascular smooth muscle. The net effect is to coordinate the actions of platelets and the vasculature in response to deendothelialization of arterial vessels. Thus, prostanoids can be viewed as local hormones which coordinate the effects of circulating hormones and other agents (e.g., collagen) that activate their synthesis.

3.3. Prostanoid receptors

The identification and characterization of prostanoid receptors has occurred during the past ten years [16]. These results coupled with studies of PGHS and prostanoid receptor knockout mice [5,15] have been critical in beginning to rationalize earlier results of

studies on the physiological and pharmacological actions of prostanoids that had been somewhat confusing and difficult to interpret because prostaglandins were found to cause such a wide variety of seemingly paradoxical effects.

The seminal step in understanding the structures of prostanoid receptors and their coupling to second messenger systems resulted from the cloning of receptors for each of the prostanoids by Narumiya and others [16]. Prostanoid receptor cloning began with the TXA/PGH receptor known as the TP receptor. cDNA encoding this receptor was cloned using oligonucleotide probes designed from protein sequence data obtained from the TP receptor purified from platelets. The results confirmed biochemical predictions that the TP receptor was a seven-membrane spanning domain receptor of the rhodopsin family. Subsequent cloning of other receptors was performed by homology screening using receptor cDNA fragments as cross-hybridization probes. All of these prostanoid receptors are of the G-protein-linked receptor family.

It is now clear that there are pharmacologically distinct receptors for each of the known prostanoids. In the case of PGE₂, four different prostaglandin E (EP) receptors have been identified and designated as EP1, EP2, EP3, and EP4 receptors. Based on studies with selective agonists for each of the EP receptors and their effects on second messenger production, it appears that EP1 is coupled through G_q to the activation of phospholipase C, EP2 and EP4 are coupled via G_s to the stimulation of adenylate cyclase and EP3 receptors are coupled via G_i to the inhibition of adenylate cyclase [16].

In order to determine the physiological roles of various prostanoid receptors a number of knockout mice have been developed [15,16]. These studies indicate that prostacyclin receptors are involved in at least some types of pain responses, EP3 receptors are involved in the development of fever, PGE receptors are important in allergy, EP2 and EP4 function in bone resorption and EP1 receptors are involved in chemically induced colon cancer. The availability of cloned prostanoid receptors provides a rationale and the appropriate technology to search for receptor agonists and antagonists that might provide some specificity beyond the currently available cyclooxygenase inhibitors which prevent broadly the synthesis of all prostaglandins.

The realization that the gene for PGHS-2 is an immediate early gene associated with cell replication and differentiation suggests that prostanoids synthesized via PGHS-2 may have nuclear effects. As noted above there have been several reports indicating that prostanoid derivatives can activate some isoforms of peroxisomal proliferator activated receptors (PPARs). There is evidence that PGI₂ can be involved in PPAR δ -mediated responses such as decidualization and apoptosis [5] (S.K. Dey, 2000).

4. Leukotrienes and lipoxygenase products

4.1. Introduction and overview

Leukotrienes are produced by the action of 5-lipoxygenase (5-LO) which carries out the insertion of a diatomic oxygen at carbon atom-5 of arachidonic acid yielding 5(S)-hydroperoxy eicosatetraenoic acid (5-HpETE). A subsequent dehydration reaction catalyzed by the same enzyme, 5-LO, results in formation of leukotriene A₄ (LTA₄),

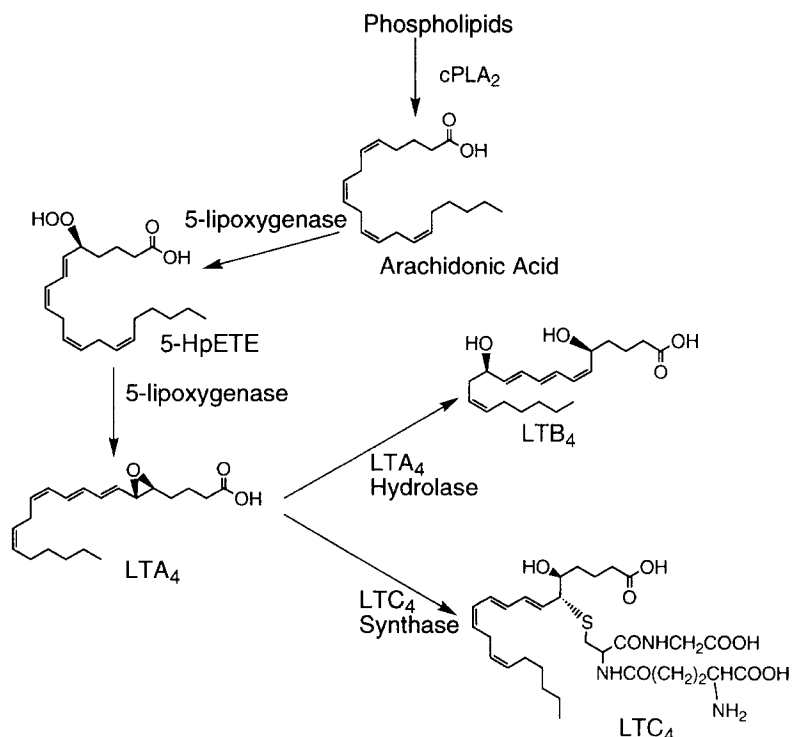


Fig. 7. Biochemical pathway of the metabolism of arachidonic acid into the biologically active leukotrienes. Arachidonic acid released from phospholipase by cytosolic (c) PLA₂ is metabolized by 5-lipoxygenase to 5-hydroperoxyeicosatetraenoic acid (5-HpETE) and leukotriene A₄ (LTA₄) which is then enzymatically converted into leukotriene B₄ (LTB₄) or conjugated by glutathione to yield leukotriene C₄ (LTC₄).

the chemically reactive precursor of biologically active leukotrienes. As was the case with prostanoid biosynthesis, leukotriene biosynthesis depends upon the availability of arachidonic acid as a free carboxylic acid as the 5-LO substrate, which typically requires the action of cytosolic phospholipase A₂ to release arachidonic acid from membrane phospholipids. Also, leukotrienes are not stored in cells, but synthesis and release from cells are rapid events following cellular activation. Interest in the leukotriene family of arachidonate metabolites arises from the potent biological activities of two products derived from LTA₄, that being leukotriene B₄ (LTB₄) and leukotriene C₄ (LTC₄) (Fig. 7). LTB₄ is a very potent chemotactic and chemokinetic agent for the human polymorphonuclear leukocyte, while LTC₄ powerfully constricts specific smooth muscle such as bronchial smooth muscle, and mediates leakage of vascular fluid in the process of edema [17]. The name leukotriene was conceived to capture two unique attributes of these molecules. The first attribute relates to those white blood cells derived from the bone marrow that have the capacity to synthesize this class of eicosanoid, for example the polymorphonuclear *leukocyte*. The last part of the name refers to the unique chemical structure, a conjugated *triene*, retained within these eicosanoids.

There are numerous other biochemical products of arachidonate metabolism formed

by lipoxygenase enzymes other than 5-LO. Other monooxygenases expressed in mammalian cells include 12-lipoxygenase, 15-lipoxygenase and much less frequently 8-lipoxygenase. These enzymes are named in accordance with the carbon atom position of arachidonate initially oxygenated even though other polyunsaturated fatty acids can be substrates. In addition, arachidonic acid can be oxidized by specific isozymes of cytochrome P-450, leading to a family of epoxyeicosatrienoic acids (EETs). Methyl terminus oxidized arachidonate as well as lipoxygenase-like monohydroxy eicosatetraenoic acid (HETE) products are also formed. In general, much more is known about the biochemical role of prostaglandins and leukotrienes as mediators of biochemical events; little is known concerning the exact role played by the other lipoxygenase products or cytochrome P-450 products. In accord with the body of information available, various pharmacological tools are available to inhibit 5-LO as well as specific leukotriene receptors.

4.2. Leukotriene biosynthesis

The arachidonate 5-lipoxygenase (5-LO, EC 1.13.11.34) is a metalloenzyme with bound iron coordinated by four histidine residues. This 77,852 Da protein (human 5-LO) catalyzes the addition of molecular oxygen to the 1,4-*cis*-pentadienyl structural moiety closest to the carboxyl group of arachidonic acid to yield a conjugated diene hydroperoxide, typical for all lipoxygenase reactions. Some details of the mechanism of 5-LO are known in that 5-LO removes the pro-*S* hydrogen atom from carbon-7 of arachidonic acid, leading to reduction of Fe(III) to Fe(II) likely in a radical type mechanism [18]. Molecular oxygen then adds to carbon-5 to yield the hydroperoxy radical. The hydroperoxy radical abstracts a hydrogen atom to yield 5(*S*)-HpETE (5(*S*)-hydroperoxy-6,8,11,14-(*E,Z,Z,Z*)-eicosatetraenoic acid). A second enzymatic activity of 5-LO, sometimes termed LTA₄ synthase activity, catalyzes the stereospecific removal of the pro-*R* hydrogen atom at carbon-10 of the 5(*S*)-HpETE through a second redox cycle followed by a sigmatropic shift of the electrons to form the conjugated triene epoxide and loss of hydroxide (Fig. 8). One unique feature of the iron redox cycle is that these reactions involve one electron rather than two electron transfers, typical of most peroxidase enzymes, in what has been called a pseudoperoxidase reaction [19]. The product of this reaction, 5(*S*),6(*S*)-oxido-7,9,11,14-(*E,E,Z,Z*)-eicosatetraenoic acid,

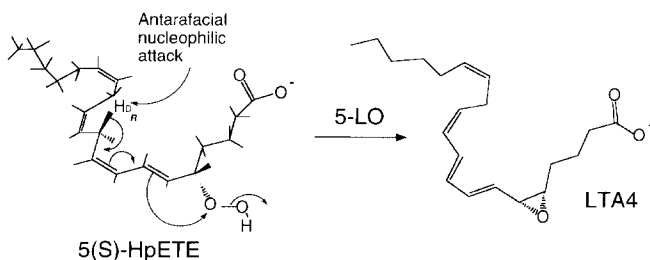


Fig. 8. Detailed mechanism of the 5-lipoxygenase reaction where the pro-*R* hydrogen from carbon-10 in 5-HpETE is removed by 5-lipoxygenase followed by sigmatropic rearrangement of electrons to form LTA₄.

is LTA₄, a conjugated triene epoxide that is highly unstable. LTA₄ undergoes rapid hydrolysis in water with a half-life of less than 10 s at pH 7.4 and also can react with proteins as well as DNA. Nonetheless within cells, LTA₄ is stabilized by binding to proteins that remove water from the immediate environment of the epoxide structure. The nonenzymatic hydrolysis products of LTA₄ include several biologically inactive and enantiomeric 5,12- and 5,6-diHETEs. However, the hydrolysis of LTA₄ catalyzed by LTA₄ hydrolase [20] produces the biologically active LTB₄, 5(*S*),12(*R*)-dihydroxy-6,8,10,14-(*Z,E,E,Z*)-eicosatetraenoic acid. A second pathway for LTA₄ metabolism is prominent in cells expressing the enzyme LTC₄ synthase [21] which catalyzes the addition of glutathione to carbon-6 of the triene epoxide yielding 5(*S*),6(*R*)-*S*-glutathionyl-7,9,11,14-(*E,E,Z,Z*)-eicosatetraenoic acid (LTC₄). LTC₄ synthase has been found to be localized on the nuclear membrane and is a unique glutathione (*S*) transferase. The formation of either LTC₄ or LTB₄ is controlled by the expression of either LTA₄ hydrolase or LTC₄ synthase by specific cell types. The human neutrophil, for example, expresses LTA₄ hydrolase and produces LTB₄ while the mast cell and eosinophil produce LTC₄, since they express LTC₄ synthase. Interestingly, cells have been found which do not have 5-LO, but do express either LTA₄ hydrolase (e.g., erythrocytes and lymphocytes) or LTC₄ synthase (e.g., platelets and endothelial cells). Studies have found that cells in fact cooperate in the production of biologically active leukotrienes through a process termed transcellular biosynthesis (J. Macclouf, 1989) where a cell such as the neutrophil or mast cell generates LTA₄ which is then released from the cell and is then taken up by either a platelet to make LTC₄ or red blood cell to make LTB₄. In spite of the chemical reactivity of LTA₄, this process is known to be highly efficient and approximately 60 to 70% of the LTA₄ produced by the activated neutrophil can be released to another cell for transcellular biosynthesis of leukotrienes (A. Sala, 1996).

4.3. Enzymes involved in leukotriene biosynthesis

4.3.1. 5-Lipoxygenase

The human 5-LO gene is unusually large and is present on chromosome 10. This gene covers more than 80 kb of DNA and has 14 exons that encode a 673 amino acid protein (without the initiator methionine residue) [22]. 5-Lipoxygenase has been purified from human, pig, rat, and guinea pig leukocytes, all with close to 90% homology. It is interesting to note that the purified or recombinant enzyme requires several cofactors for activity that include Ca²⁺, ATP, fatty acid hydroperoxides, and phosphatidylcholine in addition to the arachidonic acid and molecular oxygen substrates [17,19]. Purified 5-LO was found to catalyze the initial oxidation of arachidonic acid to yield 5-HpETE as well as the second enzymatic reaction to convert 5-HpETE into LTA₄. Recombinant human 5-LO has been expressed in osteosarcoma cells, cos-M6 cells, baculovirus-infected SF9 insect cells, yeast, and *Escherichia coli* [19].

Low concentrations of calcium ion (1–2 μM) are required for maximal activity of purified 5-LO, but the major role of calcium appears to be that of increasing lipophilicity of 5-LO in order to promote membrane association. ATP has a stimulatory effect on 5-LO at 20 nM and lipid hydroperoxides are important to initiate the 5-LO catalytic

cycle since they readily form Fe(III) within 5-LO by the pseudoperoxidase mechanism. Microsomal membranes as well as phosphatidylcholine vesicles can stimulate purified 5-LO activity since 5-LO performs the oxidation of arachidonic acid at the interface between the membrane and cytosol in a manner similar to that of cytosolic PLA₂. Calcium ions increase the association of 5-LO with phosphatidylcholine vesicles that likely recapitulates events within the cell where 5-LO becomes associated with the nuclear membrane [23] and to which arachidonic acid is presented by a second gene product which has been termed 5-LO activating protein (FLAP) [19]. In cells such as the neutrophil and mast cell where 5-LO is found in the cytosol, 5-LO is only catalytically active when bound to a membrane, typically the nuclear membrane. In fact, in some cells 5-LO is found to be constitutively associated with the nuclear membrane, likely a result of a process of cellular activation while 5-LO is found in the alveolar macrophage within the nucleus itself [23].

4.3.2. 5-Lipoxygenase activating protein (FLAP)

FLAP was found during the course of development of the drug MK-886 by workers at Merck-Frosst in Montreal [19]. They found this drug bound to a novel protein that was essential for the production of leukotrienes in stimulated, intact cells and hence, the name 'five lipoxygenase activating protein'. FLAP is a unique 161 amino acid containing protein (18,157 Da). While the role of FLAP is not entirely clear, experiments using a ¹²⁵I-labeled photoaffinity analog of arachidonic acid suggested that FLAP functions as a substrate transfer protein and in this manner stimulates 5-LO catalyzed formation of leukotrienes. Recently, it has been found that LTC₄ synthase has 31% amino acid identity to FLAP with a highly conserved region possibly involved in arachidonate binding for both proteins [19].

4.3.3. LTA₄ hydrolase

LTA₄ hydrolase catalyzes the stereochemical addition of water to form the neutrophil chemotactic factor LTB₄. LTA₄ hydrolase contains 610 amino acids (excluding the first methionine) with a molecular weight of 69,399 Da. LTA₄ hydrolase contains one zinc atom per enzyme molecule and this metal ion is essential for the catalytic activity [20]. LTA₄ hydrolase is also a member of a family of zinc metalloproteases and exhibits some protease activity. The finding of this activity led to the discovery of several drugs such as bestatin and captopril that are inhibitors of this enzyme. LTA₄ hydrolase is found in many cells including those which do not contain 5-LO and it is felt that these cells play an important role in transcellular biosynthesis of LTB₄ through cell-cell cooperation. LTA₄ is thought to be localized in the cytosol of the cell and is the only protein in the leukotriene biosynthetic cascade that is not found on the nuclear membrane following cellular activation. Therefore, in order to efficiently metabolize the chemically reactive LTA₄, either the LTA₄ hydrolase must come in close contact with the nuclear membrane during LTA₄ biosynthesis or a carrier protein must present LTA₄ to LTA₄ hydrolase in the cytosol. LTA₄ hydrolase is known to be efficiently suicide inactivated by LTA₄ when the electrophilic epoxide becomes covalently bound to the enzyme presumably within the active site (F.A. Fitzpatrick, 1990). A specific tyrosine residue has been found to be modified by LTA₄ and this residue has been implicated as the potential proton donor

in the suggested mechanism of LTA₄ hydration. The 3-dimensional structure of LTA₄ hydrolase has now been determined by X-ray crystallography [20].

4.3.4. LTC₄ synthase

The conjugation of the tripeptide glutathione (γ -glutamyl-cysteinyl glycine) to the triene epoxide LTA₄ is carried out by LTC₄ synthase (EC 2.5.1.37). This enzyme is found localized on the nuclear envelope of cells and has little homology to soluble glutathione (*S*) transferases. LTC₄ synthase does have some primary amino acid sequence homology to the recently described microsomal glutathione (*S*) transferases and FLAP [21]. LTC₄ synthase has a restricted distribution and is found predominantly in mast cells, macrophages, eosinophils, and monocytes. However, human platelets and endothelial cells have been found to express LTC₄ synthase. Purified recombinant LTC₄ synthase will conjugate glutathione to both LTA₄ and LTA₄ methyl ester with a K_m of approximately 2–4 μ M for the free acid and 7–10 μ M for the methyl ester. The K_m for glutathione is approximately 2 mM. The drug MK-886 was found to inhibit LTC₄ synthase with an IC₅₀ of approximately 2–3 μ M [19]. The gene for LTC₄ synthase is located on human chromosome 5, distal to that of cytokine, growth factor, and receptor genes relevant to the TH2 phenotype (T.D. Bigby, 2000).

4.4. Regulation of leukotriene biosynthesis

The biosynthesis of leukotrienes within cells is highly regulated and depends not only on the availability of arachidonic acid and molecular oxygen, but also on the subcellular location of 5-LO. Resting neutrophils synthesize little if any leukotrienes; however, following the elevation of intracellular calcium either through a physiological event such as phagocytosis or by pharmacological manipulation with the calcium ionophore A23187, the neutrophil produces a substantial amount of 5-LO products including leukotrienes from either endogenous or exogenous arachidonic acid. The need for an increase in intracellular calcium ion concentrations is a distinguishing feature of 5-LO that differentiates this enzyme from other lipoxygenases and cyclooxygenase.

A major determinant of 5-LO activity is the translocation of 5-LO to the nuclear membrane (Fig. 9). It has even been possible to demonstrate that the site of localization of 5-LO is at the inner nuclear membrane. The Ca²⁺-dependent translocation event is thought to bring 5-LO to the same region where FLAP and cytosolic PLA₂ translocate. Regulation of leukotriene biosynthesis is thus a process of assembly of the leukotriene biosynthetic machine at a nuclear envelope site. It is this site where arachidonic acid is released from nuclear membrane phospholipids, then converted to LTA₄ and ultimately conjugated with glutathione by nuclear membrane LTC₄ synthase. The mechanism by which 5-LO is trafficked to the nuclear membrane is as yet undefined, but the discovery of this site of leukotriene biosynthesis was unexpected and suggests novel intracellular actions of leukotrienes or of 5-LO itself within the immediate nuclear environment [23].

The activity of 5-LO is also regulated by a suicide inactivation mechanism where LTA₄ rapidly inactivates 5-LO, likely through a covalent modification mechanism [18]. Continued synthesis of leukotrienes then requires synthesis of new 5-LO. The production of 5-LO is known to be regulated at the level of gene transcription as well

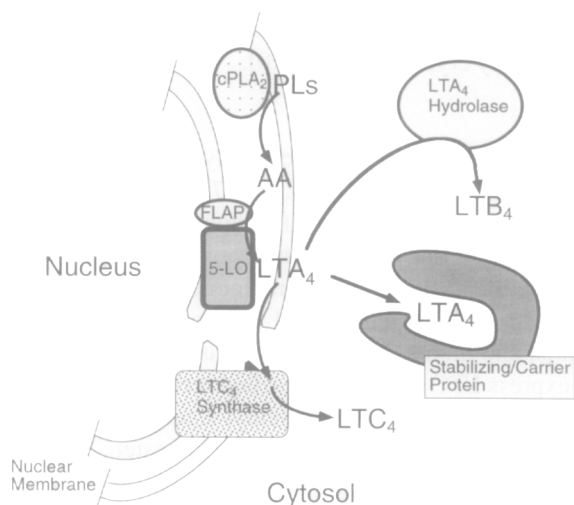


Fig. 9. Proposed model for the location of biosynthetic events occurring during leukotriene biosynthesis at the nuclear membrane of cells. Arachidonic acid is released from membrane glycerophospholipids (PLs) by translocated and nuclear membrane associated cytosolic (c) PLA₂ and is then presented to 5-lipoxygenase (5-LO) by way of FLAP. LTA₄ is either converted by nuclear membrane associated LTC₄ synthase into LTC₄ or carried to LTA₄ hydrolase possibly by a fatty acid binding protein which can stabilize the chemically reactive LTA₄ that is available for transcellular biosynthesis.

as mRNA translation in addition to the translocation mechanism [22,23]. The 5-LO gene promoter contains several consensus-binding sites for known transcription factors including Sp1 and EGR-1. In a region located 212 to 88 base pairs upstream from the translation start site of the human 5-LO gene, a highly rich G + C region is found that contains the consensus sequence or SP1 and EGR-1 transcription factors. Furthermore, fairly common variations have been found in human populations in which deletions of one or two of the Sp1 binding motifs were observed [22]. This genetic polymorphism could have substantial effects on the induction of 5-LO transcription.

Various biochemical mechanisms can also alter leukotriene biosynthesis within cells. Because of the complexity of 5-LO activation and the requirement of the Ca²⁺-dependent translocation event, modification of the signal transduction pathways are known to alter leukotriene biosynthesis. For example, elevation of cellular levels of cyclic AMP have been known for some time to inhibit leukotriene biosynthesis even when synthesis is stimulated by the powerful calcium ionophore A23187. Adenosine and A2A receptor agonists are known to inhibit production of leukotrienes in human neutrophils, most likely through an enhanced production of cyclic AMP (P. Borgeat, 1999). The product in neutrophil leukotriene biosynthesis, namely LTB₄, can also inhibit the synthesis of leukotrienes when initiated during the course of phagocytosis. These effects can be observed at 1–3 nM, are mediated through the LTB₄ receptor, and likely represent a feedback-like inhibition of leukotriene biosynthesis through inactivation of 5-LO as well as cytosolic PLA₂ (J. Fiedler, 1998). Pharmacological agents have been developed to inhibit leukotriene production through direct action on 5-LO. The drug

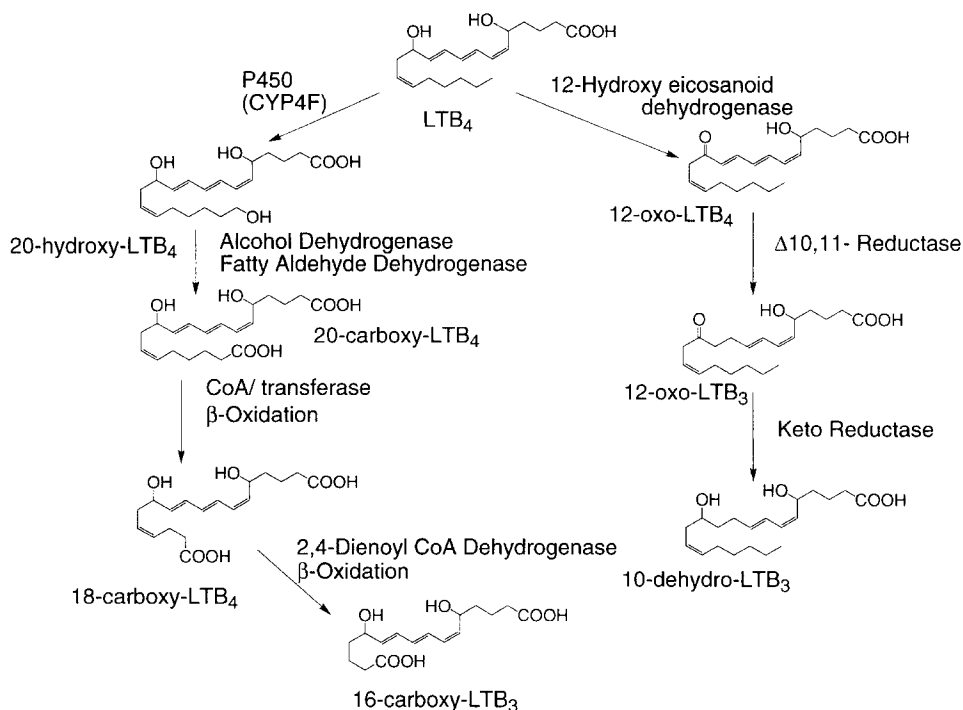


Fig. 10. Common metabolic transformations of LTB₄ either by the cytochrome P-450 (CYP4F-family members) and ω -oxidation followed by β -oxidation or by the 12-hydroxyeicosanoid dehydrogenase pathway which leads to reduction of the $\Delta^{10,11}$ double bond.

zileuton likely causes the reduction of activated 5-LO Fe(III) to the inactive 5-LO Fe(II) or prevents the oxidation of Fe(II) by lipid hydroperoxides, the pseudoperoxidase step, by serving as a competitive substrate [19]. Inhibitors of FLAP have been discussed above and include MK-886 and the related agent BAY x1005. Because of its close similarity to the FLAP protein, LTC₄ synthase can be inhibited by MK-886 albeit at higher concentrations.

4.5. Metabolism of leukotrienes

The conversion of leukotrienes into alternative structural entities is an important feature of inactivation of these potent biologically active eicosanoids. Metabolism of leukotrienes is rapid and the exact pathway depends upon whether the substrate is LTB₄ or LTC₄. LTB₄ is rapidly metabolized through both oxidative and reductive pathways (Fig. 10) [24]. The most prominent pathway present in the human neutrophil (CYP4F3) as well as hepatocyte (CYP4F2) involves specific and unique cytochrome P-450s of the CYP4F family. cDNAs encoding sixteen different proteins have now been cloned and expressed in several animal species [25] and each of these enzymes efficiently converts LTB₄ into 20-hydroxy-LTB₄. 20-Hydroxy-LTB₄ has some biological activity since it is a competitive agonist for the LTB₄ receptor. In the human neutrophil, 20-hydroxy-LTB₄

is further metabolized into 20-carboxy-LTB₄ by CYP4F3. In the hepatocyte and other tissues, 20-hydroxy-LTB₄ is further metabolized by alcohol dehydrogenase to form 20-oxo-LTB₄ then by fatty aldehyde dehydrogenase to form 20-carboxy-LTB₄ both of which reactions require NAD⁺ (Fig. 10).

A unique reductive pathway has been observed to be highly expressed in cells such as keratinocytes, endothelial cells and kidney cells. This pathway arises from an initial oxidation of the 12-hydroxy group to a 12-oxo moiety followed by reduction of the conjugated dienone and double bond $\Delta^{10,11}$. The products of the 12-hydroxy eicosanoid dehydrogenase pathway have been found to be devoid of biological activity and in certain cells represent the major pathway of inactivation.

A secondary metabolic pathway is β -oxidation which was found to predominate from the 20-carboxy terminus of 20-carboxy-LTB₄ [14]. These events take place both within the peroxisome as well as the mitochondria of the hepatocyte. β -oxidation can also occur from the C-1 carboxyl moiety of LTB₄ which eventually results in the loss of the C-5 hydroxyl group. The importance of the each oxidation processes in LTB₄ metabolism is seen in human subjects with various genetic abnormalities. Deficiencies in peroxisomal metabolism (Zellweger disease) leads to a reduction in β -oxidation and in these individuals LTB₄ and 20-carboxy-LTB₄ can be measured as urinary excretion products (E. Mayatepek, 1999). Individuals with a deficiency in fatty aldehyde dehydrogenase termed Sjogren–Larsson syndrome were found to excrete measurable levels of LTB₄ and 20-hydroxy-LTB₄ (M.A. Willemsen, 2001). None of these compounds can be measured in the urine of normal individuals even when exogenous LTB₄ is administered.

The metabolism of LTC₄ results in activation as well as inactivation of biological activity of the sulfidopeptide leukotrienes. Initial peptide cleavage reactions including γ -glutamyl transpeptidase and various dipeptidases lead to the production of LTD₄ and LTE₄, both of which are biologically active metabolites (Fig. 11). Sulfidopeptide leukotrienes can also be metabolized specifically at the sulfur atom through oxidation reactions initiated by reactive oxygen species. More specific metabolic processing of the sulfidopeptide leukotrienes include ω -oxidation by cytochrome P-450 followed by β -oxidation from the ω -terminus resulting in a series of chain-shortened products [24]. Formation of 20-carboxy-LTE₄ results in complete inactivation of the biological activities of this molecule due to poor receptor recognition. Acetylation of the terminal amino group in LTE₄ and formation of *N*-acetyl-LTE₄ is an abundant metabolite in rodent tissue. In man, some LTE₄ is excreted in urine and has been used to reflect whole body production of sulfidopeptide leukotriene *in vivo*.

4.6. Biological activities of leukotrienes

LTB₄ is thought to play an important role in the inflammatory process by way of its chemotactic and chemokinetic effects on the human polymorphonuclear leukocyte. LTB₄ induces the adherence of neutrophils to vascular endothelial cells and enhances the migration of neutrophils (diapedesis) into extravascular tissues. The biological activity of LTB₄ is mediated through two specific G-protein-coupled receptors termed BLT₁ and BLT₂ [26]. BLT₁ (human receptor 37,591 Da) is almost exclusively expressed in human polymorphonuclear leukocytes and to a much lesser extent on macrophages

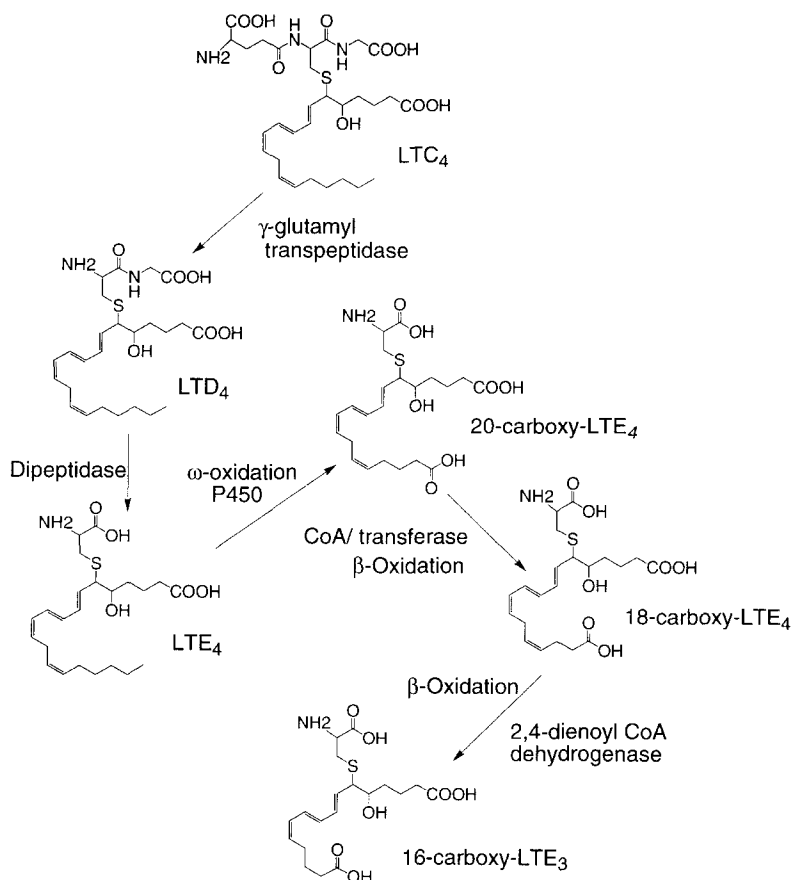


Fig. 11. Common metabolic transformations of LTC₄ to the biologically active sulfidopeptide leukotrienes, LTD₄ and LTE₄. Subsequent ω-oxidation of LTE₄ by cytochrome P-450 leads to the formation of 20-carboxy-LTE₄ which can undergo β-oxidation after formation of the CoA ester into a series of chain-shortened cysteinyl leukotriene metabolites.

and in tissues such as thymus and secretory PLeen. The human BLT gene is located on chromosome 14. The chemotactic effect of LTB₄ was shown to be mediated through the BLT₁ and BLT₂ receptors. Several specific agents have been developed by pharmaceutical companies to inhibit the LTB₄ receptor; however, none has been fully developed to be used in humans.

LTC₄ and the peptide cleavage products LTD₄ and LTE₄ have been identified as mediators causing bronchial smooth muscle contraction in asthma. These sulfidopeptide leukotrienes also increase vascular leakage leading to edema. The discovery of leukotrienes was, in fact, a result of the search for the chemical structure of the biologically active principle called 'slow reacting substance of anaphylaxis' (R.C. Murphy, 1979). Two receptors for the cysteinyl leukotriene have recently been characterized and termed CysLT₁ and CysLT₂. The CysLT₁ receptor was found to be a G-protein-coupled

receptor with seven *trans* membrane regions [27]. The CysLT₁ receptor has limited distribution in tissues with the most prominent being in smooth muscle of the lung and small intestine and both LTD₄ and LTC₄ activate this receptor. Several drugs are now available for inhibition of the CysLT₁ receptor in human subjects. These are montelukast (2–5 nM), pranlukast (4–7 nM), and zafirlukast (2–3 nM). Interestingly, the gene encoding the human CysLT₁ receptor is located on the X chromosome [27].

4.7. Other lipoxygenase pathways

Numerous lipoxygenases occur within the plant and animal kingdoms and these enzymes have in common several aspects. First, iron is an essential component of the catalytic activity of these enzymes and is held in place through histidine residues rather than by heme. Furthermore, these enzymes catalyze the insertion of molecular oxygen into polyunsaturated fatty acids, predominantly linoleic and arachidonic acids, with the initial formation of lipid hydroperoxides. In general, the overall biological activities of the lipoxygenase products are incompletely known and the significance of 12- and 15-lipoxygenase in man remains undefined.

4.7.1. 12-Lipoxygenase

Two different enzymes termed 12-lipoxygenase (12-LO) catalyze the formation of 12-hydroperoxyeicosatetraenoic acid from arachidonic acid which is subsequently reduced to 12-hydroxyeicosatetraenoic acid (Fig. 12). The human platelet expresses one 12-LO

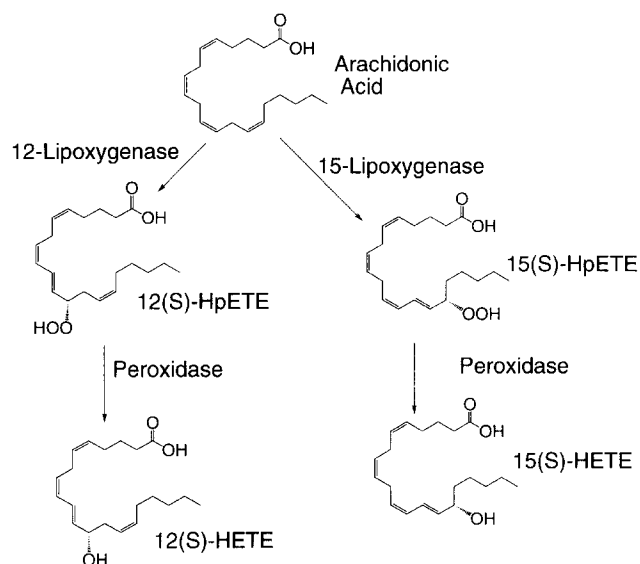


Fig. 12. Metabolism of arachidonic acid by 12- and 15-lipoxygenase pathways with corresponding stereospecific formation of hydroperoxyeicosatetraenoic acids (HpETE). Subsequent reduction of these hydroperoxides leads to the corresponding HETE at either carbon-12 or -15 which are thought to mediate biological activities of these enzymatic pathways.

type (EC 1.13.11.31) the cDNA of which has been cloned, sequenced, and found to encode a 662 amino acid protein with a molecular weight of 75,535 Da. A second 12-LO is observed in other mammalian systems including the mouse and rat and has been termed the leukocyte-type 12-LO. This latter lipoxxygenase is very similar in many respects to a 15-lipoxxygenase (15-LO) in terms of its substrate specificity and capability of forming both 12-HpETE and 15-HpETE from arachidonic acid. The human platelet 12-LO has approximately 65% identity in primary structure to that of 15-LO from human reticulocytes. In addition, there are other lipoxxygenases less well studied including an epidermal lipoxxygenase from newly differentiated keratinocytes and the lipoxxygenase that oxygenates arachidonic acid at position C-15 and C-8. Both 12/15-LO types are suicide inactivated, but the leukocyte 12-LO undergoes autoinactivation at a much higher rate. Several lines of evidence suggest that the 12-LO pathway of arachidonate metabolism plays an important role in regulating cell survival and apoptosis (A.R. Brash, 1999).

4.7.2. 15-Lipoxxygenase

The oxidation of arachidonic acid at carbon-15 is catalyzed by 15-LO, a soluble 661 amino acid containing protein with a molecular weight of 74,673 Da. Many cells express this enzyme which also efficiently oxidizes linoleic acid to 13-hydroperoxyoctadecadienoic acid and lesser extent 9-hydroperoxyoctadecadienoic acid because of broad substrate specificity as well as arachidonate to both 12-HpETE and 15-HpETE [28]. One distinguishing feature of this lipoxxygenase is that it can oxidize arachidonic acid esterified to membrane phospholipids, thus forming esterified 15-HpETE. Expression of 15-LO is enhanced by several interleukins, suggesting a role of this enzyme in events such as atherosclerosis.

The X-ray crystal structure of mammalian 15-LO has revealed two domains, a catalytic domain and a β -barrel domain [29]. The β -barrel domain may be involved in the binding of this enzyme to phospholipid membranes, the source of either arachidonate or phospholipids in the oxidation process. The catalytic domain, which contains the histidine-coordinated Fe(III), holds the arachidonic acid assisted by an ionic bond between R403 and the ionized carboxyl group of arachidonic acid. The methyl terminus of arachidonic acid is thus placed deep within a hydrophobic binding pocket in an arrangement that is likely similar for other lipoxxygenases.

Mammalian 15-LO is involved in the production of more complicated eicosanoids including the biologically active lipoxins [26]. Lipoxins are formed by the sequential reaction of both 15-LO and 5-LO acting on precursor arachidonate. For example, LTA₄ (the 5-LO product) can be converted to a lipoxin by action of 15-LO.

There are a host of biological activities initiated by these 15-LO dependent eicosanoids [28]. The unique activity of 15-LO in oxidizing intact phospholipids has been featured in several hypotheses linking the oxidation of phospholipids in atherosclerotic lesions to important role of these lipoxxygenases. The recent availability of strains of mice which have a targeted deficiency in 5-, 12-, and 15-lipoxxygenases (C. Funk, 2000) provides a powerful tool to ask specific questions concerning the role these lipoxxygenases play in host defense reactions, cellular function, and perhaps disease processes.

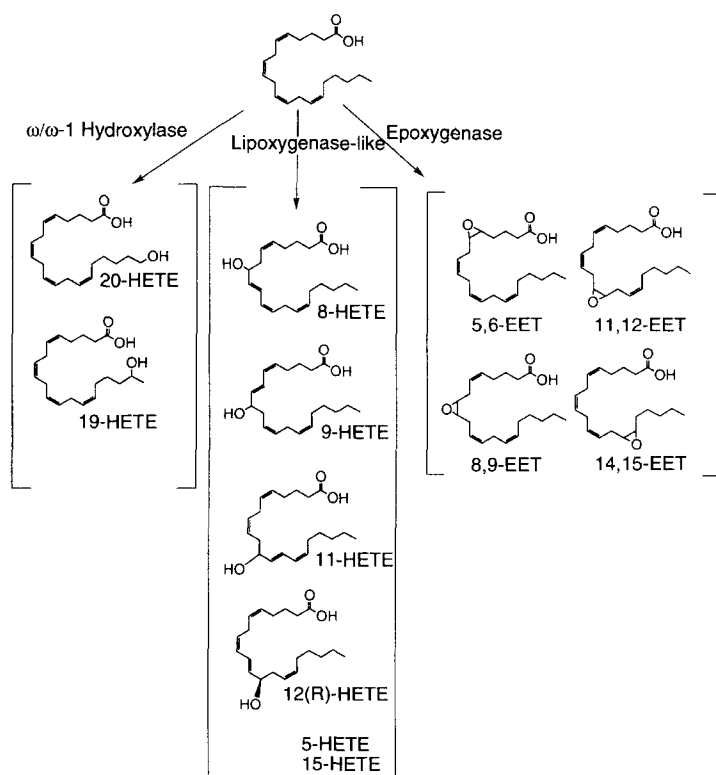


Fig. 13. Metabolism of arachidonic acid by cytochrome P-450 enzymes and the formation of three structurally distinct metabolite families. Omega-oxidation leads to a family of ω to ω -4 products of which 20-HETE (ω -oxidation) and 19-HETE (ω -1 oxidation) are indicated. The lipoxygenase-like mechanism of cytochrome P-450 metabolism leads to the formation of six different conjugated dienois, for which the structures of four are indicated. One unique biologically active lipoxygenase-like P-450 metabolite is 12(R)-HETE. The epoxygenase pathway leads to the formation of four regioisomeric epoxyeicosatrienoic acid (EETs) all of which are biologically active.

5. Cytochrome P-450s and epoxygenase pathways

Arachidonic acid can be metabolized to a series of products characterized by the introduction of a single oxygen atom from molecular oxygen and formation of three different types of initial products catalyzed by various cytochrome P-450 mixed function oxidases (Fig. 13) [30,31]. The three classes of products include a series of hydroxyeicosatetraenoic acids (HETEs) formed by an allylic oxidation mechanism resulting in a family of conjugated dienes isomeric to the reduced products of a lipoxygenase reaction. P-450 metabolites formed by this mechanism have been characterized as 5-, 8-, 9-, 11-, 12-, and 15-HETE, some of which are epimeric to the lipoxygenase catalyzed products, e.g., 12(R)-HETE. A second class of reactions involves oxidation of the terminal alkyl chain region of arachidonic acid with placement of a hydroxyl group between the terminal carbon atom (ω) through ω -1 position with formation of a family of ω -oxidized

monohydroxyeicosatetraenoic acids. Insertion of oxygen into the carbon–carbon bond results in the formation of a family of regioisomeric *cis* epoxyeicosatetraenoic acids (EETs) from which the general pathway has been named, the epoxygenase pathway. These regioisomers include 14,15-, 11,12-, 8,9-, and 5,6-EETs which can be formed either as an *R,S*, or the *S,R* enantiomer (Fig. 13).

5.1. Epoxygenase P-450 isozymes

With the availability of recombinant P-450 isozymes, it has been possible to identify specific isozymes that can metabolize arachidonic acid. EET biosynthesis can be accomplished by CYP1A, CYP2B, CYP2C, CYP2D, CYP2G, CYP2J, CYP2N, and CYP4A subfamilies [31]. For each of these, unique EET regioisomers are formed. For example CYP2C8 produces 14(*R*),15(*S*)-EET and 11(*R*),12(*S*)-EET with optical purities of 86% and 81%, respectively. However, it is likely that more than a single P-450 contributes to EET biosynthesis within a specific cell or tissue and thus the individual arachidonate epoxygenase metabolite may depend upon expression of specific P-450 isoforms. It is thought that the majority of EET biosynthesis in human and rat kidney is a result of CYP2C expression in these tissues. However, the induction of specific P-450s can greatly alter the production of specific epoxygenase products.

5.2. Occurrence of EETs

Various EETs have been measured in tissues as well as physiological fluids such as urine (G. FitzGerald, 1990). Biologically active lipids originally defined as an endothelium-derived hyperpolarizing factor and an inhibitor of Na^+/K^+ ATPase found in the thick ascending loop of Henley cells were structurally characterized as 11(*R*),12(*S*)-EET and 20-hydroxyeicosatetraenoic acid, both derived from cytochrome P-450 mediated metabolism of arachidonic acid [30]. Interestingly, the EETs can readily form CoA esters and participate in reacylation of lysophospholipids which results in the reincorporation of these oxidized metabolites of arachidonic acid into phospholipid membranes, a biochemical feature not observed for prostaglandins, thromboxanes or leukotrienes. For example, human platelets have been found to contain 14,15-EET esterified within membrane phospholipids (Y. Zhu, 1995). It is felt that the majority of EETs produced within cells become reesterified to cellular glycerophospholipids.

5.3. Metabolism of EETs

A number of metabolic pathways operate on the primary epoxygenase metabolites of arachidonic acid. Some of the more abundant pathways include CoA-dependent reesterification as mentioned above as well as β -oxidation chain-shortening. A unique pathway involves epoxide hydrolase, a cytosolic enzyme that hydrates EETs to the corresponding vicinal dihydroxyeicosatrienoic acids [30]. There is also a microsomal epoxide hydrolase that can metabolize EETs, but at a somewhat lower rate. The soluble epoxide hydrolase does have substrate specificity, both in terms of the stereochemistry of the EET as well as its position in the arachidonic acid chain. As expected, there is

nonenzymatic hydration of these epoxides especially under acidic conditions that can be accelerated during the isolation of these arachidonate metabolites. Therefore, it is sometimes difficult to distinguish between nonenzymatic and enzymatic hydration of EETs. The 5,6-EET is a poor substrate for cytosolic as well as microsomal epoxide hydrolase; however, it has been observed to be an efficient substrate for PGH synthase, leading to the formation of 5,6-epoxy-PGH₁. This reactive intermediate can subsequently be transformed into corresponding 5,6-epoxy-prostaglandins of the E, F, and I series or into an epoxy thromboxane analog. All of the EETs can also be substrates for lipoxygenases which would introduce molecular oxygen at any 1,4-*cis*-pentadienyl position not interrupted by the epoxide ring. The EETs can also be conjugated with reduced glutathione catalyzed by glutathione (*S*) transferases. Studies of the metabolism of EETs by rat or mouse liver microsomal P-450 revealed the formation of a series of diepoxyeicosadienoic acids as well as monohydroxyepoxyeicosatrienoic acids. Interestingly, the diepoxides were found to be further transformed into tetrahydrofurandiols mediated by intermediate diol epoxides formed by soluble epoxide hydrolase. The characterization of specific products of EET and PGH synthase metabolism of arachidonic acid have led to the observation of 5,6-epoxy-PGE₁ as a renal vasodilator with similar potency to that of PGE₂ and the metabolism of 8,9-EET by PGH synthase leading to 11-hydroxy-8,9-epoxyeicosatrienoic acid which is a mitogen for rat glomerular mesangial cells.

5.4. *Biological actions of EETs*

Metabolites of arachidonic acid derived from the epoxigenase P-450 pathway have been studied extensively in terms of their pharmacological properties. Potent effects have been observed in modulating various ion channels, membrane bound transport proteins, mitogenesis, PPAR α agonists, and activators of tyrosine kinase cascades [30]. EETs likely play an important role in mediating Na⁺/K⁺-ATPase and inhibiting the hydroosmotic effect of arginine vasopressin in the kidney. A picture has emerged for an important role of EETs in regulating renal vascular tone and fluid/electrolyte transport placing the EETs in the pathogenesis of hypertension.

6. *Future directions*

There is currently a reasonable understanding of the structures of PGHS-1 and -2, but many of the relationships between structure and function remain to be identified. For example, it is known that the peroxidase activities of PGHSs preferentially utilize alkyl hydroperoxides such as PGG₂ versus hydrogen peroxide; the basis for this specificity is not evident from simple observation of the structures. It will also be important to characterize further the membrane binding domain of PGHSs in the context of the interaction of these domains with specific membrane lipids and the role of this domain in governing substrate entry into the cyclooxygenase site and product exit from this site.

Our understanding of the different biological roles of PGHS-1 and PGHS-2 is only beginning to emerge. The functions of these two isozymes in apoptosis, particularly as it relates to the development of a variety of cancers, angiogenesis, respiration,

inflammation, pain and reproduction need further exploration. Of key importance is understanding the reason for the existence of the two PGHS isozymes and how coupling occurs between these enzymes, upstream lipases, and downstream synthases and receptors.

Considerable challenges remain in understanding the detailed biochemistry involved in the synthesis and release of biologically active leukotrienes. Little is known how these highly lipophilic molecules are released from cells; but even more curious is how the chemically reactive intermediate leukotriene A₄, made on the perinuclear membrane, can find its way into a neighboring cell in the process of transcellular biosynthesis. The mechanism of 5-lipoxygenase is still poorly understood; however, the detailed structure of 5-lipoxygenase (X-ray structure) may likely reveal important facets relevant to translocation events of 5-lipoxygenase as well as mechanism of suicide inactivation. Such information would be of great value in designing specific drugs as novel inhibitors of 5-lipoxygenase, a pharmacological approach highly successful for cyclooxygenase. Last, but not least, little is known concerning a potential intracellular role for leukotrienes. The currently known biological actions of leukotrienes all involve cell membrane G-protein-linked receptors, yet an understanding of why biosynthesis of these lipophilic molecules occurs deep within the cell remains a mystery.

Abbreviations

2-AG	2-arachidonyl-glycerol
BLT (1 or 2)	leukotriene B ₄ receptor (subclass 1 or 2)
cysLT (1 or 2)	cysteinyl leukotriene receptor (subclass 1 or 2) for which leukotriene C ₄ , D ₄ , and E ₄ are agonists
CYP4F(x)	cytochrome P-450 isozyme that carries out ω -oxidation of leukotriene B ₄ (subclass x)
EET	epoxyeicosatetraenoic acid
EP	prostaglandin E receptor
FLAP	5-lipoxygenase activating protein
HETE	hydroxyeicosatetraenoic acid
HpETE	hydroperoxyeicosatetraenoic acid
LO	lipoxygenase
LT	leukotriene (followed by letter to designate structural type)
NSAID	nonsteroidal anti-inflammatory drug
PG	prostaglandin (followed by letter to designate structural type)
PGHS	prostaglandin endoperoxide H synthase
PLA ₂	phospholipase A ₂

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Sphingolipids: metabolism and cell signaling

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1. Introduction

Sphingolipids were first described by Johann L.W. Thudichum in *A Treatise on the Chemical Constitution of Brain* (1884) [1]. Among the described compounds were sphingomyelin, cerebroside, and cerebrosulfatide (Fig. 1), which encompass the three categories of sphingolipids known today (phosphosphingolipids, neutral and acidic glycosphingolipids). Thudichum noted that hydrolysis of these lipids produced a compound that "... is of an alkaloidal nature, and to which, in commemoration of the many enigmas which it has presented to the inquirer, I have given the name of *Sphingosin*." Thus, this class of lipids became known as *sphingolipids*.

Thudichum, a practicing physician throughout most of his life, was searching for a better understanding of disease, but appreciated that "... to reach this goal of complete knowledge. . . the medicinal chemist must. . . not. . . carry on research by a kind of fishing for supposed disease-poisons, of which, according to my view of the subject, the attempt of the boy to catch a whale in his mother's washing-tub is an appropriate parable." Later studies fulfilled Thudichum's faith in the value of basic research when

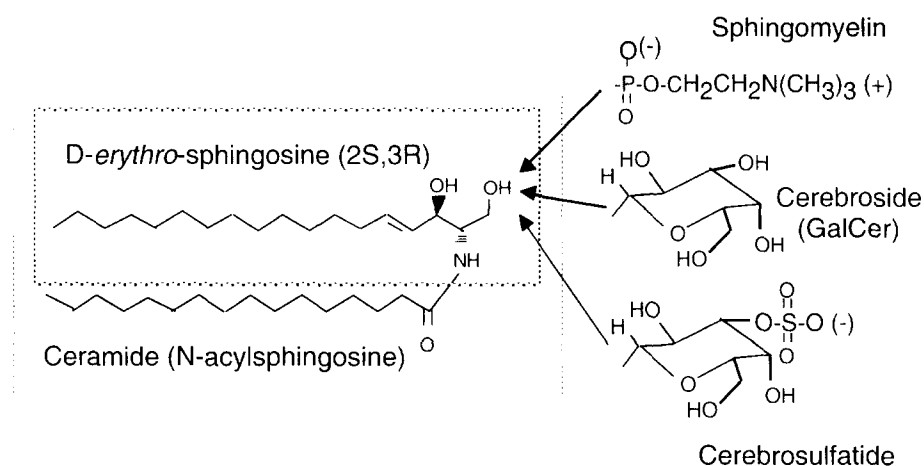


Fig. 1. Structures of sphingosine, ceramide, sphingomyelin, a cerebroside (galactosylceramide) and cerebrosulfatide from human brain.

several genetic diseases were found to have elevated amounts of sphingolipids (such as sphingomyelin in Niemann–Pick’s disease and cerebrosides in Gaucher’s disease) arising from defects in enzymes responsible for sphingolipid turnover, activator proteins for such enzymes, or lipid trafficking (for reviews, see Hakomori [2] and Schuette et al. [3]). This knowledge allowed development of methods for diagnosis of such sphingolipid storage diseases (or ‘sphingolipidoses’), screening of families at risk, and, for at least Gaucher’s disease, some degree of correction of the disorder by enzyme replacement. Progress is also being made using inhibitors of sphingolipid synthesis, and gene replacement offers promise for the future.

For many years, the only diseases associated conclusively with sphingolipids involved defective sphingolipid turnover. Disruption of sphingolipid biosynthesis is now known to be the major mechanism of action of mycotoxins (fumonisins and alternaria toxins) that cause a wide spectrum of diseases of plants and animals (A.H. Merrill, 2001). And in 2001, genetic defects in sphingolipid biosynthesis were shown to cause the most common hereditary disorder of peripheral sensory neurons (hereditary sensory neuropathy type I) (J.L. Dawkins, 2001; K. Bejaoui, 2001). It is certain that additional genetic diseases due to abnormal sphingolipid biosynthesis will be found, and knockout mice defective in the biosynthesis of glucosylceramide and other glycolipids have severe defects, especially in the developing nervous system (T. Kolter, 2000). There are also indications that sphingolipids and sphingolipid analogs may be useful for prevention and treatment of disease, e.g., gangliosides (R. McKallip, 1999) and α -galactosylceramide (M. Taniguchi, 1997 and 1998) have potent effects as modulators of the immune system, ceramide-coated balloon catheters limit neointimal hyperplasia after stretch injury in carotid arteries (R. Charles, 2000), and dietary sphingolipids protect against colon tumorigenesis (E.M. Schmelz, 2001). These probably reflect just a few of the ways sphingolipids are relevant to pathology, nutrition and medicinal chemistry.

1.1. Biological significance of sphingolipids

Sphingolipids are found in essentially all animals, plants, and fungi, as well as some prokaryotic organisms and viruses. They are mostly in membranes, but are also major constituents of lipoproteins. The functions of sphingolipids are still being discovered, but there are at least three, i.e., structure, recognition and signal transduction, which have been summarized diagrammatically in Fig. 2.

1.1.1. Biological structures

Some glycosphingolipids and sphingomyelins tend to cluster rather than behave like typical ‘fluid’ membrane lipids. This behavior arises from the mostly saturated alkyl sidechains, which allow strong van der Waals interactions, and the ceramide hydroxyls, amide bond and polar headgroups that are capable of hydrogen bonding and dipolar interactions (it is common for sphingolipids to have phase transition temperatures $> 37^{\circ}\text{C}$). Sphingolipids contribute to the formation of regions of the plasma membrane termed ‘rafts’ and ‘caveolae’ ([5,6], Chapter 1), which are enriched in growth factor receptors, transporters and other proteins, especially proteins with a glycosylphosphatidylinositol-lipid anchor. Sphingolipids contribute to the stability of other types

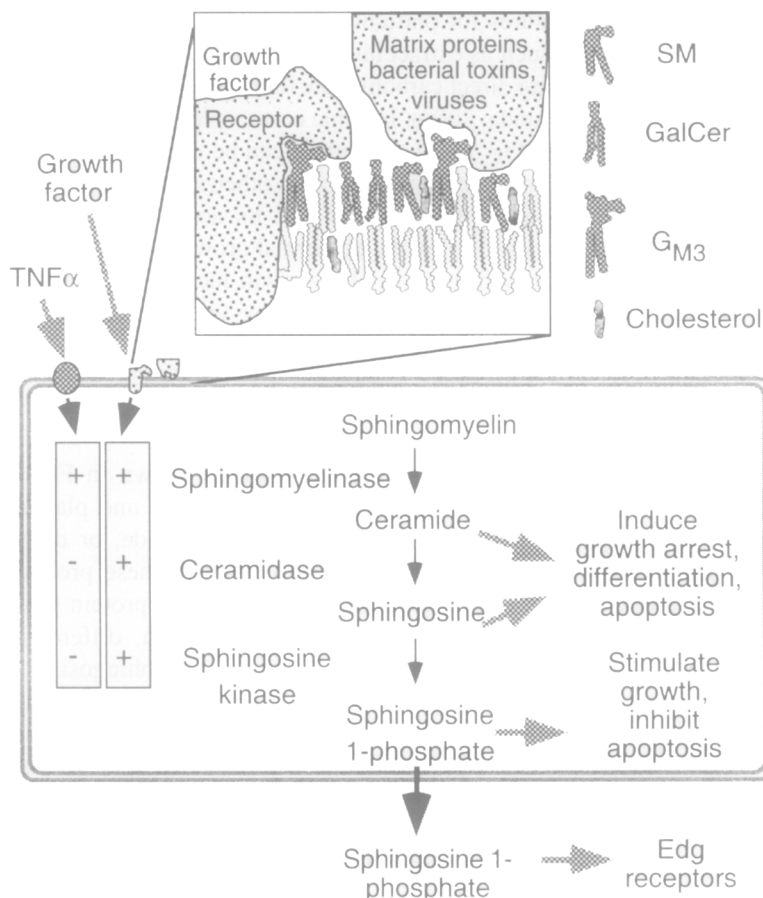


Fig. 2. Schematic representation of sphingolipid functions. Shown in the upper inset are sphingolipid- (and cholesterol-) enriched regions of the plasma membrane ('rafts'), and sphingolipids serving as ligands for extracellular proteins and receptors on the same cell. The lower diagram of a cell illustrates how agonists (such as tumor necrosis factor- α , TNF- α , and growth factors) can activate combinations of sphingolipid metabolizing enzymes to produce bioactive products that affect the shown cell behaviors. Sphingosine 1-phosphate is also secreted as an agonist for some members of the Edg family receptors (now named SIP).

of biological structures, such as the lamellar bodies that maintain the permeability barrier of skin (P. Wertz, 2000) and lipoproteins (S.L. Schissel, 1996). However, not all sphingolipids are so ordered, and some, such as sphingosine 1-phosphate and sphingosylphosphorylcholine (lysosphingomyelin), are sufficiently polar to exist in aqueous environments (Y. Yatomi, 1997).

1.1.2. Biological recognition

Membrane sphingolipids are located predominantly on the outer leaflet of the plasma membrane, the lumen of intracellular vesicles and organelles (endosomes, Golgi membranes, etc.), and in as yet undefined locations in mitochondria and nuclei. The complex

carbohydrate moieties are often signatures for particular cell types, and mediate interactions with complementary ligands, such as extracellular matrix proteins and receptors (S.I. Hakomori, 2000) (Fig. 2), including direct carbohydrate–carbohydrate binding with headgroups on neighboring cells (K. Handa, 2000). In some cases, sphingolipids interact with proteins on the same cell surface (Fig. 2). Such binding can be used to control the location of the protein (for example, in membrane rafts with other signaling proteins) as well as to modify the conformation of the receptor and its activity [7]. This is exemplified by the binding of ganglioside G_{M3} by the epidermal growth factor receptor, which makes the receptor refractory to activation by this growth factor (E.J. Meuillet, 2000; A.R. Zurita, 2001). Sphingolipids are also recognized by viruses, bacteria and bacterial toxins as a means of both attachment and entry into the cell via membrane trafficking (K.A. Karlsson, 1992; C.A. Lingwood, 1999).

1.1.3. Signal transduction

The sphingolipid backbones are members of a signaling paradigm shown in Fig. 2, wherein receptor activation by agonists such as tumor necrosis factor- α and platelet-derived growth factor induce sphingomyelin turnover to elevate ceramide, or downstream metabolites (sphingosine or sphingosine 1-phosphate) [8–10]. These products activate or inhibit multiple downstream targets (protein kinases, phosphoprotein phosphatases and others) that control cell behaviors as complex as growth, differentiation and programmed cell death (apoptosis). Because ceramide and sphingosine 1-phosphate often have opposing signaling functions (e.g., induction versus inhibition of apoptosis; inhibition versus stimulation of growth), Sarah Spiegel has proposed that cells utilize a ceramide/sphingosine 1-phosphate ‘rheostat’ in deciding between growth arrest/apoptosis versus proliferation/survival (S. Spiegel, 1999). Sphingosine 1-phosphate can be released from cells and serve as an agonist for SIP receptors [10], hence, this compound serves as both a first and second messenger!

The field of sphingolipid signaling is still relatively young and has many new facets that reveal the biochemical ‘logic’ of using such complex molecules to control cell behavior. For example, sphingomyelin turnover not only produces ‘signaling’ metabolites, but also alters the structure of membrane domains that depend on the presence of this lipid (furthermore, when ceramide accumulates, its biophysical properties can profoundly affect membrane structure and the behavior of associated receptors and other proteins) [5,6]. A similar paradigm can be envisioned for glycosphingolipids. Thus, sphingolipid ‘signaling’ is an ensemble of changes in membrane structure and dynamics, the production (and removal) of bioactive metabolites, and the activation and/or inhibition of downstream targets.

1.2. Structures and nomenclature of sphingolipids

More than 300 different types of complex sphingolipids have been reported, and this does not include differences in the ceramide backbone. It has become necessary to develop a system of nomenclature for sphingolipids so that individual species can be referred to in a logical manner [11]. Nonetheless, there is still considerable variability in the names that are used for these compounds. For example, ‘sphingosine’ is still

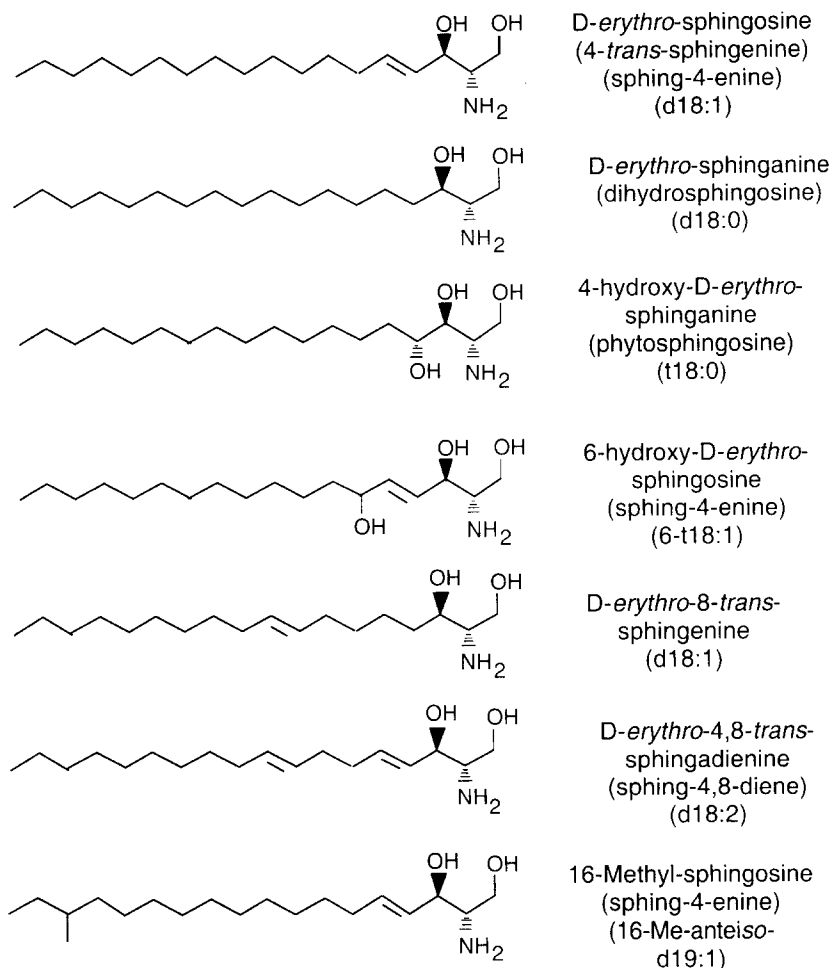


Fig. 3. Structures of some of the long-chain (sphingoid) bases that have been found in sphingolipids. Abbreviations for these compounds are shown in parentheses.

in common usage although the names recommended by the IUPAC are (*E*)-sphing-4-ene or (2*S*,3*R*,4*E*)-2-aminooctadec-4-ene-1,3-diol (by their recommendation, dihydro-‘sphingosine’ is sphinganine, and 4-hydroxysphinganine (also called phytosphingosine) is (2*S*,3*S*,4*R*)-2-aminooctadecane-1,3,4-triol). This chapter uses the most familiar names: sphingosine, sphinganine and 4-hydroxysphinganine.

Sphingosine is the prevalent backbone of most mammalian sphingolipids; however, over 60 different species of long-chain bases have been reported [12] and include compounds (Fig. 3) with (1) alkyl chain lengths from 14 to 22 carbon atoms, (2) different degrees of saturation at carbons 4 and 5, (3) a hydroxyl group at positions 4 or 6, (4) double bonds at other sites in the alkyl chain, and (5) branching (methyl groups) at the ω -1 (iso), ω -2 (anteiso), or other positions. Sphingoid bases are abbreviated by

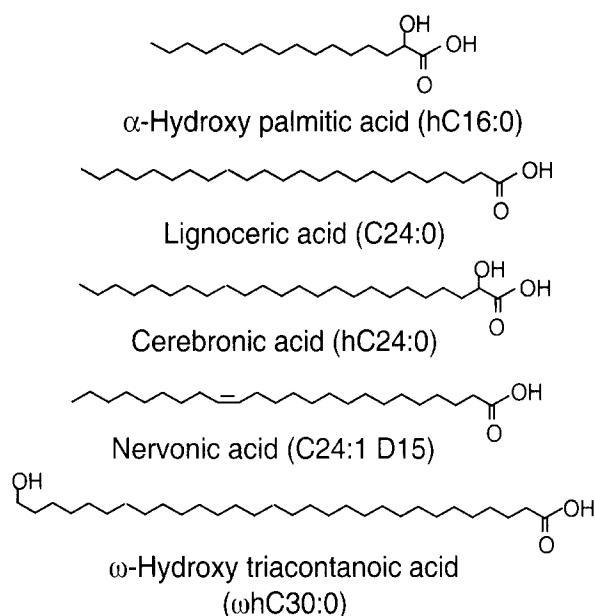


Fig. 4. Structures of representative fatty acids, including α - and ω -hydroxy fatty acids, that are found in mammalian sphingolipids. Abbreviations for these compounds are shown in parentheses.

citing (in order of appearance in the abbreviation) the number of hydroxyl groups (d and t for di- and tri-hydroxy, respectively), chain length and number of double bonds as shown in Fig. 3.

The majority of the sphingoid bases in cells are *N*-acylated with long-chain fatty acids to produce ceramides(s) (Fig. 1), although *O*-acylated (A. Abe, 1998), phosphorylated-(sphingosine 1-phosphate) and *N*-methylated-(*N,N*-dimethylsphingosine) derivatives also exist. The fatty acids of ceramide vary in chain length (14 to 30 carbon atoms), degree of unsaturation (but are mostly saturated), and presence or absence of a hydroxyl group on the α - or ω -carbon atom. Structures and abbreviations for some fatty acids are shown in Fig. 4.

Most sphingolipids have a polar headgroup at position 1 (Figs. 1 and 5). Sphingolipids are often grouped based on the headgroups into the phosphosphingolipids and glycosphingolipids; however, these categories are not mutually exclusive: the major sphingolipids of yeast are ceramide phosphorylinositols. Glycosphingolipids are classified into broad types on the basis of carbohydrate composition. Neutral glycosphingolipids contain uncharged sugars such as glucose (Glc), galactose (Gal), *N*-acetylglucosamine (GlcNAc), *N*-acetylgalactosamine (GalNAc), and fucose (Fuc). Acidic glycosphingolipids contain ionized functional groups such as phosphate, sulfate (sulfatoglycosphingolipids), or charged sugar residues such as sialic acid (*N*-acetylneuraminic acid) in gangliosides or glucuronic acid in some plant glycosphingolipids. Further classification can be made on the basis of shared partial oligosaccharide sequences, sometimes referred to as 'root structures' as summarized in Table 1.

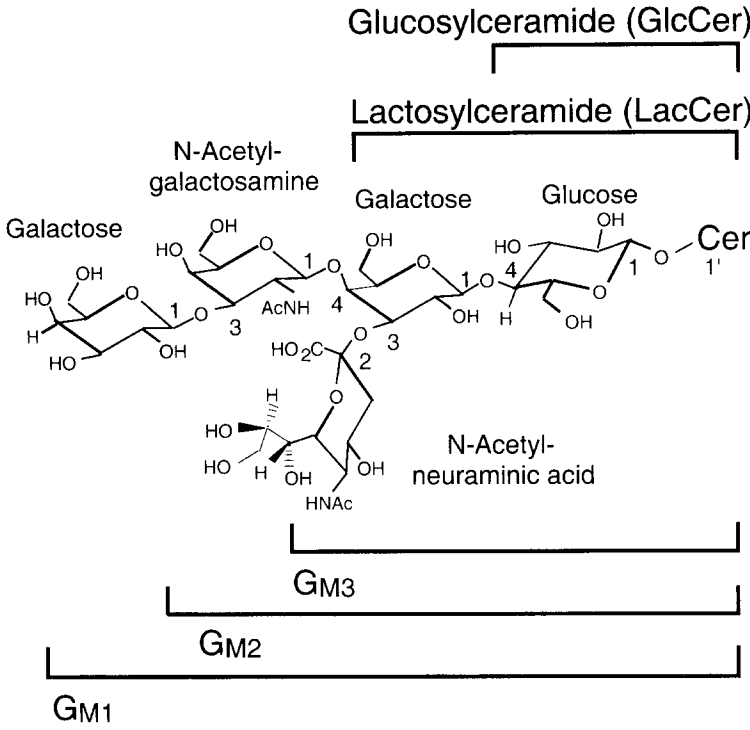


Fig. 5. Structures of some of the common neutral glycosphingolipids (GlcCer and LacCer) and gangliosides G_{M1} , G_{M2} and G_{M3} . ‘Cer’ refers to the ceramide backbone [for another useful overview structure, for ganglioside G_{D1} , see G. van Echten-Deckert (1999)].

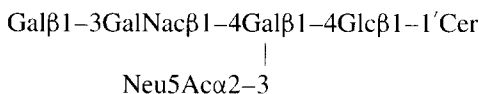
Table 1
Nomenclature for classification of glycosphingolipids

Root name	Abbreviation	Partial structure ^a			
		IV	III	II	I
Ganglio	Gg	Gal β 1–3GalNac β 1–4Gal β 1–4Glc β 1–1’Cer			
Lacto	Lc	Gal β 1–3GlcNac β 1–3Gal β 1–4Glc β 1–1’Cer			
Neolacto	nLc	Gal β 1–4GlcNac β 1–3Gal β 1–4Glc β 1–1’Cer			
Globo	Gb	GalNac β 1–3Gal α 1–4Gal β 1–4Glc β 1–1’Cer			
Isoglobo	iGb	GalNac β 1–3Gal α 1–3Gal β 1–4Glc β 1–1’Cer			
Mollu	Mu	GalNac β 1–2Man α 1–3Man β 1–4Glc β 1–1’Cer			
Arthro	At	GalNac β 1–4GlcNac β 1–3Man β 1–4Glc β 1–1’Cer			

^a Roman numerals define sugar positions in the ‘root’ structure.

Gangliosides are often denoted by the ‘Svennerholm’ nomenclature [11] that is based on the number of sialic acid residues (e.g., G_{M1} refers to a monosialo-ganglioside) and the relative position of the ganglioside upon thin-layer chromatography (thus, the order of migration of the series of monosialogangliosides in Fig. 5 is $G_{M3} > G_{M2} > G_{M1}$). By

commonly used nomenclatures, the same compound might be called ganglioside G_{M1} , II^3 - α -*N*-acetylneuraminosyl-gangliotetraosylCer, II^3 - α -Neu5NacGg₄Cer, or depicted as:



Note that the Roman numeral and Arabic superscript refer to the sugar in the root structure (cf. Table 1) that is substituted (counting from the ceramide toward the non-reducing end) and the position of that substitution, respectively.

A number of sphingolipids are referred to by their historic names as antigens and blood group structures, such as Forssman antigen (IV^3 - α -GalNac-Gb₄Cer), a globopentosylceramide that is found in many mammals (but it is unclear if humans express this antigen) and the Lewis blood group antigens, which correspond to a family of $\alpha 1$ -3-fucosylated glycan structures (Lewis x, sialyl Lewis x, etc.). For more information on these aspects of glycosphingolipidology see Varki et al. [13].

2. Chemistry and distribution

This section will summarize some of the properties of sphingolipids. More information is available in Merrill and Hannun [14].

2.1. Sphingoid bases

A distinctive feature of sphingoid bases is that they can bear a net positive charge at neutral pH, which is rare among naturally occurring lipids. Nonetheless, the pK_a of the amino group is low for a simple amine (between 7 and 8) (A.H. Merrill, 1989), which means that a portion is uncharged at physiologic pH. This may help explain why sphingoid bases can readily move among membranes and across bilayers (in the uncharged state), unless transmembrane movement is impeded by acidic pH, such as in lysosomes.

Structural elucidation and quantitation of long-chain bases is possible using a variety of analytical techniques, including gas chromatography and high-performance liquid chromatography, mass spectroscopy, and nuclear magnetic resonance spectroscopy [13,14].

2.2. Ceramides

Ceramides per se are mostly found in small amounts in tissues, with the notable exception of the stratum corneum, where they are major determinates of the water permeability barrier of skin (P. Wertz, 2000). Many ceramides (even as part of complex sphingolipids) migrate on thin-layer chromatography as multiple bands due to the presence of at least several types of sphingoid bases and fatty acids. The molecular species can be analyzed by a number of techniques, such as gas chromatography, high-performance liquid chromatography (HPLC), or hydrolysis (or methanolysis) followed by analysis of the sphingoid bases and fatty acids [14]. However, the most information

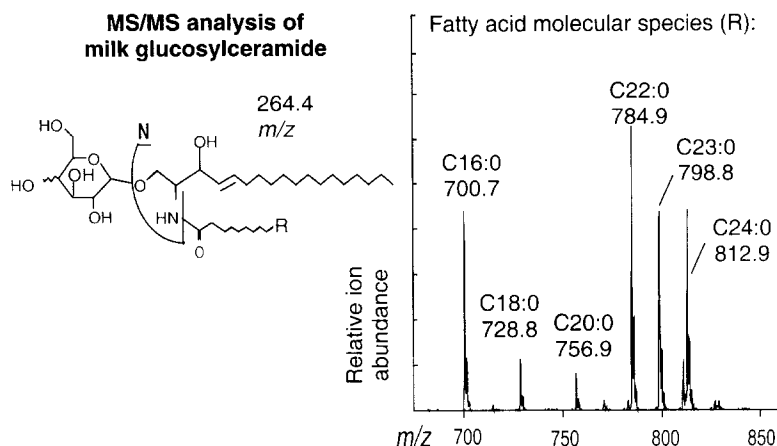


Fig. 6. Major fragmentation sites of a monohexosylceramide upon electrospray tandem mass spectrometry (left panel) and (right panel) a typical precursor ion spectrum (obtained with a bovine milk extract) monitoring m/z 264.4 (a signature fragment obtained with sphingolipids with a sphingosine backbone) over the range m/z 675–875. The labeled signals represent the various amide-linked fatty acids on the milk glucosylceramides. For more information see Sullards and Merrill [15].

is obtained by combining HPLC with electrospray tandem mass spectrometry (ESI MS/MS) [15]. In this method, the ceramides are separated as classes (free ceramides, sphingomyelins, glucosylceramides, etc.) by HPLC and the eluant is introduced directly into the ionizing chamber of the mass spectrometer, where the solvent is rapidly evaporated under high vacuum and the compounds are suspended in the gas phase as individual charged species. These 'parent' ions are separated by the first MS, then allowed to collide with a gas (such as N_2) to produce fragments that are separated by the second MS. Besides high sensitivity, the advantage of this instrumentation is the ability to focus on the compounds of interest in crude mixtures. For example, glycosylceramides containing sphingosine will fragment to m/z 264.4 (Fig. 6); therefore, the second MS can be set to detect m/z 264.4 and the first MS to identify the parent ions that produce this fragment. This is illustrated for the glucosylceramides in a milk lipid extract in Fig. 6. Quantitation is achieved by spiking the sample with an internal standard with a chemical composition sufficiently similar to the unknowns for them to fragment with similar efficiencies, as described in the legend to Fig. 7. More accurate and sensitive quantitation can be obtained using a specialized MS/MS technique known as multiple reaction monitoring (MRM), in which the mass spectrometer is programmed to maximize the time spent detecting specific precursor/product ion transitions, and the detection of each individual molecular species can be optimized with respect to ion formation and decomposition.

Use of such methods allows more accurate and facile quantitation of multiple sphingolipid species in cells and other biological materials. Fig. 7 gives a typical analysis of the sphingolipids of NIH 3T3 cells, which contain (nmol per 10^6 cells): SM (2.7), GlcCer (0.31), Cer (0.082), sphingosine (0.017), sphingosine-1-phosphate (0.011), sphinganine (0.160), and sphinganine-1-phosphate (<0.001). For compari-

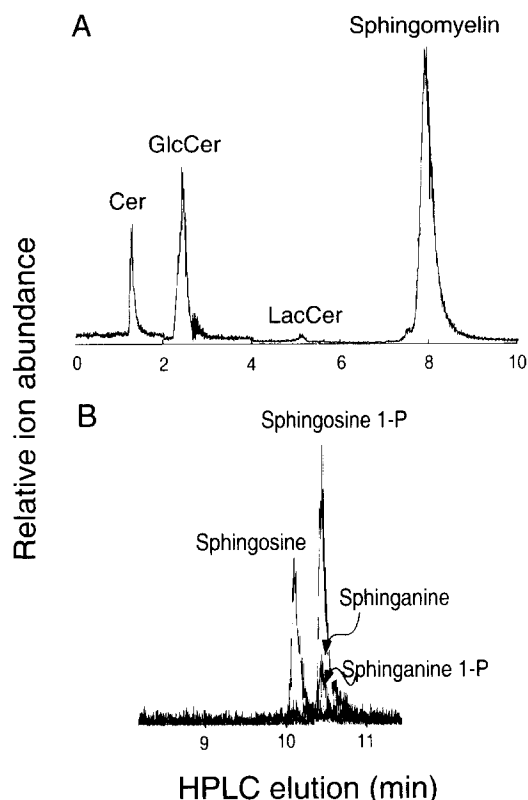


Fig. 7. A high-performance electrospray tandem mass spectrometry (HPLC–ESI–MS/MS) total ion chromatogram of endogenous levels of the complex sphingoid bases ceramide (Cer), glucosylceramide (GlcCer), lactosylceramide (LacCer), and sphingomyelin (SM) from NIH 3T3 cells (upper panel A). The cells were treated with base to remove glycerolipids and the organic solvent-soluble compounds were separated on a normal phase column to produce the profile shown. The amounts of each species can be quantified by comparison with spiked internal standards (Cer, GlcCer and SM with C12 fatty acid [15]). The elution profile in panel B is the extracted ion chromatogram for the free sphingoid bases from these cells (separated by reversed-phase chromatography prior to ESI–MS/MS) from NIH 3T3 cells (dC20:0, dC20:1 and dC17:1-1-phosphate are used as internal standards) [15]. Panel C demonstrates that this methodology can also be used to analyze lysosphingolipids (lysosphingomyelin and psychosine) and *N*-methyl sphingosines (these were not detected in NIH 3T3 cells so the data are for mixtures of standards).

son, human monocytes contain (nmol per 10^6 cells): SM (1.2), GlcCer (0.032), Cer (0.027), sphingosine (0.024), sphingosine-1-phosphate (0.007), sphinganine (0.007), and sphinganine-1-phosphate (<0.001) (M.C. Sullards, unpublished). The amounts of the sphingolipid backbones (ceramides and sphingoid bases) are often small, but can be comparable to other sphingolipids in the cells (cf. GlcCer and Cer in monocytes).

Many studies add ceramides or ceramide analogs to cells in culture, enzyme assays, or other aqueous environments, but delivery is hampered by the hydrophobicity of long-chain ceramides, which have low critical micelle concentrations ($\leq 10^{-10}$ M). This is often circumvented by using short-chain ceramides (e.g., *N*-acetyl- or *N*-hexanoyl-

sphingosine) or by delivering long-chain ceramides in detergents, liposomes, or organic solvent mixtures (C. Luberto, 2000). Fluorescent ceramide analogs, such as *N*-[6-(7-nitrobenz-2-oxa-1,3-diazol-4-yl)amino]hexanoylceramide (NBD-ceramide) and boron dipyrromethane difluoride ceramide (BODIPY-ceramide) are readily taken up by cells and have proven very useful in studies of sphingolipid transport and metabolism (A. Dagan, 2000; R.E. Pagano, 2000).

2.3. Phosphosphingolipids

Sphingomyelin is the major phosphosphingolipid in mammalian tissues. The biophysical properties of sphingomyelins have been described in Section 1.1.1 with respect to their role in rafts and caveolae [5] (see also Brugger, 1999 and 2000).

Animals additionally produce ceramide phosphorylethanolamines (M.N. Nikolova-Karakashian, 2000) and ceramide phosphate (S. Bajjalieh, 2000). Fungi such as *Saccharomyces cerevisiae* and *Sporothrix schenckii* contain ceramides derivatized with inositol phosphate with or without further addition of mannose and other carbohydrates (R.C. Dickson, 1999; C.V. Loureiro y Penha, 2001).

2.4. Glycosphingolipids

Plants and fungi often contain glycosphingolipids with relatively simple carbohydrate structures [16,17] whereas animals have a wide variety of simple to complex sphingolipids [18–20], including higher-order globosides, gangliosides and sulfatides.

2.4.1. Neutral glycosphingolipids

Glucosylceramide (Glc β 1–1'Cer, or GlcCer), galactosylceramide (Gal β 1–1'Cer, or GalCer), and lactosylceramide (Gal β 1–4Glc β 1–1'Cer, or LacCer) are the most common neutral glycosphingolipids in higher organisms (Figs. 1 and 5). The glycosidic linkage to ceramide is of the β configuration in these lipids. The *gala* type, such as galabiosylceramide (Gal α 1–4Gal β 1–1'Cer), is found primarily in kidney and pancreas. Other organisms utilize additional sugars in neutral glycosphingolipids; for example, the freshwater bivalve *Hyriopsis schlegelii* has Man β 1–1'Cer and Man β 1–2Man β 1–1'Cer, and Man β 1–4Glc β 1–1'Cer occurs in plants. More complex neutral glycosphingolipids are generally derived from LacCer or Man β 1–4Glc β 1–1'Cer. Numerous compounds also contain the oligosaccharide sequence –Gal β 1–4GlcNAc β 1–3– as seen at the non-reducing end of the paragloboside Gal β 1–4GlcNAc β 1–3Gal β 1–4Glc β 1–1'Cer and a Lewis blood group-specific antigen from human adenocarcinoma, Gal β 1–4[Fuc α 1–3]GlcNAc β 1–3Gal β 1–4Glc β 1–1'Cer.

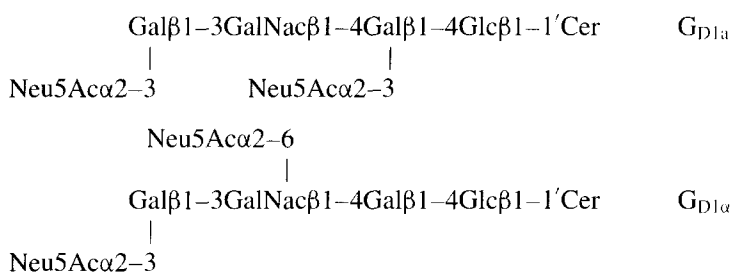
As discussed in Section 1.1.1, neutral glycolipids can form aggregates that may be important in establishing regions of the membrane with unique properties (see also T.E. Thompson, 1985).

2.4.2. Acidic glycosphingolipids

2.4.2.1. Gangliosides. Gangliosides are found in all cells of vertebrates, but in especially high amounts in the central nervous system. While diverse in structure, they

have in common one or more units of an acidic sugar called *N*-acetyl-neuraminic acid (more commonly called 'sialic acid') attached via α -glycosidic linkages to other sugars (Fig. 5). Sialic acids may have *N*-acetyl or *N*-glycolyl (i.e., hydroxyacetyl) groups at C5 (and less often C7, C8 or C9), and are distinguished by the names *N*-acetylneuraminic acid (Neu5Ac) and *N*-glycolylneuraminic acid (Neu5Gc). The simplest gangliosides contain one sialic acid and galactose or glucose, such as Neu5Gc α 2-6Glc β 1-1'Cer (found in human brain). Most gangliosides are derivatives of LacCer (Fig. 5). The addition of one sialic acid gives G_{M3} ganglioside (Neu5Ac α 2-3Gal β 1-4Glc β 1-1'Cer), which is found in many biological sources. Gangliosides frequently contain a string of two or three sialic acid residues, attached to each other in α 2-8 glycosidic linkages, examples of which are G_{D3} (Neu5Ac α 2-8Neu5Ac α 2-3Gal β 1-4Glc β 1-1'Cer) and G_{T3} (Neu5Ac α 2-8Neu5Ac α 2-8Neu5Ac α 2-3Gal β 1-4Glc β 1-1'Cer).

Gangliosides were initially classified into a few related series, such as the *ganglio* and *neolacto* types (Table 1), with the sialic acids on one or both root galactose residues of the *ganglio* type as, for example, in ganglioside G_{D1a} (shown below) from human brain. However, additional kinds of naturally occurring structures (shown below) illustrate the structural diversity that can be obtained with glycosphingolipids of even the same carbohydrate composition.



The extensive carbohydrate chains and (poly)anionic charge make many gangliosides highly amphiphilic (with critical micelle concentrations of ca. 10^{-8} M); many will partition into the aqueous phase when extracted with organic solvents.

2.4.2.2. Phosphorus-containing glycosphingolipids. As described in Section 2.3, fungi, plants, and protozoa contain sphingolipids in which ceramide is attached to an oligosaccharide via a phosphodiester linkage to *myo*-inositol and other carbohydrates.

2.4.2.3. Sulfatoglycosphingolipids. Over a dozen sulfated glycosphingolipids have been isolated from vertebrates echinoderms and microorganisms [21]. Cerebrosulfatide (3'-sulfo-Gal β 1-1'Cer, or galactosylceramide-1³-sulfate) (Fig. 1) was the first such substance to be isolated, and is the major sulfoglycolipid of brain, kidney, the gastrointestinal tract and endometrium, and is a major glycolipid of mammalian male germ cells. Sulfatides are thought to be involved in neuronal cell differentiation, myelin formation and maintenance (F.B. Jungalwala, 1994; J.L. Dupree, 1999), and other processes (D. Mamelak, 2000). Two glucuronyl (GlcA) sulfatoglycolipids (3-*O*-SO₃-GlcA β 1-3Gal β 1-4GlcNAc β 1-3Gal β 1-4Glc β 1-1'Cer and 3-*O*-SO₃-GlcA β 1-3(Gal β 1-4GlcNAc β 1-3)₂Gal β 1-4Glc β 1-1'Cer) occur in the peripheral nervous system.

2.5. *Lysosphingolipids*

Lysosphingolipids lack the amide-linked fatty acid of the ceramide backbone, which makes them highly water-soluble. A lysoglycosphingolipid with one hexose (e.g., glucosyl- or galactosyl-sphingosine) is colloquially referred to as a 'psychosine'. It has long been suspected that the appearance of psychosines, which are toxic to cells in culture, in some sphingolipid storage diseases implicates them in the pathology [22]. The cause of the toxicity is not known, but lysosphingolipids tend to form micelles and disrupt membranes, as well as inhibit protein kinase C (Y.A. Hannun, 1987) and bind to a recently discovered family of psychosine receptors (D.S. Im, 2001). Novel lysosphingolipids with alkyl groups on the sugar have been discovered and termed plasmalopsychosines (K.K. Sadozai, 1993; T. Hikita, 2001).

2.6. *Sphingolipids covalently linked to proteins*

At least two examples of covalent attachments between sphingolipids and proteins have been found. Structural proteins of the cornified cell envelope of the skin permeability barrier are covalently attached to ω -hydroxyceramides and ω -hydroxyglucosylceramides through the ω -hydroxyl groups (T. Doering, 1999; M.E. Stewart, 2001). And for some organisms, such as yeast, ceramides replace diacylglycerols in the covalently attached phosphatidylinositolglycan-linkage (Chapter 2) that is used to attach the proteins to membranes (A. Conzelmann, 1995).

2.7. *Sphingolipids in food*

Unlike other lipids, for which dietary consumption has long been known, the per capita consumption in the United States has only recently been estimated to be 115–140 g/year, or 0.3–0.4 g/day [23]. The amounts in any given food vary considerably, from a few $\mu\text{mol/kg}$ in fruits and some vegetables to ca. 2 mmol/kg (1–2 g/kg) in dairy products, egg and soybeans. Most foods of mammalian origin (beef, milk, poultry, etc.) have sphingomyelins, cerebroside, globosides, gangliosides, sulfatides, etc., that are comprised of ceramide backbones with sphingosine (d18:1 Δ^4) or 4-hydroxysphinganine (t18:0). The complex sphingolipids of plants are mainly cerebroside (mono- and oligohexosylceramides, with glucose the most common hexose), and ceramides that have relatively little d18:1 Δ^4 and mostly d18:1 Δ^8 , d18:2 $\Delta^{4,8}$, t18:0 and t18:1 Δ^8 sphingoid bases with α -hydroxy fatty acids.

3. *Biosynthesis of sphingolipids*

The general pathways for sphingolipid metabolism are reasonably well characterized [24] and their regulation is beginning to be understood through the development of specific inhibitors and identification of the genes for many of these enzymes.

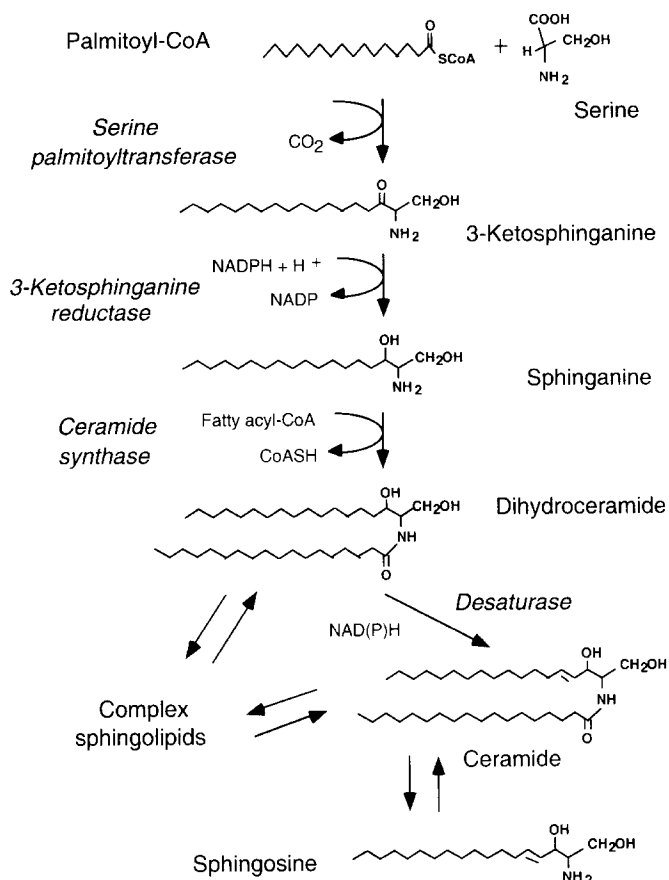


Fig. 8. Biosynthetic pathway for ceramide and sphingosine. Note that free sphingosine is formed after ceramide turnover, not as an intermediate of de novo biosynthesis.

3.1. Sphingoid bases and ceramide

3.1.1. Synthesis of the long-chain base backbone

Ceramide synthesis begins with the condensation of palmitoyl-CoA and L-serine (Fig. 8), catalyzed by the pyridoxal phosphate-dependent enzyme serine palmitoyltransferase. The reaction proceeds with overall retention of configuration of C2 of serine via the proposed mechanism shown in Fig. 9.

As would be predicted from this mechanism, serine palmitoyltransferase undergoes 'suicide' inhibition by β -halo-L-alanines and L-cycloserine. More potent and selective inhibitors have been isolated from microorganisms and are listed in Table 2. One is available commercially (myriocin or ISP-1) (Fig. 10), and is illustrated in Fig. 9 as the likely adduct that it forms with serine palmitoyltransferase.

For mammals and yeast, at least two gene products (termed SPTLC1 and SPTLC2, or sometimes SPT1 and SPT2) are necessary for activity (R.C. Dickson, 2000) and appear to be physically associated (K. Gable, 2000; K. Hanada, 2000). Human genetic

Table 2
Inhibitors of sphingolipid metabolism

Enzyme	Inhibitor	Reference
Serine palmitoyltransferase	Cycloserine	M. Lev, 1984
	β -Fluoroalanine	K.A. Medlock, 1988
	Sphingofungin C	M.M. Zweerink, 1992
	Lipoxamycins	S.M. Mandela, 1994
	ISP-1/myriocin	Y. Miyake, 1995
	Viridofungin A	S.M. Mandela, 1997
Ceramide synthase	Fumonisin B ₁ , B ₂ , etc.	E. Wang, 1991
	Alternaria toxin	A.H. Merrill, 1993
	Australifungins	S.M. Mandela, 1995
	<i>N</i> -acylaminopentols	H. Humpf, 1998
Glc : ceramide synthase	PDMP (D- <i>threo</i> -1-phenyl-2-decanoylamino-3-morpholino-1-propanol) and related analogs	N.S. Radin, 1990; A. Abe, 1995;
	<i>N</i> -butyldeoxynojirimycin	J.S. Shayman, 2000
		F.M. Platt, 1997
Lactosylceramide synthase	Epoxy-glucosylceramide	C. Zacharias, 1994
Sphingomyelin synthase	PDMP (at very high concentrations)	A. Abe, 1995
	D609	C. Luberto, 1998
Inositolphosphorylceramide synthase	Khafrefungin	S.M. Mandala, 1997
	Aureobasidin A	M.M. Nagiec, 1997
	Rustimicin	S.M. Mandala, 1998
Sialidase	2-Deoxy-2,3-dehydro- <i>N</i> -acetylneuraminic acid	P. Meindl, 1969; S. Usuki, 1988
Glucocerebrosidase	Conduritol B-epoxide	G. Legler, 1977; S. Mahdiyyoun, 1992
Acidic sphingomyelinase	Phosphatidylinositol 4',5'-bisphosphate, adenosine 3',5'-diphosphate, adenine-9-beta-D-arabinofuranoside 5'-monophosphate	L.E. Quintern, 1987
	SR33557 ((2-isopropyl-1-(4-[3- <i>N</i> -methyl- <i>N</i> -(3,4-dimethoxy- β -phenethyl) amino]propyloxy)-benzene sulfonyl))indolizine	J.P. Jaffrezou, 1992
Neutral sphingomyelinase	3- <i>O</i> -methyl-sphingomyelin	M.D. Lister, 1995
	Glutathione	B. Liu, 1997
	Difluoromethylene sphingomyelin	T. Yokomatsu, 2001
	Scyphostatin and analogs	T. Izuhara, 2001; C. Arenz, 2001
Ceramidase	<i>N</i> -Oleoyl-ethanolamine	M. Sugita, 1975
	D-MAPP	A. Bielawska, 1996
Sphingosine kinase	D- and L- <i>threo</i> -sphingosine (and sphinganine)	B.M. Buehrer, 1993
	S-15183a and b	K. Kono, 2001
Sphingosine 1-phosphate lyase	4-Deoxypyridoxine-5' phosphate	W. Stoffel, 1969;
		P. Van Veldhoven, 1993

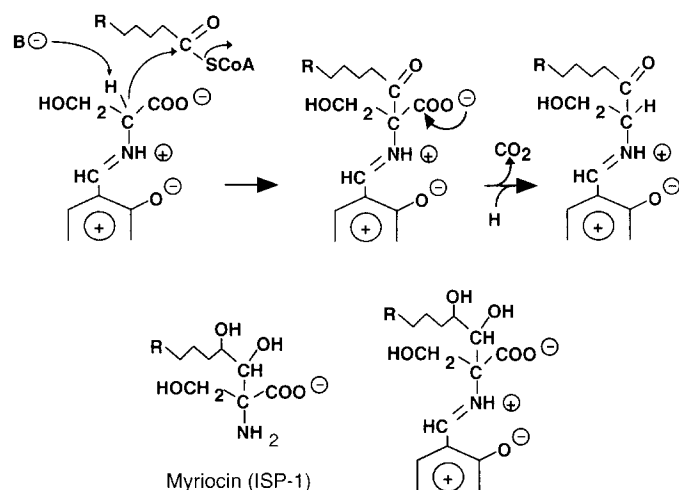
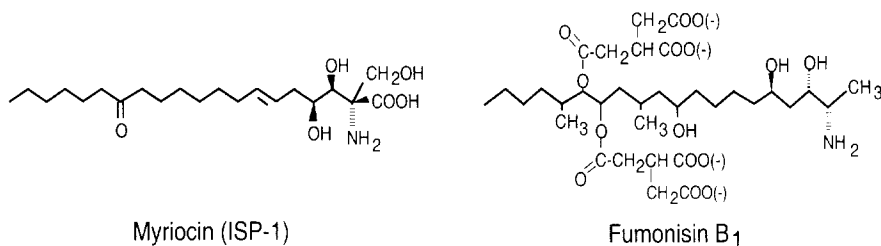


Fig. 9. A probable reaction mechanism for serine palmitoyltransferase (modified from K. Krisnangkura, 1976) and the adduct that presumably accounts for the high-affinity inhibition by myriocin (ISP1) (for the full structure of myriocin see Fig. 10).



Inhibitor of serine palmitoyltransferase

Inhibitor of ceramide synthase

Fig. 10. Representative inhibitors of sphingolipid biosynthesis that have been isolated from microorganisms. For more inhibitors and references for these compounds, see Table 2.

defects in SPTLC1 have been shown to cause hereditary sensory neuropathy type I. In yeast, an 80-amino acid polypeptide has been shown also to affect activity (K. Gable, 2000). Relatively little is known about the structure of serine palmitoyltransferase due to its membrane association; however, a soluble, homodimeric enzyme is produced by *Sphingomonas* (H. Ikushiro, 2001) and should be amenable to structural and mechanistic studies.

Sphingoid base synthesis can be suppressed by addition of lipoproteins or free sphingoid bases to cells in culture, and studies with a phosphorylated but poorly degraded analog (*cis*-4-methylsphingosine) indicates that sphingoid base 1-phosphate(s) down-regulate serine palmitoyltransferase activity (G. van Echten-Deckert, 1997). Regulation at a transcriptional level has been seen mostly with cells in culture (reviewed in Linn et al. [25]) but sometimes in vivo (R.A. Memon, 2001) and include endotoxin and

cytokines, UV irradiation, retinoic acid, corticosteroids, and phorbol esters. Transcriptional regulation appears to involve mainly changes in SPTLC2 mRNA. There is also post-translational activation of serine palmitoyltransferase in response to etoposide in mammalian cells (D.K. Perry, 2000) and heat shock in yeast (G.M. Jenkins, 2001). The heat shock response is interesting, not only with respect to the downstream pathways that are affected, such as amino acid transport (M.S. Skrzypek, 1998) and ubiquitin-dependent proteolysis (N. Chung, 2000), but also because the long-chain bases that are elevated are predominantly eicosasphingamines (i.e., 20-carbon atoms in length) (R.C. Dickson, 1997).

The next step of sphingoid base synthesis, the reduction of 3-keto-sphinganine (Fig. 8), is apparently rapid *in vivo* because the 3-keto-intermediate is rarely detected. The gene for this reductase has been identified in yeast (T. Beeler, 1998).

3.1.2. Synthesis of the *N*-acyl-derivatives of sphingoid bases

As shown in Fig. 8, free sphinganine is acylated to dihydroceramides by ceramide synthase(s), which utilize a wide variety of sphingoid bases and fatty acyl-CoAs, and may be a family of isoenzymes (E. Wang, 2000). Ceramide can also be made by reversal of ceramidase (S. El Bawab, 2001); however, this reaction appears to account for relatively little ceramide synthesis under normal physiological conditions. Recent studies (I. Guillas, 2001) have identified yeast (Lag1P and Lac1P) and mammalian (K. Venkataraman, 2002) genes that encode ceramide synthase or are obligatory for ceramide synthase activity.

Microorganisms produce a number of inhibitors of ceramide synthase (Table 2). Fumonisin (Fig. 10), which are produced by some species of *Fusaria*, were discovered as causes of human esophageal cancer, equine leukoencephalomalacia, and porcine pulmonary edema (W.F.O. Marasas, 2001), and are now known to induce multiple pathologies, including liver and kidney toxicity and carcinogenicity, immunosuppression (and in some cases immunostimulation), and birth defects (A.H. Merrill, 2001). Fumonisin not only block complex sphingolipid formation, but also cause sphinganine to accumulate, which is both key to fumonisin toxicity (E.M. Schmelz, 1998) and provides a useful biomarker for exposure of organisms to this mycotoxin. There can also be elevation of sphinganine 1-phosphate [15], which might explain how fumonisins can be toxic and mitogenic for different cell types.

The last step of ceramide synthesis is the insertion of the 4,5-*trans*-double bond into the sphingoid base backbone, which occurs at the level of dihydroceramide (Fig. 8) (J. Rother, 1992). Therefore, free sphingosine *per se* is not an intermediate of sphingolipid biosynthesis *de novo*. The genes for desaturases have been characterized in plants (P. Sperling, 2000) and mammals (P. Ternes, 2002). For 4-hydroxysphingamines (phytosphingosine), insertion of the 4-hydroxyl group occurs at the level of sphinganine in yeast (M.M. Grilley, 2000) and plants (P. Sperling, 2000).

3.2. Sphingomyelin and ceramide phosphorylethanolamine

Sphingomyelin is synthesized by transfer of phosphorylcholine from phosphatidylcholine to ceramide, liberating diacylglycerol. This reaction links glycerolipid and

sphingolipid signaling pathways, although it is not known if cells frequently capitalize on this relationship for signaling purposes. Most *de novo* sphingomyelin synthesis occurs in the Golgi apparatus (in liver); however, synthesis also occurs in the plasma membrane of many cell types and is probably a major site as well. Relatively little is known about the regulation of sphingomyelin biosynthesis, but it has interesting features such as stimulation by phorbol esters, 25-hydroxycholesterol, and Brefeldin A (G. Hatch, 1992; N. Ridgway, 1995), increases during development of the lung (C.A. Longo, 1997), decreases in aging (S.A. Lightle, 2001), and changes very early in the development of colon cancer (P.K. Dudeja, 1986).

Ceramide phosphorylethanolamine is synthesized from phosphatidylethanolamine and ceramide in a reaction analogous to sphingomyelin synthesis, and once formed can be methylated to sphingomyelin (M. Malgat, 1986; M.N. Nikolova-Karakashian, 2000). Inositolphosphoceramides are formed by transesterification (from phosphatidylinositol) (R.C. Dickson, 1999; A.S. Fischl, 2000) by transferases that share a conserved structural motif across yeast and pathogenic fungi, and which resembles somewhat a motif of lipid phosphatases (S.A. Heidler, 2000).

3.3. Neutral glycosphingolipids

Pathways for the biosynthesis of the different root glycosphingolipids (Figs. 11 and 12) appear complex, but are achieved by surprisingly few glycosyltransferases that commit precursors and intermediates to predictable products based on the specificities of the enzymes. The enzymes transfer a specific sugar from the appropriate sugar nucleotide (e.g., UDP-Glc, UDP-Gal, etc.) to ceramide or the non-reducing end of the growing carbohydrate chain attached to ceramide. The glycosyltransferases often recognize mainly the carbohydrate portion of the acceptor glycosphingolipid; however, ceramides with α -hydroxy-fatty acids are preferentially incorporated into GalCer (and sulfatides), whereas those with non-hydroxy-fatty acids are used to make GlcCer (I. van Genderen, 1995).

GalCer and GlcCer are synthesized with inversion of the configuration of the

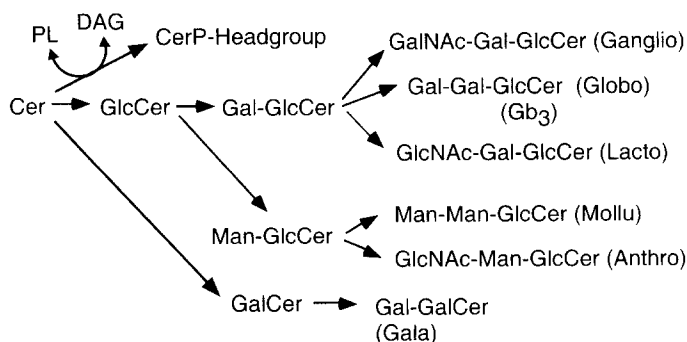


Fig. 11. Biosynthesis of phosphosphingolipids and 'root' glycosphingolipids. For phosphosphingolipids, the headgroup is transferred from a phosphoglycerolipid, such as phosphatidylcholine for sphingomyelin. For glycosphingolipids, the headgroup is transferred from a UDP-sugar.

The synthesis of GlcCer and GalCer can be inhibited by structural analogs of ceramide such as PDMP (Table 2), which decreases cellular levels of neutral glycosphingolipids and gangliosides and elevates ceramide, causing cell cycle arrest (C.S.S. Rani, 1995). GlcCer synthesis appears to be a major determinant of survival of tumor cells (by removal of ceramide and affecting multidrug resistance) (Y.Y. Liu, 2001; A. Senchenkov, 2001). But, the inability to make detectable glycolipids due to deficient GlcCer synthase is not lethal to some cell lines, although it slows growth and results in an elongated fibroblastic morphology (S. Ichikawa, 1994).

Additional glycosyltransferases are involved in the synthesis of the other neutral glycolipids as well as the addition of neutral sugars to gangliosides (Figs. 11 and 12), and many of the genes for these enzymes have been identified, including lactosylceramide synthase (T. Takizawa, 1999), globosylceramide (Gb3) synthases (Y. Kojima, 2000; J. Keusch, 2000), β -1,3-galactosyltransferases (M. Amado, 1999), β 1,3 *N*-acetylglucosaminyltransferase (Lc3 synthase) (T. Henion, 2001) as well as numerous fucosyltransferases for Lewis $\times$ antigens. Labs are already beginning to engineer surface glycosphingolipids of desired composition by transfecting cells with combinations of these and other enzymes (E.G. Prati, 2000).

Synthesis of GlcCer occurs on the cytosolic aspect of the endoplasmic reticulum and/or early Golgi membranes (A.H. Futerman, 1991; D. Jeckel, 1992), whereas more complex neutral glycosphingolipids (beginning with LacCer) are made in the lumen of the Golgi apparatus (H. Lannert, 1994). Therefore, GlcCer and the sugar nucleotides must undergo transbilayer movement to the lumen of the Golgi for the synthesis of more complex sphingolipids. Sphingolipids are generally thought to reach their membrane locations via trafficking from Golgi; however, newly synthesized GlcCer can also be transported to the plasma membrane via a non-Golgi pathway (D.E. Warnock, 1994).

3.4. Gangliosides

The general pathway for the synthesis of gangliosides has evolved to the scheme depicted in Fig. 12 [24]. Gangliosides are synthesized by the stepwise transfer of neutral sugars and sialic acids by membrane-bound glycosyltransferases that are located in the regions of the Golgi apparatus that generally correspond to the order in which the sugars are added. For example, the sialyltransferase catalyzing the synthesis of G_{M3} ganglioside is in the *cis* Golgi, whereas the enzymes involved in terminal steps in ganglioside synthesis are localized in the more distal *trans* Golgi network. Ganglioside biosynthesis can also involve the introduction of *O*-acetyl groups on sialic acid and *N*-deacetylation to produce a free amino group on position 5 of sialic acid. Gangliosides are incorporated into the outer leaflet of the plasma membrane by vesicle-mediated transport.

There has been considerable progress in identification of the genes responsible for the key reactions in Fig. 12, allowing these relationships to be tested by transfecting cells with the cDNA for enzymes of this pathway and determining the types of glycosphingolipids that are made; for example (a) transfection of GalNAc-transferase cDNA into Chinese hamster ovary cells, which normally make mainly G_{M3} , produced cells that now synthesize mainly G_{D2} , whereas transfection of cells that are defective in

sialylation yielded G_{A2} (M.S. Lutz, 1994), (b) transfection of several cell lines with a GalNAc-transferase yielded G_{A2} , G_{M2} (the preferred product when both G_{M3} and LacCer were available), G_{D2} , GalNAc sialylparagloboside and GalNAc G_{D1a} (S. Yamashiro, 1995), and (c) transfection of the cDNA for G_{D3} synthase (sialyltransferase II in Fig. 12) into Neuro2a cells increased G_{D3} and G_{Q1b} (N. Kojima, 1994). The findings to date support the view that ganglioside synthesis can be viewed as 'combinatorial' reactions (Fig. 12) catalyzed by a conservative number of key enzymes such that the ultimate composition is determined by the relative activities of these enzymes and the availability of their substrates [27].

Regulation of ganglioside biosynthesis involves both transcriptional and post-transcriptional factors. Transcriptional control of key glycosyltransferases appears to account for many of the developmentally regulated, tissue-selective variations in ganglioside amounts and types in mammalian organs, including large changes with oncogenic transformation (R.W. Ledeen, 1998; S. Hakomori, 1998). The biosynthesis of gangliosides is also controlled through post-translational modification of glycosyltransferases (R.K. Yu, 2001); several sialyltransferases (X. Gu, 1995) are down-regulated by protein kinase C in cell-free and intact cell systems and *N*-acetylgalactosaminyl-transferase can be up-regulated by protein kinase A in cultured cells.

3.5. Sulfatoglycosphingolipids

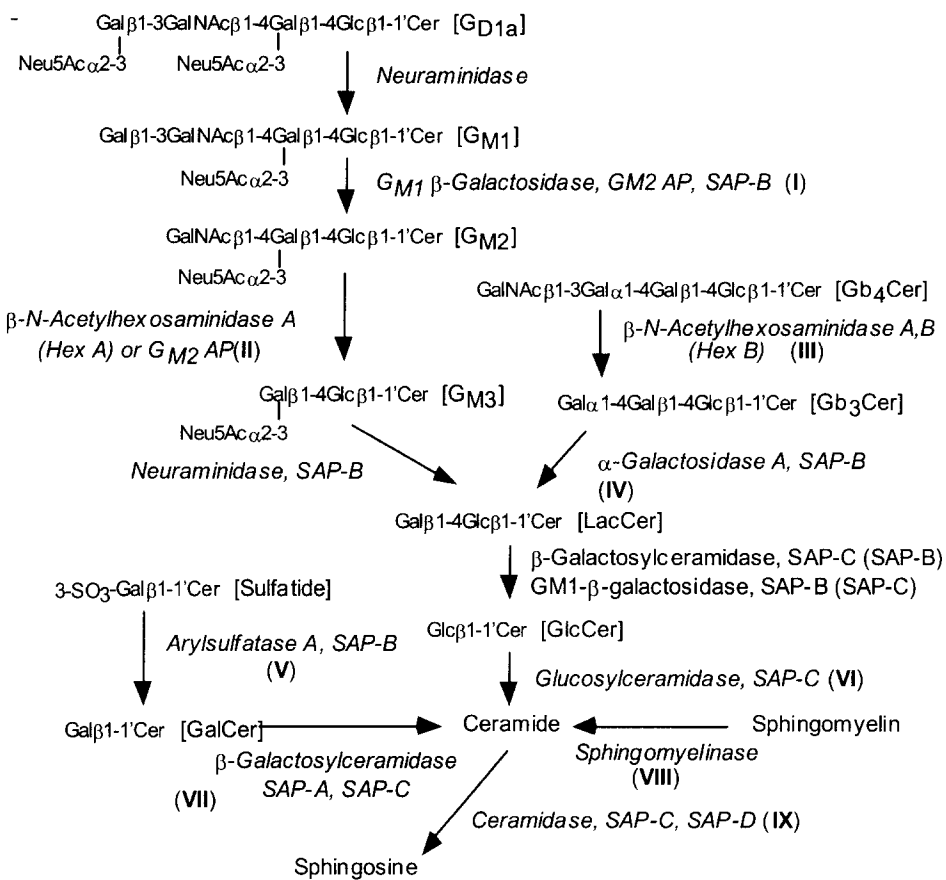
Sulfatide (3'-sulfo-galactosylceramide) synthesis is catalyzed by GalCer sulfatotransferase (3'-phosphoadenylylsulfate : galactosylceramide 3'-sulfotransferase), which utilizes the activated sulfate donor 3'-phosphoadenosine-5'-phosphosulfate (PAPS). The cDNA encoding the sulfotransferase has been cloned (T. Honke, 1997). Regulation of sulfatide biosynthesis appears to reside in the activity of this sulfatotransferase.

4. Sphingolipid catabolism

Complex sphingolipids are lost from cells by (1) membrane internalization, recycling, and degradation, (2) hydrolysis to release bioactive products that participate in cell signaling, and (3) release from the cells by secretion or shedding.

In general, sphingolipids are internalized with endocytic vesicles, sorted in early endosomes, and recycled back to the plasma membrane (often with remodeling of the sphingolipid) or transported to lysosomes where they are degraded by specific acid hydrolases. Given that lysosomal membranes are rich in sphingolipids, it has been unclear why they, too, do not undergo hydrolysis if the endocytosed membranes and lysosomal membranes are simply fused. This dilemma was solved when it was shown (W. Furst, 1992; W. Mobius, 1999) that endocytosed sphingolipids (and presumably other components) become invaginated into intraendosomal vesicles that are delivered into the lumen of the lysosome. Thus, hydrolytic enzymes contact the sphingolipids to be digested in the lumen rather than as part of the lysosomal membrane, which is additionally protected by an elaborate glycocalyx that lines the inner leaflet.

The pathways for sphingolipid catabolism (Fig. 13) converge on ceramide; nonethe-



Inherited disorders of sphingolipid catabolism:	
I. Generalized gangliosidosis	VI. Gaucher disease
II. Tay-Sachs disease (Sandhoff AB variant)	VII. Globoid cell leukodystrophy;
III. Sandhoff disease	Krabbe disease
IV. Fabry disease	VIII. Niemann Pick disease
V. Metachromatic leukodystrophy	IX. Farber disease

Fig. 13. Catabolism of complex sphingolipids and associated diseases [3,24].

less, cells contain at least small amounts of other lysosphingolipids, such as galactosylsphingosine (psychosine) and sphingosylphosphorylcholine, implying that such compounds may arise from as-yet-to-be-characterized enzymatic activities analogous to phospholipase A₂, or by de novo biosynthesis.

4.1. Sphingomyelin

The lysosomal hydrolysis of sphingomyelin to ceramide and phosphocholine is catalyzed by acid sphingomyelinase, a water-soluble, lysosomal glycoprotein that interacts with a sphingolipid activator protein (SAP-C) and anionic lipids such as bis(mono-acylglycero)phosphate (T. Linke, 2001). Individuals may display a number of different molecular defects that cause insufficient lysosomal sphingomyelinase activity and result in accumulation of sphingomyelin in reticuloendothelial cells scattered throughout the spleen, bone marrow, lymph nodes, liver, and lungs. These are called Niemann–Pick disease Types A and B, and sphingomyelin also accumulates in Type C Niemann–Pick disease, but not due to genetic defects in sphingomyelinase per se. Screening for human Niemann–Pick disease is used to identify carriers and for prenatal diagnosis of affected fetuses. Acid sphingomyelinase-deficient mice have been generated (K. Horinouchi, 1995) as models for Niemann–Pick disease.

Sphingolipid turnover in other cellular compartments involves both acid sphingomyelinase and sphingomyelinases with neutral to alkaline pH optima. Acidic sphingomyelinase is also secreted, and the secreted form requires supplemental Zn^{2+} for activity (I. Tabas, 1999). Neutral sphingomyelinases have been found in other cellular compartments, including the nuclear membrane, and at least one form resides in sphingolipid-enriched microdomains and is inhibited by the caveolin-scaffolding domain (R.J. Veldman, 2001). These are probably involved in cell signaling.

A sphingomyelinase D is found in the venom of brown recluse spiders, *Corynebacterium pseudotuberculosis* (which commonly infects sheep), *Vibrio damsela* (an aquatic bacterium that causes wound infections in humans) and the human pathogen *Arcanobacterium haemolyticum*. This enzyme, which yields ceramide 1-phosphate and choline, is able to produce much of the tissue damage caused by these organisms (A.P. Truett, 1993).

4.2. Glycosphingolipids

Glycosphingolipids are catabolized by the stepwise hydrolysis of the terminal monosaccharides through the concerted action of a series of specific exoglycosidases (Fig. 13). For the in vivo degradation of glycolipids with short oligosaccharide headgroups, i.e. of less than four carbohydrate residues, there is often a requirement for sphingolipid activator proteins (or saposins) SAP-A, -B, -C or -D. A number of inherited diseases are caused by mutations in the structural genes for these enzymes that result in reduced enzymatic activity, loss of the appropriate targeting signals for transport to lysosomes, or alteration of the domains that interact with other subunits of the enzyme and/or activator proteins (Fig. 13) [24].

That sphingolipidoses can result from several types of genetic defects is exemplified the $\text{G}_{\text{M}2}$ gangliosidoses, which arise from mutations of β -*N*-acetylhexosaminidase (Hex A or Hex B) (Fig. 13) or $\text{G}_{\text{M}2}$ activator proteins (R.A. Gravel, 1995) [24]. Hex A is an $\alpha\beta$ heterodimer and degrades negatively charged and uncharged substrates, whereas Hex B is the $\beta\beta$ homodimer and cleaves mainly *N*-acetylgalactosamine residues from uncharged substrates such as $\text{G}_{\text{A}2}$, globotetraosylceramide and oligosaccharides.

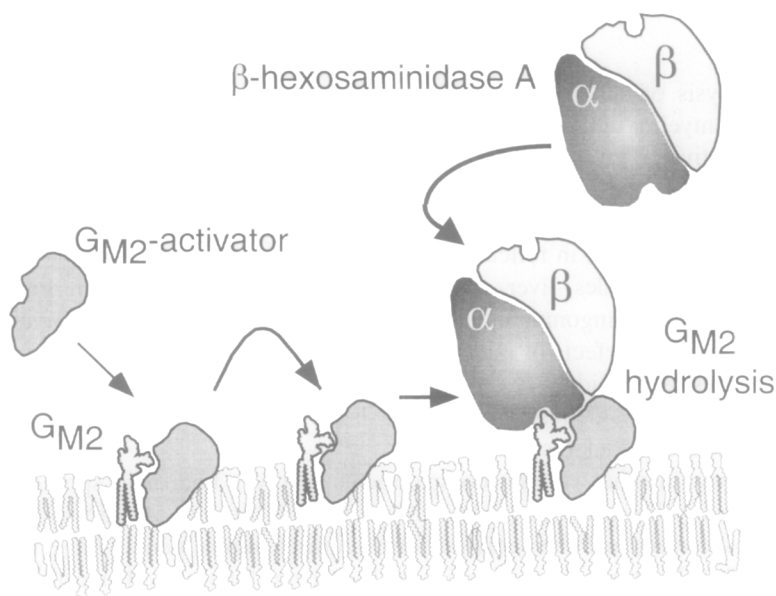


Fig. 14. A model for the role of G_{M2} activator protein (SAP) in the hydrolysis of ganglioside G_{M2} by β-hexosaminidase A (subunits α and β). For more information, see Schuette et al. [3] and Kolter and Sandhoff [24].

Therefore, mutations in the structural gene for the α subunit result in partial or complete loss of Hex A activity (Tay–Sachs disease), and mutations in the β subunit affect both Hex A and Hex B (Sandhoff disease). The phenotypes of these two gangliosidoses are similar, but they are easily distinguished by measuring Hex A and Hex B activities and by the accumulation of Gb₄Cer in Sandhoff disease but not in Tay–Sachs disease (G_{M2} accumulates in both disorders). The AB variant has normal levels of both Hex A and Hex B activity when measured *in vitro*, but there is a defective (or absent) G_{M2} activator protein. The G_{M2} activator protein is a membrane active protein that also binds ganglioside G_{M2} as well as structurally related gangliosides forming complexes (usually in a 1 : 1 molar ratio) that present them to hexosaminidase A as illustrated in Fig. 14.

4.3. Ceramide

The major pathway for catabolism of the ceramide backbone is shown in Fig. 15.

In lysosomes, ceramides are hydrolyzed to free sphingoid bases and long-chain fatty acids by a ceramidase that has an acidic pH optimum. The lysosomal ceramidase is a water-soluble glycoprotein that hydrolyzes membrane-bound ceramide in an interfacial reaction that needs the stimulation by SAP-D and anionic phospholipids such as the lysosomal bis(monoacylglycerol)phosphate (A. Klein, 1994; T. Linke, 2001). The human acidic ceramidase is a heterodimeric enzyme of 40 kDa and 13 kDa subunits synthesized as a single precursor polypeptide of approximately 53–55 kDa and targeted to the lysosome via the mannose 6-phosphate receptor (K. Ferlinz, 2001). Additional